Aalto-yliopisto
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Aalto University

Master's Programme in Accounting

"*You Will Be Replaced!*" *for the 1000th time*

A Task-Level Analysis of Generative AI Exposure in the Record-to-Report Process

Vinh Le

Master's thesis
2026

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Title of thesis *"You Will Be Replaced!" for the 1000th time:* A Task-Level Analysis of Generative AI Exposure in the Record-to-Report Process

Programme Master's of Science in Economics and Business Administration

Major Accounting

Thesis supervisor Prof. Jukka Sihvonen

Date 16.10.2026 **Number of pages** 99+35 **Language** English

Abstract

Generative AI (GenAI) is widely predicted to transform accounting work, yet evidence on how it is actually used in accounting remains scarce and is drawn largely from capability ratings, job postings, and self-reported surveys. This thesis measures observed GenAI exposure in the Record-to-Report (R2R) process at the task level, using five releases of the Anthropic Economic Index, a privacy-preserving record of real Claude usage. The study asks not only how much of R2R is exposed to GenAI, but what the location, composition, and effectiveness of that exposure imply for the reallocation of accounting work between people and machines. Three research questions guide the analysis: where GenAI exposure is concentrated within R2R, at what level of expertise, and how effective is that exposure.

Although almost every R2R task is theoretically exposed, observed exposure is realised in only a minority of process time and concentrates in a narrow core of tasks, most visibly in Record and Close & Consolidation. The expertise level of exposure is bimodal, with a low-expertise conforming regime in Record and a high-expertise anchoring regime in Master Data Management and Analysis & Business Insight. Once task success is accounted for, Claude effectively performs only about a quarter of process time, with the most automated subprocesses converting least reliably into performance. Success depends on platform-specific burdens—human-only task time on Claude.ai and input context on 1P API.

The thesis contributes evidence from the layer between perception and outcome that accounting research has so far left unobserved, using records of real-world usage data. Exploring exposure by task within a single accounting process offers early evidence of how work is being redistributed between humans and machines along the Record-to-Report process.

Keywords Generative AI, accounting, Record-to-Report, Anthropic Economic Index, labour reallocation, task-based approach

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Preface and acknowledgements

It's hard to believe this thesis was started 9 months ago. October has been just a blink and poof I am, apparently, done with my Master's thesis. It's like something has been chasing me that I completed it so early. Or maybe it was me chasing something with the never-ending wonderings and ponderings at the age of 21, what good could I do? What great could I be? What great should I be?

Or maybe when you enjoy something, it does not feel like working. A funny observation I have gathered throughout my thesis process, was that I did not dread it. This might be pretentious, and such acknowledgement might simply target the very people who I want dearly to impress, so, I'm going to say it: "I enjoy research". I adore the process of reading and patching pieces together, to come up with a gap, and feel like I'm on top of the world and nobody knows anything better than I do. I adore how the next day I would also realize I'm the dumbest person in the world, and I'm so incompetent that I would never be able to make something out of this. I love to design, to experiment, to choose, to wreck my brains out trying to figure out a formula, a sentence, a problem that seems non-existent the day before but determines the whole trajectory of my thesis the day after! To fear, to question, to explore, to be curious, to fail, to learn, and to grow. Somehow, this excites me all very much, the prospects of expanding infinitely, quote my brain, and perhaps I told quite a few people already, "You change the topic; you're learning something new! You change the methodology; you're exposing yourself again to the endless land of knowledge." The more you know, the less you know. I haven't been feeling much like learning lately, but maybe, as a point of reiterating, I have been learning so much, so immersed, that it feels like I'm just, floating.

This could not have happened without my supervisor, Jukka Sihvonen. I remember being the only student in your cohort not knowing what to do, and phrased such lost-of-direction in a manner fancy enough that it cackled us all up. It was Jukka who introduced me to the topic, and that moment indefinitely sparked something in me, in how I was able to execute the whole project with so much learning, struggling, and floating, all unfolding to culminate to something I am proud of myself to pull through, and grateful that I have the opportunity, the possibility to pull through at all.

To mom and dad and my brother, who have always been the most stable foundation that makes me feel so safe, so grounded, so gentle with how I carry myself in the thesis. To the day when I was crying to you that I had an 88/100, but all you could think about was how to send me the food that I mentioned. Oh, to be surrounded and nurtured in so much love.

To adventure ever to come.

Espoo, October 16th, 2026
Vinh

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Symbols and abbreviations

Symbols

t	an individual O*NET task
Sp	an R2R subprocess
α_t	automation factor for task t
β_t	theoretical exposure indicator for task t (Eloundou et al., 2024)
$\tilde{r}_t$	task-level observed exposure
$R_{o,sp}$	subprocess-level observed exposure
$R_{o,p}$	process-level (pooled) observed exposure
N	conversation volume used to scale the concentration score
w_t	task time-fraction weight
w_t^{norm}	time-fraction weight renormalised to relevant scope
K_t	set of platforms (Claude.ai, 1P API) reporting a task-success reading for task t
s_{tk}	share of task t's conversations on platform k classified as successful
$success_t$	task-level success rate (%), zero for untouched tasks
α	Krippendorff's alpha
β	slope coefficient of a fitted line
R^2	coefficient of determination
ρ	Spearman's rank correlation coefficient
Δ	change from Release 4 to Release 5

Operators

Σ	summation over the indicated index
ln()	natural logarithm
$\times$	multiplication

Abbreviations

1P API	First-party Application Programming Interface
ACCA	Association of Chartered Certified Accountants

AEI	Anthropic Economic Index
AI	Artificial Intelligence
AICPA	American Institute of Certified Public Accountants
API	Application Programming Interface
APQC	American Productivity and Quality Center
BLS	Bureau of Labor Statistics
CI	Confidence Interval
ESG	Environmental, Social, and Governance
GenAI	Generative Artificial Intelligence
GPT	Generative Pre-trained Transformer
IPO	Initial Public Offering
LLM	Large Language Model
OEWS	Occupational Employment and Wage Statistics
OLS	Ordinary Least Squares
O*NET	Occupational Information Network
R2R	Record-to-Report
RMSE	Root Mean Square Error
RPA	Robotic Process Automation
SOC	Standard Occupational Classification

1 Introduction

Accounting is consistently cited among the occupations most exposed to Generative Artificial Intelligence (GenAI). Exposure frameworks place the work of accountants, auditors, and tax preparers almost entirely within the reach of large language models (Eloundou et al., 2024), general-purpose models now perform at or above the standard of the major accounting certification exams (Eulerich et al., 2024), and professional bodies document applications across invoice processing, reconciliation, reporting, and forecasting (Brisbourne, 2024). Yet, the evidence on how far that exposure has materialised remains confined to a few kinds of data. Across the economy, exposure is predicted from capability ratings of occupational tasks, inferred from job postings and firm-level proxies, or self-reported in surveys (Acemoglu et al., 2022; Bick et al., 2026; Hampole et al., 2025), measures that disagree most where exposure is highest (Gimbel et al., 2026) and that diverge from the usage they are meant to approximate (Handa, Stern, et al., 2025; Massenkoff & Huang, 2026). In accounting specifically, the few attempts to observe actual use, similarly, rely on AI-related job postings (Böke et al., 2024), AI detection in corporate disclosures (Blankespoor et al., 2026), or the logs of a single accounting platform (Choi & Xie, 2026). What remains missing is evidence of where and in what manner GenAI is actually being used across the full span of accounting work.

This thesis addresses that gap by tracing observed GenAI exposure through an accounting-specific process, Record-to-Report (R2R), the end-to-end cycle through which an organisation maintains its financial architecture, records its transactions, reports its results, and interprets them to plan the next period. It does so with real-world Claude usage data using the Anthropic Economic Index dataset and interprets that usage through the task-based framework of labour economics. In such framework, technology reallocates tasks between labour and capital rather than replacing occupations as a whole (Acemoglu & Autor, 2011; Acemoglu & Restrepo, 2019), where the consequences of exposure depend on how concentrated it is across tasks (Hampole et al., 2025) and on the expertise of the tasks it reaches (Autor & Thompson, 2025). The study asks not only how much of R2R is exposed to GenAI, but what the location, composition, and effectiveness of that exposure imply for the reallocation of accounting work between people and machines. Three research questions guide the analysis:

- **RQ1:** Where is GenAI exposure concentrated within the Record-to-Report process?
- **RQ2:** What is the expertise level of such exposure and concentration?
- **RQ3:** How effective is such exposure?

The empirical basis is the Anthropic Economic Index, a series of periodic releases that map privacy-preserving records of real Claude conversations onto O*NET occupational tasks (Appel et al., 2026; Handa, Tamkin, et al., 2025). Five data releases are consolidated into a single panel and restricted to 91 O*NET tasks across the six R2R subprocesses: Master Data Management, Record, Close & Consolidation, Report, Analysis & Business Insight, and Planning & Budgeting. On this panel, the study builds a multi-dimensional exposure profile following the logic of the research questions. Observed exposure measures how much of each subprocess's time is carried by tasks that are not only theoretically capable but actually used, and used in an automated rather than augmented manner (RQ1). Exposure concentration captures whether that exposure spreads across a subprocess or narrows onto a few tasks (RQ1). Expertise composition locates those exposed and concentrated tasks across the subprocess cognitive complexity (RQ2). Effective exposure discounts exposure by the rate at which Claude succeeds with the tasks it is given (RQ3). Together, these constructs move from where exposure sits, to what it is made of, to whether it converts into performance.

The study found that the R2R process is fully exposed to GenAI in capability, narrowly exposed in use, and effectively exposed for about a quarter of its time. Although almost every R2R task is theoretically within GenAI's reach, automated, observed exposure is realised for only a limited share of process time and is heterogeneous across subprocesses, ranging from production-scale automation, exploratory and largely augmentative engagement, to near-absent adoption. The mode of use is sorted by channel, with the API acting as an automation engine and the chat interface as an augmentation environment. Where exposure exists, it concentrates on a narrow core of tasks rather than diffusing across the process, and its expertise composition is bimodal rather than central. At the low-expertise end, exposure conforms across a bounded block of tasks, while at the high-expertise end, exposure is anchored on a few tasks that drive concentration inside subprocesses that are otherwise left largely untouched.

Measured against success, the picture shifts again. While Claude touches close to half of the in-scope tasks, it successfully performs only about a quarter of the time the process requires, with subprocesses that look most automated converting least reliably into performance. Success depends less on the expertise a task requires, but more on the operational burden it places on each channel—the length of the task in interactive use and the volume of context in programmatic use. The exposure that does exist sits either in an augmented mode that does not substitute or in an automated mode that does not yet succeed.

Interpreted in accordance with theoretical frameworks adopted from labour economics studying AI exposure, GenAI has the potential to reshape

accounting not by simply displacing professionals, but by altering the mix of tasks they perform and by magnifying the value of expertise. The findings of this study demonstrate where in the process, in which mode, and on what condition that reshaping is currently taking place. The study contributes, first, by occupying the layer between perception and outcome that accounting research has so far left unobserved. Grounding exposure in recorded usage rather than in capability ratings, occupational proxies, or self-reported signals, and resolving it task by task while aggregating it to a process central to the finance function, allow the task-based account of technological change to be examined within a single domain of accounting, at a granularity at which its mechanisms can be observed directly rather than inferred from occupation-level results. Yet, this real-world usage data itself also complicates these very mechanisms. Followed across releases, whether a task is automated or augmented, and how that choice unfolds as reliability is revealed, marks a technology regime that departs from the automation waves on which existing theory rests, in a manner that only such periodic data can provide.

In this light, task, process, capital, and labour fit together in a single frame, in which the nature of the technology allows augmentation, the nature of the human allows adjacent tasks to be taken up, and the nature of the process allows both to be seen as movement along it. This study provides early evidence of how work is being redistributed between humans and machines along the Record-to-Report process.

The remainder of the thesis proceeds as follows. Chapter 2 reviews the literature, points out current research gap in AI exposure study in both the broad economy and within accounting specifically, and derives the research questions. Chapter 3 describes the data, the construction of the exposure measures, and the descriptive statistics. Chapter 4 reports the results, addressing RQ1 in Section 4.1, RQ2 in Section 4.2, and RQ3 in Section 4.3. Chapter 5 discusses what the findings imply for the distribution of labour and capital along the process. Chapter 6 concludes with the contributions, limitations, and directions for future research.

2 Literature review

2.1 Generative AI: Definition

GenAI refers to a class of machine learning systems capable of producing novel, contextually coherent outputs, such as text, code, and structured reasoning, by learning latent representations from large bodies of unstructured data (Feuerriegel et al., 2023). This represents a categorical departure from prior automation waves: whereas rule-based systems and earlier machine learning models required explicit feature engineering to approximate a target function f(x), GenAI models inductively learn what f(x) should be from data alone, extending automation reach into non-routine cognitive tasks that earlier technologies could not touch (Brynjolfsson & Mitchell, 2017). This study adopts GenAI as its primary unit of analysis rather than AI more broadly, for three reasons. First, the central empirical dataset consists of Claude usage observations, which is a large language model-based system. Second, the core exposure framework employed by Eloundou et al. (2024) is explicitly scoped to GPT-class models. Last, current GenAI models have evolved beyond static text generation into systems with agentic capabilities that encompass autonomous tool-use, multi-step reasoning, and programmatic execution via APIs (Wang et al., 2024), enabling them not merely to retrieve or classify information but to interpret, contextualise, and act upon it. Therefore, GenAI is an appropriate unit of analysis that can meaningfully engage the full range of cognitive tasks under study.

Global investment in artificial intelligence signals both market confidence and transformative economic expectation: worldwide AI spending reached $1.76 trillion in 2025 and is forecast to climb to $2.52 trillion in 2026, a 44% year-over-year increase (Gartner, 2026). Such capital deployment is not without empirical grounding in performance outcomes. Evidence from the GDPval framework by Patwardhan et al. (2025) suggests that frontier model performance on complex knowledge-work tasks is improving in roughly linear fashion over time, with leading models approaching industry-expert quality on a growing range of deliverables. The labour market implications of this trajectory are correspondingly large: Eloundou et al. (2024) predict that approximately 80% of the U.S. workforce could have at least 10% of their tasks affected by large language models, and Anthropic's own productivity research indicates Claude task-level efficiency gains of up to 80% (Tamkin & McCrory, 2025). Critically, these capability trends are beginning to translate into observed adoption: Bick et al. (2026), find that 32% of workers reported using GenAI for work, with 10% using it every workday and 17% on some but not all workdays, representing an adoption trajectory that has outpaced that of the personal computer and the internet at comparable points in their diffusion histories. Indeed, Bick et al. (2026) further demonstrate that estimated GenAI exposure scores are highly correlated with adoption, with ρ

= 0.65 using the Eloundou et al. (2024) exposure measure and ρ = 0.71 using Felten et al. (2021). In one of their releases, the Anthropic Economic Index even found that actual automation usage among workers exceeded augmentation usage (Appel et al., 2025), underscoring the accelerating and potentially restructuring impact of these technologies on the nature of work.

Heightened observed exposure does register in workers' economic anxiety. Massenkoff and Huang (2026) found that for every ten-percentage-point increase in occupational AI exposure, perceived job threat from their survey rises by 1.3 percentage points, with workers in the top exposure quartile reporting concerns three times as often as those in the bottom. Nevertheless, actual macro-level labour market consequences remain muted. Acemoglu et al. (2022) find that while a one-standard deviation increase in AI exposure is associated with a roughly 14% decline in non-AI vacancy postings at the establishment level, there are currently no discernible aggregate labour market consequences regarding overall employment or wage growth at the industry or occupation levels. Similarly, Hampole et al. (2025) find that a one-standard-deviation increase in an occupation's mean task exposure to AI reduces its within-firm employment share by 14.5% over five years. However, this direct displacement is offset by a 7.5% reallocative increase from concentrated exposure alongside firm-level productivity growth. This productivity effect is further affirmed by Kharazian et al. (2026) findings that firms adopting AI tend to grow faster, but this is almost entirely driven by high-intensity adopters. Concurrently, these high-intensity AI adopters actually saw total headcount increase by 10.2% and entry-level hiring rise by 12% in the 24 months following adoption, directly aligning with Hampole et al. (2025) findings while contradicting predictions that AI disproportionately displaces junior workers by Brynjolfsson et al. (2025). Indeed, Mertens et al. (2026) noted that rather than AI capabilities advancing in abrupt jumps that render complex tasks suddenly automatable, which they term the "crashing waves" view, improvement follows a "rising tide" pattern, where continuous, broad gains across tasks of varying complexity diffuse gradually rather than displace suddenly. Taken together, these findings establish a consistent pattern that AI is actively substituting for human labour in specific tasks, yet its macro-economic footprint remains limited by within-firm reallocation, firm-level productivity outcome that allows them to hire more, and gradual capability diffusion.

2.2 Generative AI in Accounting: Application, Adoption, and Outcomes

The application of GenAI within accounting is emerging as a central theme in both academic and practitioner discussion, reflecting the technology's expanding capability that marks its distinction from earlier automation technologies.

The literature converges on a broad but increasingly well-mapped picture of where generative AI and large language models can be applied within accounting. At the transactional end, LLMs present significant opportunities including optimising current processes, extracting new information, and updating accounting measurements, where Li and Vasarhelyi (2024) specifically demonstrated this through LLM integration via user interface, API, and RPA pipelines for routine accounting tasks. Moving up the value chain, Li et al. (2025) identify LLM applicability across financial reporting, ESG disclosure, financial analysis, and risk management, while also projecting forward-looking applications such as intelligent knowledge bases, automated budget management, contract management, and cost management. Jones and Free (2026) extend this picture further into the analytical domain, synthesising evidence that ML-based models frequently outperform traditional statistical approaches on prediction on range of prediction tasks, and that LLMs can enhance internal analytics by improving forecasting speed and insight. Abbas (2026) comprehensively positions AI as a technology that transforms both the operational and structural dimensions of accounting, automating basic accounting tasks and decision-making processes across planning, performance measurement, and control.

A study by Eulerich et al. (2024) further demonstrated GenAI's accounting capability by testing ChatGPT's performance on the CPA, CMA, CIA, and EA certification exams. While an earlier version of ChatGPT (GPT-3.5) could not pass any major accounting certification exam with an average score of only 53.1%, the addition of a more capable model, few-shot training, and reasoning and acting capabilities brought GPT-4 performance to an average score of 85.1%. That a general-purpose language model can perform at or above professional certification standards implies that GenAI possesses the domain knowledge and reasoning capacity to engage meaningfully with accounting tasks, supporting the stance of GenAI as a technology with substantial reach across the accounting function.

Real-world adoption patterns broadly reflect this theoretical positioning. Choi and Xie (2026), in a panel survey of 277 professional accountants conducted in late 2024, found that over one-third had at least partially integrated GenAI into their work, with use concentrated in data entry, administrative work, and business communication, with each reported by roughly 44% of users, consistent with GenAI's greater effectiveness in routine, text-based work. In the reporting domain, Blankespoor et al. (2026) provide direct textual evidence of GenAI's penetration into corporate disclosures, finding that in 2024, approximately 4.3%, 2.5%, and 4.5% of new text in risk factors, MD&A, and IPO business descriptions, respectively, was generated by GenAI. A study of Claude Code usage found that Business and Financial Operations is the second-largest occupational group among

inferred users, trailing only computer and mathematical occupations (Hitzig et al., 2026).

Industry and professional body evidence corroborates findings established in the academic literature. The ACCA (2023) and CPA.com (2024) both document a wide application landscape spanning invoice processing, report generation, risk assessment, scenario modelling, policy drafting, and forecasting. The ACCA's Smart Alliance report by Brisbourne (2024) positions GenAI as distinct from prior automation waves such as RPA precisely because its language and reasoning capabilities reach into open-ended, judgment-heavy work that rule-based tools cannot handle. Via surveys and interviews with over 900 accounting leaders already using AI, they documented actual adoption patterns across accounting functions: data analysis and reporting at 33%, financial planning and analysis at 31%, invoice/payment processing at 28%, reconciliations at 20%. Case studies presented within the same report also demonstrate live deployments across collections reconciliation, accounts payable optimisation, document analysis, and ESG reporting. Indeed, whether viewed through the lens of academic literature or practitioner evidence, GenAI's application potential is not confined to any single layer of accounting activity.

Beyond adoption patterns, a growing body of evidence speaks to the outcomes of GenAI deployment in accounting. Böke et al. (2024) find that accounting-related AI investments are associated with faster 10-K filings by 5.38 days, improved readability by 3.01 units, reduced audit fees of approximately $424,960, and a 1.1 percentage point reduction in forecast errors. At the task level, Choi and Xie (2026) report that AI-assisted accountants achieve approximately 17.5 percentage points higher classification accuracy relative to a baseline mean of 74.7%, and complete tasks roughly 10.2 seconds faster per correct transaction on an efficiency-adjusted basis. Nearly half of surveyed accountants independently attribute improvements in timeliness and accuracy to GenAI, and firms observe that each additional month of GenAI use is associated with an expanded client base and reduced time on data entry, suggesting meaningful capacity reallocation.

Despite these benefits, there exists a recurring set of concerns across technical, organisational, and behavioural dimensions. The most commonly raised issues centre on output reliability, data confidentiality, and the adequacy of governance frameworks (Brisbourne, 2024; Choi & Xie, 2026), while firms deploying GenAI in external reporting face the additional risks of proprietary information leakage and opportunistic manipulation of narrative tone (Blankespoor et al., 2026). Barriers such as integration complexity, unclear solution fit, and fear of job displacement further slow institutional adoption (Brisbourne, 2024). Perhaps most consequentially, these risks persist even when human review is built into the workflow, with Choi and Xie

(2026) finding that accountants are significantly more likely to adopt minority AI suggestions even when those suggestions are less accurate, demonstrating how misplaced trust can actively propagate errors rather than catch them.

2.3 Advancing from Prediction to Observed Exposure in Accounting

2.3.1 The Limitations of Predictive Exposure Methods and the Case for Observed Usage

As GenAI's reach across the economy widens, so too does the urgency to move beyond documenting its scale and toward understanding precisely how it is being integrated into work. A growing body of literature has sought to anticipate this through predictive modelling, most notably exemplified by Eloundou et al. (2024) who assess LLM exposure by systematically evaluating whether tasks described in O*NET occupational rubrics could, under idealised conditions, be performed by a language model. Their result yielded a striking estimate that roughly 80% of the U.S. workforce could have at least 10% of their tasks affected, with occupations such as accountants, auditors, and tax preparers facing effectively full exposure. However, such approach fundamentally captures potential capability rather than realised use, and it cannot capture whether practitioners are actually delegating that task to AI, in what mode, at what frequency, or with what degree of success. As a result, there is a structural gap between what AI systems are theoretically capable of and what they are demonstrably doing in real-world contexts, which is particularly pertinent in a technological landscape often characterised by forward-looking and, at times, overhyped claims about future impact rather than grounded evidence of present application. A more grounded frame recognises that AI capability exists across concentric circles that encompass current capabilities, the capabilities frontier, and future potential (Brisbourne, 2024). Predictive exposure estimates like that of Eloundou et al. (2024) operate at the outer rings of this framework, projecting what frontier or future AI could accomplish, while the economy remains somewhere earlier on Gartner's Hype Cycle (O'Leary, 2008), still well short of the productivity plateau at which the most valuable and cost-effective uses of AI have been identified and stabilised.

Responding to this limitation, two broad strands of empirical work have emerged that anchor exposure measurement in observed behaviour rather than annotated potential. The first, rooted in labour economics, infers AI adoption through occupational proxies constructed from job postings and firm-level data. Acemoglu et al. (2022) inferred establishment-level AI adoption indirectly, weighting each firm's occupational composition derived from Burning Glass job postings against external task-suitability indices. Hampole et al. (2025) achieved greater granularity by using workforce data

and text embeddings to map firm-level AI deployments onto the specific tasks workers perform and derive occupational mean exposure and exposure concentration, capturing not only direct substitution effects but also reallocative shifts across the task portfolio. The second strand measures adoption more directly through surveys. Bonney et al. (2024) took a more direct route via the U.S. Census Bureau's Business Trends and Outlook Survey, capturing real-time self-reported estimates of current and anticipated business AI use. Humlum and Vestergaard (2024) shifted the lens to individual workers, surveying 18,000 Danish employees and linking responses to administrative records to identify the institutional frictions that shape adoption in practice. Most recently, Bick et al. (2026) drew on the nationally representative Real-Time Population Survey with over 10,000 respondents to produce the first systematic U.S. estimates of generative AI usage at work.

While these empirical studies represent meaningful progress, a shared limitation persists: whether through theoretical exposure frameworks, occupational adoption proxies from postings, or periodic user surveys, these methods offer only indirect or temporally bounded snapshots rather than explicit observation of how AI is integrated into real work context. Such limitation went beyond methodological to be present empirically, where Gimbel et al. (2026) found that agreement among the seven leading exposure metrics, spanning both actual and theoretical, are the weakest among the most exposed occupations. A parallel disagreement emerges once exposure measures are set against self-reported worker experience, with Lindenlaub et al. (2026) comparing worker-reported AI use from the German DiWaBe employee survey against those prominent exposure measures and finding only a weak relationship between the two. Such divergence is also apparent when comparing self-reported results against directly observed usage data. As part of the Anthropic Economic Index series, both Handa, Stern, et al. (2025) and Massenkoff and Huang (2026) documented participants' descriptions of their own AI use diverging from usage patterns recorded from actual Claude usage. These findings trace a widening gap along the same axis, from exposure metrics disagreeing among themselves, to surveys diverging from exposure measures, to self-reports diverging from the usage data they are meant to approximate. Existing attempts in the current literature lack visibility into how AI is being used. This is compounded by a broader apprehension gap in terms of 'exposure', where Doshi and Moore (2026) note the rapid and uncertain trajectory of GenAI capabilities raises questions about not only where AI is being adopted, but also how it is being used and what that implies for human economic activity.

A small but growing body of work has since moved beyond these indirect measures to record actual usage directly from the logs of AI providers and adopting firms. Google's AI & Economy ATLAS (Iscenko et al., 2026) maps

15 million de-identified interactions across the Gemini App, Google AI Mode, and the Gemini API onto O*NET occupations and tasks, finding that while adoption spans occupations covering just above 88% of US employment, AI is used for only 21% of the tasks in the median occupation with any usage, and attempts at end-to-end automation account for less than 10% of conversations in non-routine cognitive work. Chatterji et al. (2026) link ChatGPT Enterprise account records to usage, worker roles, message-level task classifications, and public-company financial data across more than 1,500 organisations and 17 million messages, documenting that active use within adopting firms spans job functions and seniority levels and is most intensive among early-career workers. At the level of a single firm, Hallman et al. (2026) observe 713,564 prompts and responses from nearly 4,000 back-office employees of KPMG, including its Accounting & Finance function, and use them to measure not whether but how skilfully employees use GenAI. These studies share the methodological move this thesis builds on, treating the record of actual interactions, rather than projected capability, self-report, or proxy as the primary evidence of how GenAI enters work. Yet, each answers a different question—Iscenko et al. (2026) and Chatterji et al. (2026) describe economy- or enterprise-wide diffusion, and Hallman et al. (2026) characterise the user rather than the tasks.

Against this backdrop, the Anthropic Economic Index is one of the earliest and most sustained of these provider-side streams, and it is distinct in three respects that matter for this study: it is released periodically and publicly, its usage is mapped directly onto O*NET tasks, and from Release 4 onwards it is annotated with economic primitives describing how, and how successfully, each task is performed (Appel et al., 2026; Handa, Tamkin, et al., 2025). Grounded in anonymised real-world conversations from Claude.ai and enterprise-level API deployments, this series offers an empirical window into actual AI integration rather than predicted or self-reported usage. Conversations are processed through Clio, Anthropic's privacy-preserving pipeline, which clusters interactions thematically and maps them to O*NET occupational categories, enabling systematic comparison with existing exposure frameworks. Critically, the Index's analytical scope extends beyond simple usage counts, where its economic primitives introduced in Release 4 (Appel et al., 2026) provide a multidimensional characterisation of AI deployment that speaks directly to mode, not merely incidence, of adoption. These features establish the Anthropic Economic Index as a foundational empirical resource for understanding GenAI usage in terms of actual task performance, grounding the analysis not in survey responses or occupational proxies, but in the observed reality of how GenAI are engaged with in accomplishing work.

2.3.2 Real-World Exposure in Accounting: Existing Attempts and Remaining Gaps

In accounting specifically, the case for moving beyond predictive exposure methods is particularly compelling. Accounting occupations are consistently cited among the most theoretically exposed to GenAI, yet there have been only a handful of attempts to measure real-world GenAI usage, each methodologically distinct and each carrying its own limitations.

Choi and Xie (2026) represent the most similar attempt to this thesis in measuring real-world usage. By triangulating a panel survey of 277 professional accountants with proprietary platform logs containing hundreds of thousands of transaction-level records from a GenAI-based accounting platform serving 79 SME clients, alongside a framed field experiment that randomly assigns accountants to AI-assisted or manual classification tasks, their rigorous multi-method design links what accountants believe and expect about GenAI's costs and benefits to how they actually use it, and ultimately to how it shapes productivity, time allocation, and reporting quality. Their survey reveals that accountants report using GenAI for financial analysis, quality assurance, and advisory services, which is precisely the judgment-intensive tasks where GenAI's language and reasoning capabilities are most distinctively applicable. Yet, these uses are not visible in the platform logs, making the evidence for their productivity and reallocation findings heavily weighted toward the most automatable slice of the accounting workflow. More fundamentally, platform logs and framed experiments share a structural limitation: both observe AI exposure through a fixed interface, recording either what a software environment registers or what occurs when a researcher-controlled platform is handed to participants. Neither can capture the organic variation in how professionals actually seek out, frame, and repurpose AI assistance across their workflows, which is precisely the dimension that matters for understanding how GenAI is reshaping the full scope of accounting work.

While Choi and Xie (2026) attempt to capture actual usage via granular, micro-level platform interaction logs, Böke et al. (2024) represents an alternative empirical effort approaching the measurement challenge from the macro-firm level by operationalising AI investment in accounting through approximately 76 million job postings drawn from Revelio Labs. By constructing skill dictionaries across accounting subfields and exploiting historical exposure to research universities as an instrument, they estimate how firms' AI investment affects traditional labour demand and work outcomes. This macro-level design is appropriate given the study's focus on measuring macro labour outcome, but the proxy itself introduces a structural limitation in the context of actual AI usage studies. A job posting measures corporate intent and labour-demand composition, not whether the technology is successfully integrated, which tasks are automated, or where human-AI bottlenecks occur in practice. In this sense, Böke et al. (2024)

provide a measure of strategic commitment rather than operational execution, and the task-level mechanics of adoption remain unobserved.

Blankespoor et al. (2026) applied GPTZero, a commercial AI-detection tool, to actual public filings to find statistically significant GenAI usage across all five disclosure types. AI detection applied to actual public filings provides an observable, non-self-reported signal of real usage; however, the scope of this study is deliberately restricted to the disclosure-writing task, leaving the upstream subprocesses of transaction recording, subledger close, or consolidation entirely outside the frame. Brisbourne (2024) offers a complementary industry perspective, drawing on interviews and a survey of more than 900 finance leaders already using AI. The study methodological character is, however, self-selected and perception-based: the sample is restricted to those already using AI, which introduces adoption-readiness bias and precludes inference about non-adopters. Periodic surveys of this kind produce static snapshots and cannot reflect the dynamic relationship between advancing model capabilities and shifting patterns of real-world use. Hallman et al. (2026) come closest to native interaction data in an accounting setting, observing actual prompts from a large professional-services firm whose back office includes an Accounting & Finance function; their focus, however, is the sophistication of the user rather than which accounting tasks are exposed, and their sample is confined to one firm's internal tools.

Across these approaches, a common limitation emerges: none can capture the effective task horizon of GenAI as it is actually experienced by users, a dynamic interplay between model capability and real-world behaviour that only native interaction data can reveal. Systematic empirical evidence of how GenAI is actually being integrated into accounting practice remains scarce. This study responds directly to that gap by measuring observed exposure, drawing on the Anthropic Economic Index dataset grounded in real Claude interaction data and operationalised through economic primitives (Appel et al., 2026). The contribution is not simply to document another instance of GenAI usage in accounting, but to identify where GenAI is actually being used and how the nature of that use varies.

2.4 The Limitations of Occupation-Based Analysis and the Task-Level Approach

2.4.1 From Occupation to Task: Labour Economics Foundation

What would constitute the appropriate unit of analysis for understanding how technology reshapes the organisation of professional work? Is the "accountant" occupation a meaningful unit of analysis for studying GenAI exposure?

The occupational category of "accountant" is notably broad: Scarlata et al. (2011) show that a single O*NET code subsumes CPAs, staff accountants, cost accountants, business analysts, and accounting supervisors, and Bryant et al. (2011) further found that financial accounting work requires measurably more creativity than auditing, taxation, or managerial accounting. These studies demonstrate that even within a single profession, sub-domains differ substantially in their underlying task content. If meaningful task heterogeneity exists even across accounting sub-disciplines within the same occupational code, then any exposure estimate applied at the occupation level risks obscuring precisely the variation that matters most for understanding where and how GenAI is restructuring the profession. This motivates a finer unit of analysis.

The theoretical foundation for the task-level approach is rooted in a fundamental critique of the canonical production function, which treats labour and capital as aggregate inputs with fixed, static roles and thus cannot account for the fact that technological change reallocates specific tasks between humans and machines rather than displacing workers as a whole (Acemoglu & Autor, 2011). Autor et al. (2003) offered a preliminary departure from this canonical tradition, demonstrating in a simple production function framework how information technology can simultaneously displace labour from routine, codifiable tasks while complementing labour in non-routine, problem-solving, and complex communication tasks. Acemoglu and Autor (2011) then operationalised this insight more formally, proposing that production is best modelled as a continuum of discrete tasks, each of which can in principle be assigned to labour or capital depending on comparative advantage: in equilibrium, any given task shall be assigned only to the least-cost factor, and such assignment is determined dynamically by shifting economic and technological forces rather than by fixed institutional roles. Autor (2013) further consolidates this position when advocating for a task approach to labour markets, arguing that what capital displaces is not workers but specific codifiable task categories, and that the boundary between labour tasks and capital tasks is permeable and shifting. Consequently, understanding technological impact requires looking not at occupations as a whole, but at the specific tasks and the conditions of the different factors, either labour or capital, that are central to the task itself.

Autor (2015) operationalises the task-based framework in a general context of automation, arguing that automation has not produced mass unemployment because technology simultaneously substitutes for some tasks and raises the productivity value of others, and focusing only on what is lost misses the central economic mechanism by which automation affects labour demand. Autor (2015) illustrates this vividly with bank tellers: as cash-handling tasks were automated, the remaining work shifted toward

relationship banking, which is a higher-value, non-routine function that technology augmented rather than displaced. While the occupation persisted, its task composition changed. This is precisely the pattern that occupation-level analysis cannot detect. Acemoglu and Restrepo (2019) then conceptualise this into a general framework with three mechanisms on how tasks are shifted between labour and capital in the context of automation: automation shifts the task content of production adversely for labour through a displacement effect, as capital takes over tasks previously performed by labour; it simultaneously contributes to the demand for labour in non-automated tasks through a productivity effect; it may generate a reinstatement effect when new tasks are created in which labour has comparative advantage.

This shift in the task content of production i.e. whether a task is allocated to labour or capital is the mechanism that produces intra-occupation heterogeneity in exposure: not all tasks within an occupation are equally codifiable (Autor et al., 2003), so not all shift at the same rate or in the same direction. Raisch and Krakowski (2021) formalise such distinction into automation, where humans hand over a task to a machine with little or no further involvement, and augmentation, where continued close interaction between humans and machines allows for complementing machine capabilities with human reasoning. Because any given occupation contains a bundle of tasks varying along this automation-augmentation axis, occupations will almost never be uniformly automated or uniformly augmented, but split. This means intra-occupation heterogeneity in AI exposure is not an empirical anomaly but a structural prediction based on the task framework. An occupation containing both routine and non-routine tasks will necessarily show divergent exposure across its range of tasks involved.

Empirical evidence in the specific context of AI corroborates this theoretical prediction. Arntz et al. (2017) provide the most direct confirmation, where accounting for the full spectrum of tasks within occupations rather than treating occupations as homogeneous units drops the automation risk of US jobs from AI from 38 to 9%, illustrating how occupation-level aggregation systematically overstates exposure by averaging over profoundly heterogeneous task portfolios. Beyond task count, what matters more pertains to which task being automated. Hampole et al. (2025) demonstrate that while higher mean task-level AI exposure reduces employment growth in occupation-firm pairs, greater concentration of that exposure in a subset of tasks counteracts this effect, as workers can reallocate effort toward unaffected tasks and potentially enhance overall productivity. Autor and Thompson (2025) deepen this argument by introducing the dimension of expertise, suggesting that automation of any given task either raises or lowers the average expertise level of the remaining tasks, as

expertise level differs among tasks within an occupation. Appel et al. (2026) extend it further with the dimension of effectiveness, weighting each covered task by its share of working time and by the rate at which AI actually completes it. They found that even 90% task coverage need not translate into large job impacts if AI fails on key covered tasks or misses the most time-intensive ones.

The fact that these patterns all manifest in real labour market consequences confirms that intra-occupational task heterogeneity is not merely a measurement artifact but an actual mechanism, strengthening the proposition of what defines an occupation's exposure profile is not a uniform property of the role, but an emergent outcome of which specific tasks are exposed, how expertise is distributed across those tasks, and whether those tasks are actually performed successfully.

2.4.2 Task-Level Analysis in Accounting Research

The accounting literature has begun to utilise task-level frameworks, but has not yet adopted task-level analysis in a level of granularity similar to those established in labour economics literature. Most existing empirical studies use the occupation as their functional unit. Böke et al. (2024), for instance, while documenting heterogeneous effects "across accounting functions," operationalise this heterogeneity through keyword buckets in job vacancy postings rather than explicit task definition. Postings that use audit vocabulary are classified as audit-adjacent, postings that use tax vocabulary as tax-adjacent. Choi and Xie (2026) move closer to a task-level understanding, conceptualising the accountant's role as encompassing five interdependent tasks of data collection, data processing, accounts categorisation, confirmation and review, and financial reporting to observe how GenAI affects task allocation and reallocation across this chain. However, this taxonomy operates at a level of abstraction that compresses considerable within-category heterogeneity, and the study's unit of observation remains the individual accountant and client rather than a task within the accounting process.

The practical consequence of this aggregation is that both studies surface evidence of GenAI's effects without identifying the structural mechanism through which those effects arise, in a manner consistent with the task-based labour economics framework that particularly targets AI’s impact. These nuances of concentration (Hampole et al., 2025), expertise (Autor & Thompson, 2025), and effectiveness (Appel et al., 2026) are all, by definition, invisible at the occupation level by Böke et al. (2024), and insufficiently resolved at the five-task abstraction of Choi and Xie (2026). In accounting, where the profession spans from the highly deterministic to the deeply judgement-intensive, the concentration is precisely what determines where GenAI deployment can vary significantly. Adopting a task-level approach is

thus an analytical frame consistent with the purpose of this thesis, identifying where within the accounting process it is actually doing so, and how the character of that effect varies across structurally distinct task contexts.

Operationalising a task-level analysis requires a structured, validated source of occupational task data, and O*NET is the natural instrument for this purpose. Autor (2013) explicitly endorses O*NET as the best available database for task-level empirical work, and it has since become the standard input for AI exposure analyses. For accounting research specifically, Scarlata et al. (2011) validate O*NET's suitability on multiple grounds: it is federally funded at scale, developed with a rigorous representative sampling plan that contrasts favourably with the small convenience samples typical of accounting studies, and its job descriptions for accounting occupations demonstrate high convergent validity with within-profession competency frameworks developed by the AICPA.

2.5 The Record-to-Report as the Unit of Analysis

While task literature establishes that tasks are the right unit, aggregation is still necessary to move implications beyond a flat list of task descriptors, and the question is therefore not whether to aggregate but at what level. Existing literature defaults to occupation as the container, yet, as established in the previous section, within-occupation task heterogeneity is large enough to overturn occupation-level conclusions. For a broad labour market analysis this may be acceptable, but when the focus narrows to a single functional domain of accounting, the occupation lumps together tasks that range from fully routine to highly judgment-intensive that produces precisely the kind of misleading average that task literature warns against. Moreover, Falk and Tsoukalas (2026) argue that work is not a list of discrete tasks but a chain of interdependent decisions; evidence from accounting specifically also shows that the role is moving towards so-called high-value tasks (Awyong et al., 2025; Choi & Xie, 2026; Rautiainen et al., 2024). This is consistent with the broader macroeconomics task-reallocation logic (Acemoglu & Restrepo, 2019; Appel et al., 2026; Autor & Thompson, 2025; Hampole et al., 2025) in which AI exposure does not eliminate roles so much as hollow them out and redistribute what remains between humans and autonomous systems. When such redistribution is ongoing, the occupation becomes an increasingly arbitrary institutional container, and aggregating to it risks measuring the residue of a past division of labour rather than the emerging one.

A more technical consideration reinforces the choice to aggregate: individual task-level counts derived from Clio's O*NET classification by Anthropic are sensitive to the granularity and wording of task descriptors, with some tasks broadly worded and systematically overcounted and others narrowly worded and undercounted. Empirical validation of Anthropic's Clio methodology corroborates this reasoning. Tamkin et al. (2024) hand-

validated 150 conversation classifications and found accuracy declining with granularity: 95.3% correct at the top level of its task hierarchy, 91.3% at the middle level, and 86% at the base ONET task level. The effect is also visible in the correlation between direct and cluster-based assignment, which is weak at the task level (minimum Pearson r = 0.47, Spearman = 0.22, Kendall's Tau-b = 0.18) but strengthens markedly once assignments are aggregated to occupations (minimum Pearson r = 0.95, Spearman = 0.95, Kendall's Tau-b = 0.82). In other words, uncertainty compounds at the task level and narrows with aggregation, allowing these wording characteristics to partially cancel, producing a more stable signal than any single task count. Therefore, while tasks remain the absolute levers underlying GenAI exposure, aggregating them surfaces a meaningful pattern at a level broad enough to be interpretable, trading the exact task-level accuracy for a directional accuracy of exposure patterns.

This study, therefore, adopts Record-to-Report (R2R) as its unit of aggregation. R2R is a practitioner-defined process framework that has become the standard operating model for finance and accounting shared services globally. The process is an end-to-end financial management cycle through which an organisation captures, validates, and communicates its financial reality. Bedford et al. (2025) establish that most finance functions conduct a broad range of activities for two primary objectives: an efficiency objective, focused on standardising and automating routine activities to minimise resource consumption, and an effectiveness objective, focused on decision-making and value-adding support for management. Drawing on APQC (2026) Process Classification Framework on Managing Financial Resources and Doxey (2021), this study thus employs an extended version of the R2R process that encompasses both transactional and decision-support subprocesses, unfolding across six sequential phases: Master Data Management, Record, Close, Report, Analysis and Business Insight, and Planning and Budgeting. This extended framing is necessary to align with the full scope of objectives the finance function serves (Bedford et al., 2025), as well as with the documented breadth of GenAI applications in accounting established in the preceding literature review.

The cycle's opening phase Master Data Management is concerned with establishing the structural integrity of the financial architecture itself: maintaining the chart of accounts, reporting entities, cost and profit centres, currency rates, and access governance, without which no downstream process can produce reliable outputs. The Record phase then governs the ongoing accumulation and initial processing of transactions as they occur during the fiscal period, handling journal entries, subledger postings, accruals, and transaction-level reconciliations in accordance with accepted accounting principles. Once individual transactions are captured, Close & Consolidation shifts the unit of work from the transaction to the period: it

finalises and aggregates recorded activity into group-level financial results, closing subledgers, executing general ledger accounting, reconciling account balances, and calculating tax liability.

The second half of the cycle shifts from bookkeeping toward the communication and interpretation of financial results. The Report phase serves the organisation's disclosure obligations, producing management reports, external financial statements, and regulatory and tax filings, translating ledger into structured outputs for internal and external stakeholders. Analysis and Business Insight moves beyond disclosure toward judgment, interrogating the reported numbers, identifying variances, developing KPIs, and providing the interpretive layer that converts financial data into actionable intelligence for decision-makers. Finally, Planning and Budgeting closes the governance loop by directing that intelligence forward, translating analytical findings into long-range strategic plans, annual budgets, rolling forecasts, and performance targets that will in turn govern the next iteration of the cycle.

The R2R framework resolves the principal analytical problem posed by occupation-level aggregation in a manner that is both theoretically grounded and practically relevant. The process, unlike the occupation, is functionally stable: the R2R cycle must be completed regardless of which combination of human and AI agents executes it, making it a more durable and analytically tractable unit for understanding how GenAI is restructuring the finance function. More critically, R2R is a more structurally coherent container than the occupation for accounting work because it is sequenced and causally ordered, where each subprocess has a defined function, a defined input/output relationship, and a structurally different task composition. It spans the full cognitive spectrum of accounting work, from the highly structured and deterministic to the judgement-heavy and language-oriented. While prior automation waves such as Robotic Process Automation and rule-based AI could only penetrate the structured end of this spectrum, GenAI has a capability profile that can be applied across both sides, as established in the preceding review of GenAI applications in accounting. R2R is, therefore, a particularly fertile ground for investigating GenAI exposure. It is a single, bounded process that nonetheless contains the full structural variety of accounting work, meaning that exposure analysis conducted at the subprocess level can reveal not just whether GenAI affects financial work, but how the nature of that effect shifts as tasks move across the cognitive spectrum.

Despite the analytical advantages of process-level granularity, existing accounting literature has not examined GenAI exposure through the architecture of the R2R cycle. Böke et al. (2024) infer functional location from keyword occupational labels in vacancy postings rather than from the logical architecture of the accounting process. Choi and Xie (2026), despite

also following the accounting workflow encompassing five interdependent tasks of data collection, data processing, accounts categorisation, confirmation and review, and financial reporting, only implicitly follows a process logic. Their study also halt the cycle at the reporting stage, leaving the strategic and interpretive subprocesses of analysis, business insight, and planning entirely outside their scope, which is precisely the judgment-intensive tasks where GenAI's language and reasoning capabilities are distinctively applicable. Blankespoor et al. (2026), as noted, restrict their scope to a single downstream disclosure subprocess. None of these studies maps GenAI exposure or usage onto the process structure that practitioners actually navigate and organisations actually govern. For this thesis, R2R, therefore, is a unit of aggregation that is analytically optimal for observing human-AI redistribution of work, tailored to accounting-specific literature, and directly actionable for the finance leaders and standard-setters who manage the R2R cycle.

2.6 Research Questions

Informed by empirical models of AI task exposure in the general labour market, this study applies the same lens to a single accounting R2R process via task-level Claude usage data from the Anthropic Economic Index. Accordingly, this study is guided by three research questions.

> **RQ1:** *Where is GenAI exposure concentrated within the Record-to-Report process?*

This question follows Hampole et al. (2025) finding that AI exposure clusters unevenly across an occupation's tasks, and rather than exposure's average magnitude, this concentration is what determines its labour-market consequence. Applied to R2R, RQ1 asks whether observed Claude usage is spread across the process or concentrated in specific tasks and subprocesses.

> **RQ2:** *What is the expertise level of such exposure and concentration?*

This question draws on Autor and Thompson (2025) argument that the consequence of concentrated exposure depends on the expertise level of the tasks affected. RQ2 asks what expertise level characterises the tasks and subprocess currently exposed, and whether concentrated exposure within R2R skews toward automation or augmentation.

> **RQ3:** *How effective is such exposure?*

This question builds on Appel et al. (2026) finding that task coverage overstates job impact unless each covered task is successfully completed. Both preceding questions, and the frameworks they draw on, rest on the assumption that an exposed task is a task taken over by capital. GenAI, however, does not fit that assumption neatly. It can be used in more than an automated mode, augmenting a worker who remains responsible for the task

rather than replacing them (Raisch & Krakowski, 2021), and exposure that is not effective does not remove the task at all, as the work retreats back to the human. Knowing where exposure sits and at what expertise level, therefore, says little about whether machines substitute for human, unless it is also known whether the exposed tasks are actually being completed. Real-world usage records are well placed to supply this missing dimension, since the Anthropic Economic Index annotates not only which tasks are attempted but whether Claude completed them successfully. Applied to R2R, RQ3 asks how effective observed exposure is.

The three questions trace a single line of inquiry from location to composition, to effectiveness: RQ1 asks where exposure sits, RQ2 asks how it is composed, and RQ3 asks whether it converts into performance. Together, they ask whether a story built from occupation-level labour data holds at the level of a single accounting process, using a dataset that captures actual usage rather than exposure inferred from job postings, surveys, or theoretical task ratings. The questions also inform each other rather than standing as separate probes. A high concentration score means little without knowing whether it sits on senior judgment work or junior data entry, and neither means much without knowing whether the exposed tasks are actually being completed. Rather than testing hypotheses, such interdependence across the three research questions aligns with the study's exploratory character, translating Hampole et al. (2025), Autor and Thompson (2025), and the success-weighted logic of Appel et al. (2026) into a literature-informed sequence operationalised quantitatively on the Anthropic Economic Index dataset to understand AI exposure on a process central to the finance function.

3 Research material and methods

3.1 Data Preparation

3.1.1 Time-Fraction Elicitation

Task time-fraction was elicited to serve as the weighting basis for every subprocess- and process-level construct in the study, defined as a measure of how much of a worker's time each task occupies. Following Tamkin and McCrory (2025) methods for eliciting the same figure for each occupation's tasks, three models, Claude Sonnet 4.6, GPT-5.4-mini, and Gemini-3.1-flash-lite, were each queried with an occupation's complete O*NET task list and asked to estimate weekly hours per task independently, with the prompt explicitly instructing models not to force totals to sum to 40. The full elicitation prompt is provided in Appendix C.1. Estimates were elicited for all 212 register tasks, and only the in-scope tasks enter the analysis. The mean of the three models' estimates was taken as the final time-fraction figure of each task. This was then renormalised within the relevant scope by dividing each task's raw hours by the summed raw hours across all tasks in that scope to yield the time-fraction weight in a process-based framing, instead of an occupation-based one.

The time-fraction weight is needed because the constructs in this study are meant to describe *how much of the work in a subprocess* is exposed, not *how many of its task statements* are. A count-based measure would let infrequent tasks carry the same weight as central, time-intensive ones, which is precisely the distortion Appel et al. (2026) warn against when showing that even high task coverage on a count basis need not translate into large job impacts if the most time-intensive tasks remain outside AI's reach. O*NET's own task ratings do not supply this. It provides *Task importance*, which is rated on an ordinal 1 to 5 scale, and *Task frequency* in seven categorical bands, from "yearly or less" to "hourly or more". Neither of them can be summed into a share of working time, and neither is comparable across occupations once tasks from different occupations are pooled into one subprocess. Hampole et al. (2025) can weight by importance relative to an occupation because their unit is also the occupation, but once tasks are regrouped by process, an ordinal importance score from one occupation cannot meaningfully be added to that of another. An elicited estimate of hours, by contrast, is cardinal, can be renormalised within any scope, and allows every construct in Section 3.2 to be computed on the same basis and compared directly. The elicitation follows the precedent of Tamkin and McCrory (2025), and the use of three models from different providers guards against the idiosyncrasies of any single one.

This subsequently creates the issue that pooling these weights across occupations may conflate importance to the subprocess with importance

relative to that worker's full job. A Chief Executive small time-share on a planning task does not necessarily mean the task is unimportant to the Planning & Budgeting subprocess itself, only that they have many other responsibilities competing for their total time budget. However, genuinely central tasks to a subprocess should surface as high time-shares for other specialist occupations such as Budget Analysts or Financial Managers, whose job scope is narrowly concentrated on R2R activity, while CEO involvement tends to be more peripheral. Accordingly, a Chief Executive, or other occupations included in the study, small weight on a task might actually be appropriately small in the subprocess context, not distorted. Moreover, Tamkin and McCrory (2025) relies on the same unaddressed generalisation, and their downstream numbers still hold up against external benchmarks. Extending the logic from worker to subprocess here follows that established precedent within the methodology being adapted.

Cross-model agreement across the three models' hours estimates returns Krippendorff's $\alpha = 0.681$ on an ordinal level, across all 212 task-occupations rows. Models agreeing on how many hours a task would take, numerically, matters less than on its ranks relative to the tasks around it. An ordinal α of 0.681 indicates that this ordering holds reasonably consistently even where their absolute estimates diverge, which, according to an inspection of highest disagreement tasks, was mostly driven by Gemini consistently overshooting its estimates compared to the other two models (Table D6, Appendix D).

3.1.2 Task-Level Education Requirement and Expertise

RQ2 asks whether the tasks carrying observed exposure and concentration skew toward low- or high-expertise work, following Autor and Thompson (2025) argument. Testing this requires a measure of task expertise that is independent of exposure, observed usage, and concentration constructs that have already been annotated from the Clio pipeline. The Anthropic Economic Index itself publishes a field that seems a direct fit, human education years– a 0 to 20 scale estimating the schooling needed to understand a user's prompt. This is, however, unsuitable, since it varies with how a given user happened to phrase that specific request rather than with any fixed property of the task.

The Bureau of Labor Statistics (U.S. Bureau of Labor Statistics, 2026a), however, assigns education requirements only at the occupation level, where each SOC code receives one label for its typical education level required for entry. This is too broad for a task-level analysis, so a task-level construct of education requirement is estimated by training a Ridge regression to predict occupation-level education from the semantic content of task text. Under this approach, a task within a low-education occupation can still score high if its semantic content resembles tasks that typically occur in high-education occupations, and vice versa. Task statements are embedded using all-mpnet-

base-v2, and BLS's typical-education categories are mapped to conventional years of schooling (no formal educational credential 10, high school diploma 12, some college or postsecondary non-degree award 13, associate's degree 14, bachelor's degree 16, master's degree 18, doctoral or professional degree 20).

O*NET task statements (National Center for O*NET Development, 2026) are merged with BLS education assignments by SOC code, yielding 18,217 tasks across 893 occupations after dropping rows lacking a matched education category or task text. The training pool excludes all tasks belonging to the 10 occupations from which the 212 R2R tasks are drawn. Ridge's regularisation strength is selected via grid search under GroupKFold cross-validation (five folds; Table D5, Appendix D), with folds grouped by occupation so that every task from a given occupation lands entirely on one side of a train or test split: O*NET task statements from the same occupation are often textually similar, so a model could see two tasks from the same occupation in both training and testing where one is mirroring the other; hence, if left ungrouped, the model might not need to have learned anything material about what constitutes education requirement, but simple pattern-matching on occupation-specific phrasing can already inflate its performance. A naive random KFold is run in parallel to detect this exact gap between the group and ungrouped figure. The final model is fit once on the full training pool and applied to the 212 held-out R2R tasks. Task-level predictions on the training pool are then averaged back to the occupation level and compared against the actual BLS values to ensure that the model is still anchored to the established ground truth.

Grouped cross-validation returns R^2=0.570 with a standard deviation of 0.034, RMSE=1.852 years, and Spearman ρ=0.751. The corresponding ungrouped random KFold returns R^2=0.608, a leakage gap of 0.038, indicating that grouped performance is not substantially depressed by removing same-occupation information and that the embeddings carry task-content signals rather than occupation phrasing artefacts. Occupation aggregation returns R^2=0.753, confirming the model didn't lose the training signal when resolving occupation-level labels down to task-level predictions. These outcomes support treating the model as a reasonable estimate of task-level education requirement ranking, which is sufficient for positioning tasks along an education-requirement axis to distinguish deskilling from upskilling patterns in Claude coverage to be touched on by the study.

3.1.3 Task Master Compilation

The task master consolidates every task-level construct used downstream into a single table, built on top of the final R2R task register (Appendix A, Table A2). Occupations associated with each task were also included. Task value is defined as the hourly wage of the occupation performing the task,

merged from the BLS Occupational Employment and Wage Statistics (U.S. Bureau of Labor Statistics, 2026b) and matched to each task's O*NET occupation. Task time-fraction (Section 3.1.1) and ridge-predicted education years (Section 3.1.2) were also included. Beta construct from Eloundou et al. (2024) was merged last. All of these were matched onto the task master by task ID.

3.1.4 Anthropic Economic Index Panel

The dataset central to this study draws on the Anthropic Economic Index (AEI), a series of reports Anthropic has published periodically since 2025 profiling how people use Claude across the economy, based on real interactions with its products (Anthropic, 2026a). The underlying usage data is generated through Clio, Anthropic's privacy-preserving analysis pipeline (Tamkin et al., 2024). Rather than researchers viewing individual raw transcripts, Clio uses AI models to extract characteristics and patterns from conversations such that it protects the identity of individual users. Because the data originate in real users' conversations, their use raises questions of confidentiality and research ethics that warrant explicit treatment. This study is a secondary analysis of data that Anthropic has already released publicly in aggregated form; the author had no access to any conversation, user account, or organisation-level record, and no contact with the individuals whose interactions underlie the data.

Under Clio, before any clustering is carried out, two quantitative thresholds are enforced in determining whether a task should be included in a release: a minimum of five distinct user accounts and a minimum of 15 separate conversations. Each data release is then assessed against the three anonymisation criteria of the European Data Protection Board's draft guidelines. The harms the model guards against include identity linkage and attribute disclosure, but also small-group disclosure and organisational disclosure, the latter treating the exposure of an identifiable organisation's non-public activity as a commercial confidentiality concern. A privacy red-team from Imperial College London's AI Security and Privacy Lab was unable to reidentify any user under this model (Anthropic, 2026b).

The study adds its own safeguards on top of this design. First, no attempt was made to link, enrich, or reidentify any record; the only external data merged onto the panel, namely O*NET task statements, BLS education and wage statistics, and the Eloundou et al. (2024) exposure ratings, are public and defined at the task or occupation level, never at the level of a user or firm. Second, restricting the panel to R2R tasks only filters cells that Anthropic has already published and never disaggregates them further, so no result in this thesis is reported at a granularity finer than a cell that has already cleared the release thresholds. Third, the LLM-assisted steps of the study process only public O*NET task text and contain no personal data. Finally, because the

data provider is also the developer of the model under study, the analysis relies solely on the public releases, and its findings and interpretations are the author's own.

Each of the five AEI releases publishes data under its own schema with different column names, different task identifiers, and different levels of granularity, from Release 1's flat file consisting of only a task-level pct to Release 5 splitting collaboration into six categories. Thus, they were compiled to a shared canonical long-format data scheme with columns referring to release, platform, task, dimension, category, variable, value (Appendix B for detailed data documentation). Rows outside R2R scope were dropped, and only in-scope tasks relevant to the study were retained. The 1P API platform additionally carries completion_tokens, cost, and prompt_tokens dimensions. Because AEI's published cost index and token counts are computed over the entire economy-wide dataset, they were re-indexed within the R2R-restricted task set to ensure downstream analysis remains internal to the process under study rather than anchored to a baseline it may not resemble. Release 6 is excluded because it does not publish the task-success facet required for effective exposure and reports the human-only-ability and multitasking facets only as single shares (Appendix B.2).

Release 4 and Release 5 additionally introduce economic primitives, where each of them is extracted per conversation by a classifier prompted for that specific primitive. They are defined as:

- Human time estimate: hours a competent professional would need to complete the task without AI assistance, assuming full domain knowledge, context, and tool access
- Human-with-AI time estimate: minutes the user spent completing the task within the conversation
- Multitasking: whether the conversation covers a single task or several.
- Human ability to complete the task alone
- Human education years: years of formal education needed to understand the user's prompts, estimated on a discrete 0 to 20 scale
- AI education years: years of formal education needed to understand the Assistant's responses, estimated on a discrete 0 to 20 scale.
- Use case: whether the conversation serves work, coursework, or personal use.

- AI autonomy: how much decision-making the Assistant exercised, on a discrete 1 to 5 scale.
- Task success: whether the Assistant completed the requested task successfully. Denotes as 0 for unsuccessful conversations and 1 for successful conversations. This is then averaged across conversations annotated to belong to a task to get a task-level success rate.

Full data documentation available in Appendix B.

3.2 Construct Building

The constructs built in this section map onto the three research questions. Observed exposure (Section 3.2.1) and concentration (Section 3.2.2) locate GenAI exposure within R2R for RQ1, and their expertise composition for RQ2 draws on the task-level expertise measure of Section 3.1.2. Effective exposure (Section 3.2.3) discounts that exposure by task success to answer RQ3.

3.2.1 Observed Exposure Construction

Different from the theoretical capability by Eloundou et al. (2024), observed exposure captures which of those theoretically automatable tasks are actually seeing automated usage in professional settings, and this construct is higher for tasks with significant usage in a work-related context, higher share of automated use, and larger share of overall subprocess composition. Replicating Massenkoff and McCrory (2026), the occupation-based pipeline is adapted to a process framing, where observed exposure is composed of 4 constructs: usage volume, automation factor, task-level exposure, and subprocess exposure.

First, for each task t, usage volume is defined as:

$$\text{WorkUsage}_t = \text{ClaudeWorkUsage}_t + \text{APIUsage}_t$$

Claude.ai conversations classified as work, plus first-party API calls. On Claude.ai, only conversations classified under the work use case are included, with education and personal use excluded, since the aim is to isolate professional usage specifically. On the API, usage is included in full and unfiltered by use case, since API traffic is treated as a signal of integration into production workflows.

The automation factor of a task is captured by α_t:

$$\alpha_t = 0.5 + 0.5 \cdot \frac{\text{ClaudeWorkUsage}_t \cdot \text{AutoShare}_t + \text{APIUsage}_t}{\text{ClaudeWorkUsage}_t + \text{APIUsage}_t}$$

The two platforms enter this term differently. API usage is counted as automation by construction, since an API call implies a workflow built

around the task rather than a human-driven exchange, pushing α_t toward 1. On the contrary, Claude.ai usage is not automated by default, thus it is scaled by its own measure AutoShare_t and consequently moves αt between 0.5 as purely augmentative and 1 as purely automotive. αt is thus structurally a measure of automation intensity conditional on any qualifying usage, not a neutral usage-exposure measure. This choice is consistent with the theoretical foundation that this thesis builds on, where task-substitution literature (Autor & Thompson, 2025; Hampole et al., 2025) treats automation, specifically, as the channel through which AI exposure produces labour-market consequences. A measure of exposure that treated automation and augmentation as equivalent would be poorly matched to that framework, since augmentative use, however frequent, does not substitute for the task in the way that theory informs. At the same time, the 0.5 floor lets the factor scale with the degree of automation while still registering augmented use as a meaningful form of exposure, reflecting the unique characteristics of GenAI compared to other automation paradigms prevalent in previous studies.

Two gates restrict when α_t is allowed to count. A volume gate requires $\text{WorkUsage}_t \geq 100$, as even after limiting to work-use conversations and API traffic, low-volume tasks could reflect one-off experimentation or Clio misclassification rather than a task genuinely integrated into professional workflows at scale. Since the threshold is rather arbitrary, both gated and ungated versions are reported not only as a sensitivity check, but also because the contrast between tasks clearing the gate and those that do not is itself informative about where in the R2R process Claude has achieved meaningful traction versus where usage remains marginal. A capability gate further requires $\beta_t \geq 0.5$, the release-invariant theoretical exposure indicator from Eloundou et al. (2024), so that task-level observed exposure is defined only within tasks the theoretical layer has already flagged as capable. Task-level observed exposure $\tilde{r}_t$ is then:

$$\tilde{r}_t = \begin{cases} \alpha_t & if\ \text{WorkUsage}_t \geq 100\ and\ \beta_t \geq 0.5 \\ 0 & otherwise \end{cases}$$

The task-level exposure scores are aggregated to the subprocess level exposure as a time-weighted mean of $\tilde{r}_t$ by each task's time fraction within the subprocess:

$$R_{o,sp} = \frac{\sum_{t \in sp} \text{task_time_fraction}_t \cdot \tilde{r}_t}{\sum_{t \in sp} \text{task_time_fraction}_t}$$

$R_{o,sp}$ is not the count share of tasks in a subprocess that show observed exposure, but the share of time spent on the subprocess that is being performed or accelerated by Claude, once automation intensity and the two gates have been applied. Alongside $R_{o,sp}$, the plain fraction of tasks in the subprocess with $\beta_t \geq 0.5$ is also reported as a comparison for how much each

subprocess is theoretically exposed on a simple task-count basis, independent of how time is distributed across each of them.

Because $R_{o,sp}$ is gated on βt, any gap between it and theoretical exposure is necessarily a gap of under-realisation, not a mismatch in either direction. The measure is also structurally biased toward API-integrated workflows, since API calls receive full automation credit while chat-based use is discounted by its own augmentation share. For R2R, this implies that standardised, high-volume subprocesses with API tooling should show higher $R_{o,sp}$, while judgment-heavy subprocesses may be capped near the 0.5 floor even where usage is frequent, simply because that usage remains augmentative rather than automative, lacking alignment with the theoretical foundation the research questions are based upon.

3.2.2 Concentration Score Construction

Hampole et al. (2025) identified concentration as a powerful countervailing force that partially offsets the labour displacement caused by AI substitution. Section 4.1.3 investigates whether this same structural property manifests at the subprocess level in R2R: given the exposure scores estimated in the preceding subquestion, do AI applications tend to target tasks unevenly within a subprocess, and does that unevenness scale with usage intensity?

The concentration measure adapts Hampole et al. (2025) original construction to the present context. Their index is the importance-weighted variance from O*NET of task-level exposure probabilities scaled by the log of a firm's total application count; the present analysis adopts the same structural logic of dispersion multiplied by log-volume but makes two substantive modifications. Because mean exposure has already been estimated in Section 3.2.1 using task-time-fraction weights rather than O*NET importance scores, task-level dispersion is computed relative to that same exposure and weighted by the same time fractions, preserving internal consistency across the analysis. The volume scaling term, similarly, replaces firm-level application count with conversation volume at the subprocess level, capturing how intensively users are actually engaging AI against these tasks.

As a reference baseline, all in-scope tasks across every subprocess are first pooled into a single group and the same time-fraction-weighted mean used for $R_{o,sp}$ is applied across this wider scope, treating the entire process as one giant subprocess and producing a single aggregate exposure figure against which subprocess-level dispersion can be benchmarked.

$$\mathrm{R}_{process} = \frac{\sum_t w_t \tilde{r}_t}{\sum_t w_t}, \qquad w_t = \text{task_time_fraction}_t$$

For each task, two parallel dispersion contributions are then computed, one relative to its subprocess mean $R_{o,sp}$ and one relative to $R_{o,p}$ with weights

re-normalised to sum to one within the relevant scope before squaring the deviation. Zero-exposure tasks are still included throughout.

$$w_t^{norm,sp} = \frac{w_t}{\sum_{t' \in sp} w_{t'}}, \qquad \text{subprocess_weighted_dispersion}_t = w_t^{norm,sp} \cdot \left(\tilde{r}_t - R_{o,sp}\right)^2$$

Per subprocess, release, and gate grouping, or at process scope per release and gate settings, the dispersion contributions are summed to a raw dispersion figure, which is then scaled by conversation volume N, where N is the task-count sum of usage within the relevant scope. A high concentration score signifies wide internal dispersion, indicating that Claude applications are intensely relevant to a small number of tasks while remaining largely irrelevant to others within the same subprocess. A low score indicates that task-level exposure clusters tightly around the subprocess mean, with no pronounced within-group imbalance.

$$\text{concentration} = \left(\sum_t \text{weighted_dispersion}_t\right) \times ln(1 + N)$$

Multiplying by ln(1+N) up-weights subprocesses where structural imbalance is also paired with genuine usage intensity, and high concentration thus also correlates with volumes of applications targeting those specific imbalances. On the contrary, smaller volume shrinks concentration further, making adoption not dense enough to have structural impacts on the process. Still, two variants of the concentration index are computed side by side throughout: concentration using raw N regardless of gate settings and concentration_gated reducing N to only gate-passing volume. In other words, N does not vary with the gate in the ungated specification, and only the dispersion sum changes when gate settings change, allowing the concentration construct to yield a clean decomposition, where any gate=ON versus gate=OFF difference in concentration is attributable entirely to which tasks enter the dispersion sum and how the reshuffled mean repositions their squared deviations, not to any simultaneous shift in the volume scalar.

3.2.3 Effective Exposure

Effective exposure is defined as the share of total task-time that Claude can perform successfully, calculated as a weighted sum of task-level success rates, where each task's weight reflects its share of the subprocess's total time.

A task is considered touched if it shows up in the usage data and recorded in the Anthropic Economic Index dataset. Subsequently, coverage is the share of tasks touched across the relevant scope under study, either subprocess or entire process level. A task appearing in usage data on both platforms counts only once toward task coverage.

For each touched task, success is the mean of both platform's percentage of conversation categorised as successful for that task and release. Success is deliberately left unweighted by usage intensity, since it's meant to capture whether Claude can do it instead of how often people happen to attempt it. If a task that occupies little of a subprocess' time happens to get a disproportionate share of conversations, weighting success by that intensity would let it pull the average in a direction that has nothing to do with how much the task actually matters to its subprocess. Tasks that do not appear in the data have success set to zero, so untouched tasks still register as zero contribution rather than being excluded.

$$success_t = \begin{cases} 100 \cdot \dfrac{1}{|K_t|} \displaystyle\sum_{k \in K_t} s_{t,k} & \text{if } |K_t| \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

where K_t denotes the set of platforms (Claude.ai, 1P API) that report a task-success reading for task t in the release, and $s_{t,k}$ = P(task_success = yes) on platform k.

Each task is then weighted by its time-fraction. In Appel et al. (2026), the original effective exposure construct weighted tasks by time-fraction multiplied by annual occurrence frequency. However, time-fraction already reflects "the amount of time a worker spends on that task per week," which itself already embeds how often that task occurs. A task that takes little time per instance but recurs frequently will already accumulate into a high time-fraction figure. Hence, directly adopting Appel et al. (2026) construct risks double-counting the same frequency information. Weighting only by time-fraction also allows for consistency and direct comparability with other constructs in the thesis.

$$w_t = \text{task_time_fraction}_t, \qquad w^{norm} = \frac{w_t}{\sum_{t' \in sp} w_{t'}}$$

Effective exposure then weights each task by its time-fraction as renormalised within the relevant scope, either on a subprocess or entire process level, by dividing each task's raw weight by the sum of raw weights across every task in that scope, whether those tasks show up in usage data or not. Accordingly, effective exposure is not a weighted mean of success among touched tasks. Since the denominator always includes all tasks within the scope, a (sub)process where Claude only reaches a handful of low-time-share tasks will show a correspondingly small effective exposure figure even if it succeeds perfectly on those few. The untouched majority shall drag the weighted sum down through their zero-success contribution, not through exclusion.

$$\text{EffectiveCoverage}_{sp} = \sum_{t \in sp} w^{norm} \cdot \text{success}_t$$

This effective exposure construct is then compared against both task coverage and observed exposure. Both comparisons are also summarised in companion difference tables, where a positive cell means Claude's demonstrated success on a subprocess exceeds how much of it is merely touched or targeted, while a negative cell means the subprocess is engaged with more than it is successfully completed.

$$\begin{aligned}\text{diff}_{sp,r} &= \text{EffectiveCoverage}_{sp,r} - \text{task_coverage_pct}_{sp,r}\text{diff}_{sp,r} \\ &= \text{EffectiveCoverage}_{sp,r} - \left(R_{o,sp,r} \times 100\right)\end{aligned}$$

Table 1 summarises the constructs built in this chapter, the level at which each is measured, the research question it serves, and how it is computed.

Table 1: Summary of constructs, their level of measurement, the research question they serve, and their computation.

Construct (symbol)	Level	RQ (section)	Computation and source
Panel A. Task-level inputs			
Task time fraction (w_t)	Task	All (weighting)	Mean of weekly hours estimated by three LLMs from each occupation's full O*NET task list, renormalised within the relevant scope (Section 3.1.1)
Task expertise (ridge years)	Task	RQ2 (4.2); RQ3 (4.3.2)	Ridge regression of BLS occupation education years on all-mpnet-base-v2 embeddings of O*NET task text, trained on tasks outside the ten study occupations (Section 3.1.2)
Task value	Task	RQ2 (4.2.2)	BLS OEWS (May 2025) mean hourly wage of the task's occupation; subprocess value is its time-fraction-weighted mean
Theoretical exposure (β_t)	Task	RQ1 (4.1.2); gate	Eloundou et al. (2024) rating; $\beta_t \geq 0.5$ marks a task as theoretically capable
Panel B. RQ1: location and concentration of exposure			
Usage share	Task, subprocess; platform	RQ1 (4.1.1)	Task share of R2R conversations per platform and release, summed to subprocess
Collaboration-mode share	Process, subprocess; platform	RQ1 (4.1.1)	Share of conversations classified as automation (directive, feedback loop) or augmentation (task iteration, validation, learning)
Automation overrepresentation index	Subprocess; platform	RQ1 (4.1.1)	Subprocess automation share ÷ process automation share
Work usage (WorkUsage_t)	Task	RQ1 (4.1.2)	Claude.ai conversations classified as work + all 1P API conversations
Automation factor (α_t)	Task	RQ1 (4.1.2)	$0.5 + 0.5 \times (\text{ClaudeWorkUsage}_t \cdot \text{AutoShare}_t + \text{APIUsage}_t) \div \text{WorkUsage}_t$
Task observed exposure ($\tilde{r}_t$)	Task	RQ1, RQ2, RQ3	α_t if $\beta_t \geq 0.5$ and, under gate = ON, $\text{WorkUsage}_t \geq 100$; 0 otherwise
Observed exposure ($R_{o,sp}$; $R_{o,p}$)	Subprocess; process	RQ1 (4.1.2); RQ2 (4.2.2); RQ3 (4.3.1)	Time-fraction-weighted mean of $\tilde{r}_t$ within scope (Section 3.2.1)

Construct (symbol)	Level	RQ (section)	Computation and source
Concentration	Subprocess; process	RQ1 (4.1.3); RQ2 (4.2.2)	$\Sigma_t w_t^{norm} (\bar{r}_t - R_o)^2 \times \ln(1 + N)$, with N ungated or gated (Section 3.2.2)
Usage Gini coefficient	Platform	RQ1 (4.1.3)	Gini coefficient and Lorenz curve of task usage shares
Panel C. RQ2: expertise level of exposure and concentration			
Task coverage	Subprocess; process	RQ2 (4.2.1); RQ3 (4.3.1)	Share of in-scope tasks appearing in the usage data on either platform
Net skill-coverage gap	Subprocess, process; platform	RQ2 (4.2.1)	Time-weighted mean expertise of all in-scope tasks minus that of covered tasks; negative values indicate coverage skewed to higher-expertise tasks
Panel D. RQ3: effectiveness of exposure			
Task success ($success_t$)	Task	RQ3 (4.3.1, 4.3.2)	100 × mean across platforms of P(task_success = yes); 0 for untouched tasks (Section 3.2.3)
Effective exposure	Subprocess; process	RQ3 (4.3.1)	$\Sigma_t w_t^{norm} \cdot success_t$; compared with task coverage and with $R_o \times 100$
Economic primitives	Task; platform	RQ3 (4.3.2)	AEI classifier means of human-only time, human-with-AI time, AI autonomy, human and AI education years, human-only ability, and use case (Appendix B)
Context-length and cost indices	Task; 1P API	RQ3 (4.3.2)	AEI prompt-token, completion-token, and cost indices re-indexed to the R2R task set and log-transformed

Note: RQ1 asks where GenAI exposure is concentrated within R2R; RQ2 asks at what expertise level that exposure and concentration sit. RQ3 asks how effective that exposure is. Gate = ON/OFF refers to the volume gate ($WorkUsage_t \geq 100$); the capability gate ($\beta_t \geq 0.5$) applies in both settings.

3.3 Descriptive Statistics

Table 2 describes the composition of the sample by subprocess, and Table 3 the distribution of the task-level variables.

Table 2 shows that the 91 in-scope tasks are spread unevenly across the six subprocesses, but broadly in line with their weight in working time. Record holds about a third of both the task count (31 tasks, 34.1%) and process time (34.0%), so the subprocess with the most task statements is not over-represented once tasks are weighted by time; the main divergences lie elsewhere, with Master Data Management carrying 8.8% of the tasks but 5.4% of process time and Report 14.3% of the tasks but 17.5% of the time. Almost every task is theoretically capable (90 of 91 have $\beta \geq 0.5$), whereas only 38 tasks appear in the Claude.ai data and 28 in the 1P API data in Release 5. Success readings are thinner still, and thinnest on the API: 31 tasks carry a task-success reading on Claude.ai against 20 on the 1P API in Release 5 (27 against 19 in Release 4), which bounds the samples of the regressions in Section 4.3.2. Finally, usage is strongly right-skewed (Table 3): among the

38 tasks with work usage in Release 5, the mean of 276 conversations is almost three times the median of 94 (skewness 4.6), and the most-used task carries 62 times the conversations of the median task on Claude.ai and 21 times on the 1P API. This skew motivates the log transformations of the usage, context-length, and cost variables in Section 4.3.2.

Table 2: Sample composition by subprocess.

Subprocess	In-scope tasks (n)	Share of process time (%)	Mean expertise (yrs)	Mean wage (USD/h)	Tasks with β ≥ 0.5 (n)	Tasks covered, R4 (C.ai / API)	Tasks covered, R5 (C.ai / API)
Master Data Management	8	5.4	14.89	66.25	8	5 / 5	6 / 5
Record	31	34.0	13.05	29.17	30	13 / 8	12 / 11
Close & Consolidation	5	4.2	13.65	33.88	5	2 / 2	2 / 2
Report	13	17.5	14.40	49.47	13	2 / 2	4 / 2
Analysis & Business Insight	12	15.0	15.19	65.52	12	5 / 5	5 / 6
Planning & Budgeting	22	24.0	15.35	59.40	22	8 / 4	9 / 2
Entire process	**91**	**100.0**	**14.28**	**47.61**	**90**	**35 / 26**	**38 / 28**

Note: In-scope tasks are the 91 O*NET tasks assigned to an R2R subprocess (Appendix A). Share of process time sums task time fractions (Section 3.1.1); mean expertise and wage are time-fraction-weighted means of ridge-estimated education years and BLS OEWS hourly wages; covered tasks are those appearing in the usage data of the respective platform and release (1P API data begin in Release 3).

Table 3: Descriptive statistics of task-level variables, in-scope R2R tasks, Release 5.

Variable	Platform	N	Mean	SD	Min	Median	Max
Task time fraction (hours/week, raw)	—	91	3.50	2.13	0.58	3.23	11.00
Ridge task expertise (years)	—	91	14.23	1.46	9.30	14.47	17.11
Task value (USD/hour)	—	91	49.44	29.48	24.55	45.56	129.63
Theoretical exposure (β)	—	91	0.66	0.25	0.00	0.50	1.00
Work usage (conversations)	Pooled	38	275.9	616.8	15.00	94.00	3616.00
Automation factor (α)	Pooled	38	0.77	0.16	0.50	0.81	1.00
Task observed exposure ($\tilde{r}$), gate = ON	Pooled	91	0.17	0.34	0.00	0.00	1.00
Task success (%)	Claude.ai	31	68.42	12.06	50.00	69.34	88.89
Task success (%)	1P API	20	54.29	12.91	24.00	53.86	79.31
Human-only time (hours)	Claude.ai	38	3.57	1.61	0.82	3.33	7.68
Human-only time (hours)	1P API	28	1.57	1.28	0.26	1.12	5.32
AI autonomy (1 to 5)	Claude.ai	38	3.23	0.29	2.35	3.25	3.66
AI autonomy (1 to 5)	1P API	28	2.72	0.26	2.06	2.77	3.21
ln(input context-length index)	1P API	28	−0.12	0.55	−1.64	−0.03	0.83
ln(output context-length index)	1P API	28	−0.15	0.54	−1.34	−0.02	1.36
ln(cost index)	1P API	28	−0.17	0.60	−1.63	−0.13	1.00

Note: Statistics are computed across tasks with a non-missing value on the indicated platform in Release 5; "—" marks task-level attributes that do not vary by platform. The Release 4 counterpart is Table D9 in Appendix D.

4 Results

4.1 Location of GenAI Exposure in the Record-to-Report Process

This section answers RQ1. Figures 1 to 5 establish where GenAI usage sits within R2R and in which collaboration mode (Section 4.1.1); Figure 6 compares observed with theoretical exposure by subprocess (Section 4.1.2), and Figures 7 to 10 examine whether that exposure is concentrated within subprocesses and across tasks (Section 4.1.3).

4.1.1 Platform Usage and Collaboration Mode

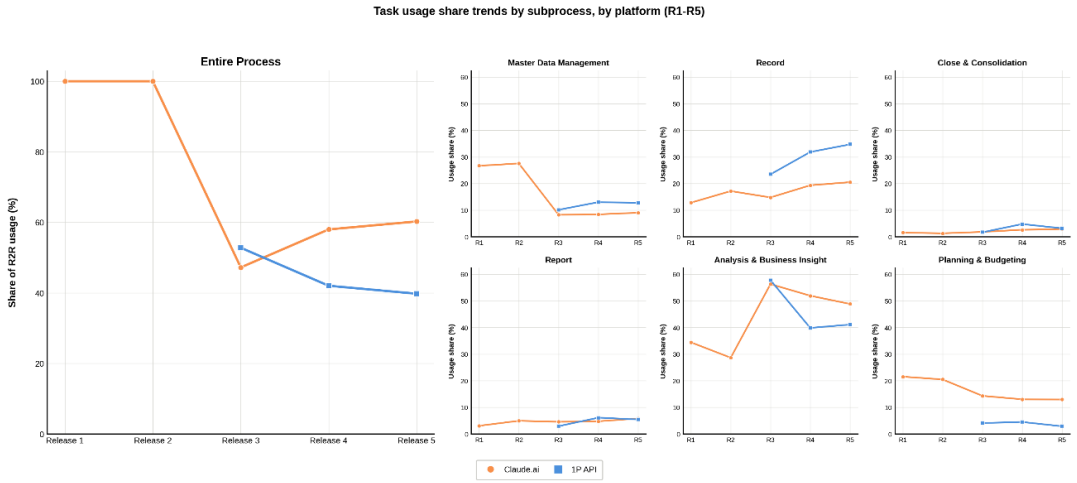


Figure 1: Task usage share by subprocess and platform across Anthropic Economic Index Releases 1 to 5. The left panel shows each platform's share of total R2R conversations; the six right-hand panels decompose that share by subprocess. Shares are obtained by summing task-level usage shares within each subprocess, so each platform's six subprocess series sum to 100% of its in-scope R2R usage; the left panel shows each platform's share of all R2R conversations. Claude.ai data begin in Release 1 and 1P API data in Release 3. The figure establishes where raw usage volume sits before any weighting by time, automation, or success, and therefore the baseline against which the exposure constructs in Section 4.1.2 should be read.

Overall, the findings central on which platform carries which type of work. 1P API functions as a programmatic automation engine, and Claude.ai functions as an interactive, judgment-assisted augmentation environment. On 1P API, automation accounts for 77 to 87% of all conversations across releases (Figure 2). Claude.ai inverts this almost entirely, where augmentation leads automation at every release (Figure 2), and the gap is stable rather than narrowing. Because subprocesses are distributed unevenly across platforms, which platform a subprocess over-indexes on may reflect how users consistently bring tasks they deem suitable with platforms' unique characteristics. That signal is clearest at the extremes. Record grows rapidly

on 1P API from roughly 23% of API volume at Release 3 to 35% by Release 5 while remaining flat on Claude.ai at 15 to 20% (Figure 1); Analysis & Business Insight dominates Claude.ai, accounting for the majority of that platform's overall volume (Figure 1) given the judgment-heavy nature of analysis work. Yet, this subprocess simultaneously commands the largest share of 1P API usage (Figures 1 and 4) in an automated manner, conforming with this platform's dominant mode. However, this signal might represent broad-based demand for analysis work across context rather than a demonstration of where that work is deemed automatable.

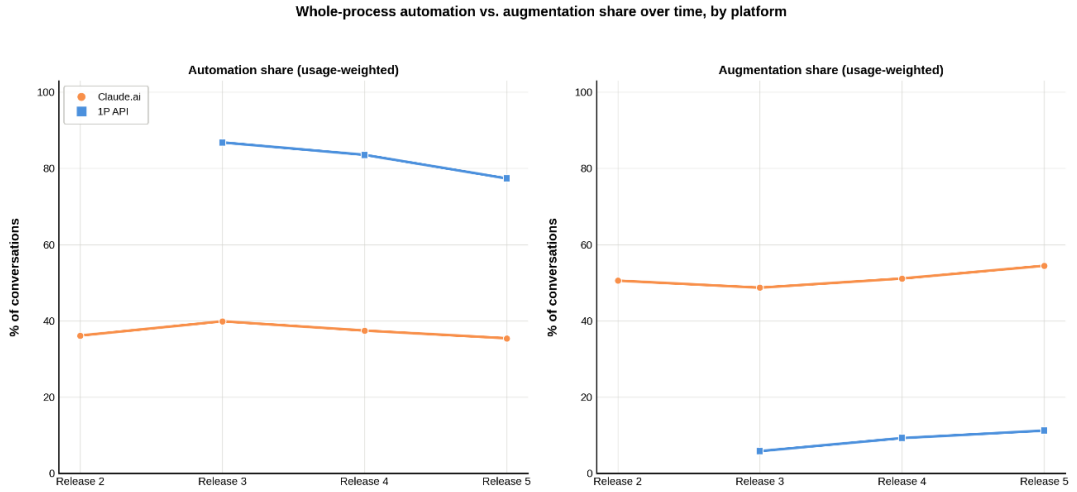


Figure 2: Whole-process automation versus augmentation share of conversations over time, by platform, Releases 2 to 5. Usage-weighted shares across all in-scope tasks; automation comprises directive and feedback-loop conversations, augmentation comprises task-iteration, validation, and learning conversations, and the remainder is unclassified. Collaboration data are unavailable in Release 1. The stable gap between platforms indicates that the mode of use is a property of the channel rather than of the release, which is why the remaining analyses treat the two platforms separately.

Within platforms, the automation overrepresentation indices reinforce such platform differences. On Claude.ai, where the automation baseline is lower, indices spread meaningfully (Figure 5, left). On 1P API, the indices compress, with most subprocesses cluster between 0.9x and 1.1x (Figure 5, right), following directly the much higher automation baseline of approximately 80%. Report's relatively high automation index on Claude.ai is notable precisely because its usage volume is low. When it does appear on the interactive platform, users are predominantly generating structured outputs rather than iterating over them. C&C is the clear outlier on both with an index of 0.7x on 1P API, 0.6x on Claude.ai, making it the subprocess most consistently resistant to routine automation.

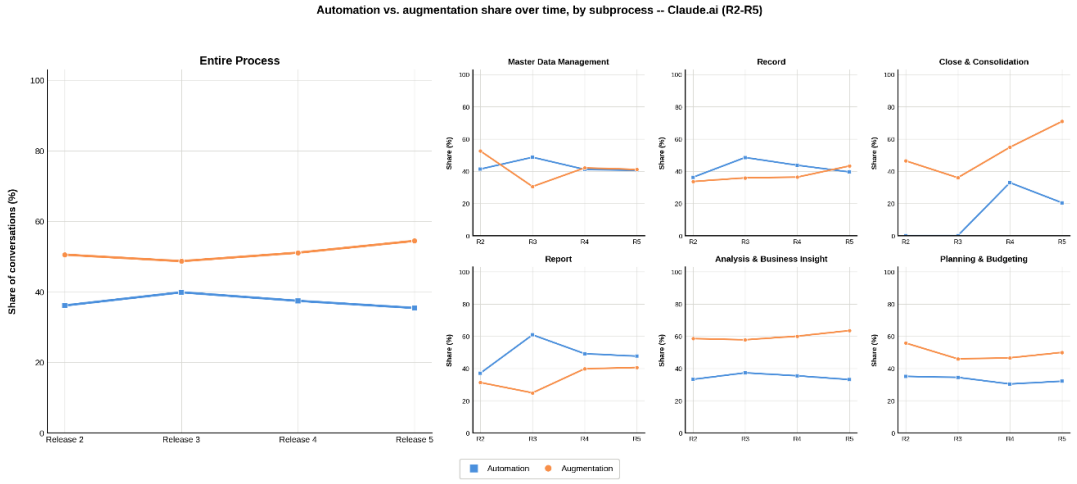


Figure 3: Automation versus augmentation share of conversations by subprocess over time, Claude.ai, Releases 2 to 5. Same measure as Figure 2, decomposed by subprocess. Shares for low-volume subprocesses, notably Close & Consolidation before Release 4, rest on few conversations and are correspondingly volatile. The persistent augmentation lead in Analysis & Business Insight and Planning & Budgeting marks the judgment-heavy end of R2R as the domain of interactive use.

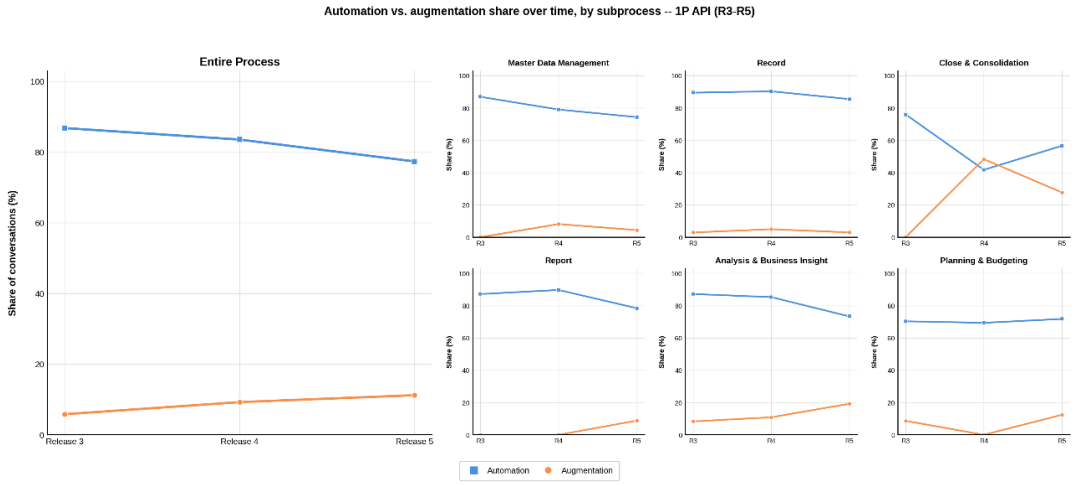


Figure 4: Automation versus augmentation share of conversations by subprocess over time, 1P API, Releases 3 to 5. Same measure as Figure 3 for the 1P API, available from Release 3. Automation dominates every subprocess; Close & Consolidation is the only one in which augmentation approaches automation (Release 4), anticipating its distinct exposure profile in Section 4.1.2.

The distribution that emerges, where codifiable subprocesses concentrated on 1P API and skewing automated, while judgment-heavy subprocesses concentrated on Claude.ai and skewing augmented, aligns with the task-content view of technological change where not all tasks within an occupation are equally codifiable (Autor et al., 2003). That it surfaces across both platforms, across five successive releases, and in both usage shares and

normalised overrepresentation indices suggests that the automation/augmentation classification is consistently demonstrating heterogeneity in exposure from the task level, as well as how users gravitate towards the appropriate tool in accordance with task-specific characteristics. In brief, Figures 1 to 5 provide the first part of the answer to RQ1: GenAI usage within R2R is located unevenly across subprocesses, and the mode of that usage is sorted by platform.

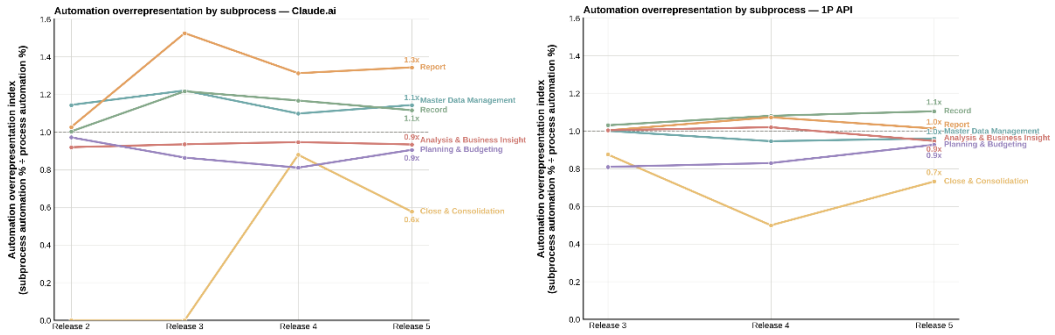


Figure 5: Automation overrepresentation index by subprocess (subprocess automation share divided by process automation share), Claude.ai (left) and 1P API (right), Releases 2 to 5 (Claude.ai) and 3 to 5 (1P API). The index divides a subprocess's automation share by the automation share of the entire process on the same platform and release; values above 1 indicate automation above the process average. Normalising by the platform baseline removes the level difference in Figure 2 and isolates the relative automation propensity of each subprocess, which is what allows the two platforms to be compared.

4.1.2 Observed Exposure by Subprocess

Task-substitution literature treats automation, specifically, as the channel through which AI exposure produces labour-market consequences (Autor & Thompson, 2025; Hampole et al., 2025). Aggregating automation and augmentation together, as Section 4.1.1 does, is, therefore, poorly matched to that framework, as augmentative use, however frequent, does not substitute for a task in the sense that substitution theory requires. Consequently, Section 4.1.2 asks how much of each subprocess's theoretical automation capability is actually materialising as automated usage, relative to the time each subprocess consumes in the process. Comparing this to theoretical capability helps surface where the gaps between those two measures lie, providing a more realistic signal on how exposed R2R actually is to AI-driven substitution.

Under Eloundou et al. (2024) β framework, the R2R process appears nearly fully automatable. β scores reach or approach 1.00 across all six subprocesses in both R4 and R5, with Record the only marginal exception at 0.97 (Figure 6). Yet when tested against actual Claude usage, exposure collapses. Even at its highest observed point of Close & Consolidation at 0.40

to 0.44 under volume gating across releases (Figure 6 for Release 5; Figure D1 in Appendix D for Release 4), realised usage achieves less than half of theoretical capability. Any gap between $R_{o,sp}$ and theoretical capability is by construction a gap of under-realisation. Three tiers emerge within such exposure profile, consisting of high production realisation (Record and Close & Consolidation), ad-hoc exploratory engagement (Master Data Management and Analysis & Business Insights), and near-absent adoption (Report and Planning & Budgeting); their persistence across R4 and R5 suggests these are not transient gaps but a durable configuration.

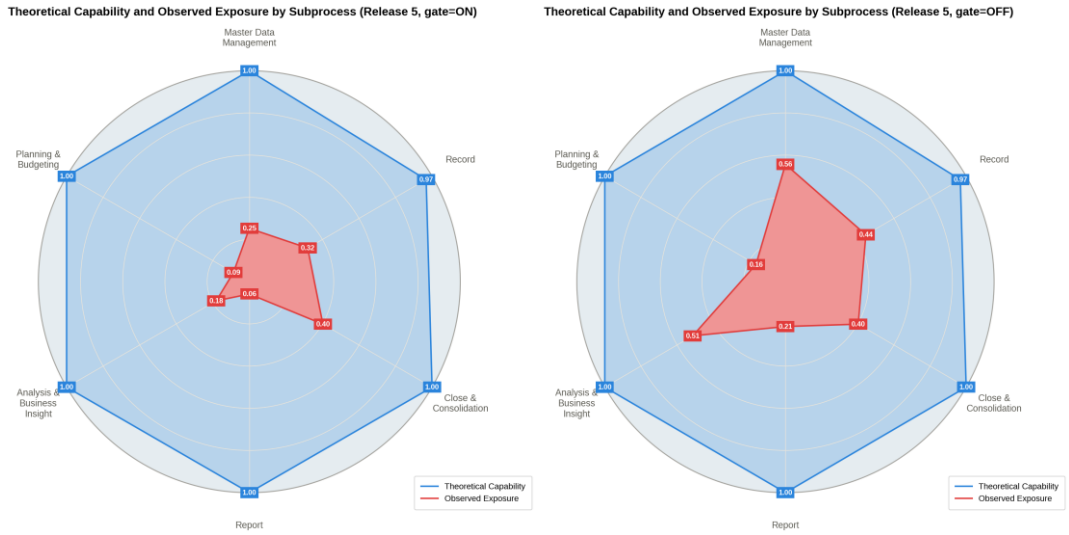


Figure 6: Theoretical capability and observed exposure by subprocess, Release 5, gate=ON (left) and gate=OFF (right). Theoretical capability is the share of a subprocess's tasks with β ≥ 0.5 (Eloundou et al., 2024); observed exposure is $R_{o,sp}$, the time-fraction-weighted mean of task-level exposure (Section 3.2.1). Under gate = ON, a task contributes only if it records at least 100 work-related conversations. Because observed exposure is defined only within capable tasks, the area between the two polygons measures unrealised capability, and its insensitivity to the gate for Report and Planning & Budgeting underlies the three-tier reading in the text.

Record and Close & Consolidation constitute a tier where observed exposure has genuinely taken hold. Under volume gating, Record grows from 0.21 to 0.32 (Figure 6; Figure D1), representing the single largest cross-release shift in the framework, while Close & Consolidation registers 0.44 in R4 and holds at 0.40 in R5. The gate ON/OFF comparison is also telling, with Close & Consolidation barely moving between conditions in either release, and Record shifting only moderately. This signals that tasks within these subprocesses individually sustain at least 100 conversations; usage is broad-based across the tasks that appear at all and not concentrated in a handful of exploratory interactions. Their usage composition directly accounts for this result. Record captures a growing share of API traffic, rising from approximately 23% of R2R API usage in R3 to 35% by R5 (Figure 1), with automation rates above 85% throughout (Figure 4). Close & Consolidation

illustrates an interesting case, where its exposure score and the persistently low automation index appear to contradict each other. This is driven by its automation intensity on 1P API being high in absolute terms even when it is low relative to that platform's baseline, which is already elevated at an automation rate of 80%. This leaves an open question, however. Close & Consolidation shows up meaningfully in automated form, yet augmentation still accounts for a large share of its usage, raising the question of whether this automation has genuinely taken hold or continues to require human involvement to complete the task. The modest decline in Close & Consolidation exposure score from Release 4 to Release 5 may be another hint in this direction, suggesting that automated usage, while occurring, may not yet be translating into automation that reliably substitutes for human efforts.

Master Data Management and Analysis & Business Insight present a different pattern with moderate observed exposure under gating, but the largest gate ON/OFF jump of any tier. Master Data Management moves from 0.25 to 0.56 in R5, and Analysis & Business Insight moves from 0.18 to 0.51 (Figure 6). Gate=OFF elevates these two to the most exposed subprocesses in the entire framework, revealing how exposure profiles of these subprocesses are driven by a long tail of low-volume, task-specific interactions that do not individually sustain production-scale deployment. Usage composition reinforces this reading, where Analysis & Business Insight accounts for 28 to 57% of R2R traffic across releases on Claude.ai (Figure 1), yet augmentation consistently outpaces automation within it, at roughly 59 to 63% augmentation share (Figure 3). This volume is thus structurally discounted in α_t even where it is large, once again reflecting exactly what $R_{o,sp}$ was constructed to capture. Master Data Management follows a similar logic, with declining usage share across both platforms and a roughly balanced automation and augmentation split on Claude.ai (Figure 3). The stability of this exposure tier from R4 to R5 confirms that while professional engagement with these tasks through Claude is real, it has remained in an augmentative manner that the measure is designed to distinguish from substitution.

Report and Planning & Budgeting sit at the floor, and this floor does not lift even when gating is removed. Report registers an exposure score of 0.06 and Planning & Budgeting of 0.09 under gate=ON (Figure 6); removing the volume threshold produces only modest upward movement for either. The gate=OFF inflation in the previous tier is largely absent here, pointing to a different failure mode different from the “experimentation yet scaled” profile present in Master Data Management and Analysis & Business Insights. Report accounts for only 3 to 5% of R2R traffic on both platforms throughout the study period; Planning & Budgeting on Claude.ai has declined from approximately 21% in R1 to 13% by R5, with API share negligible throughout

(Figure 1). Both of these subprocesses also carry augmentation-dominant Claude.ai profiles, but the volume floor is too low for such composition to matter (Figure 3). Consequently, the gap between theoretical exposure and realised $R_{o,sp}$ is widest here.

In brief, observed exposure within R2R is limited and uneven. Most subprocesses show evidence of professional engagement with the technology but remain in an exploratory or augmentative mode that $R_{o,sp}$ is designed to distinguish from substitution. That only Record is shifting meaningfully indicates that the gap between theoretical and observed exposure is not narrowing uniformly across the process, but rather concentrated only where API-integrated, automation-intensive usage has already taken root, directly informing the next sub research question investigating whether exposure concentrates within a subprocess and across the entire R2R cycle. Figure 6 thus provides the second part of the answer to RQ1: measured against capability, exposure is realised in only a minority of R2R time, and that minority sits mainly in Record and Close & Consolidation.

4.1.3 Task-Level Concentration of Exposure

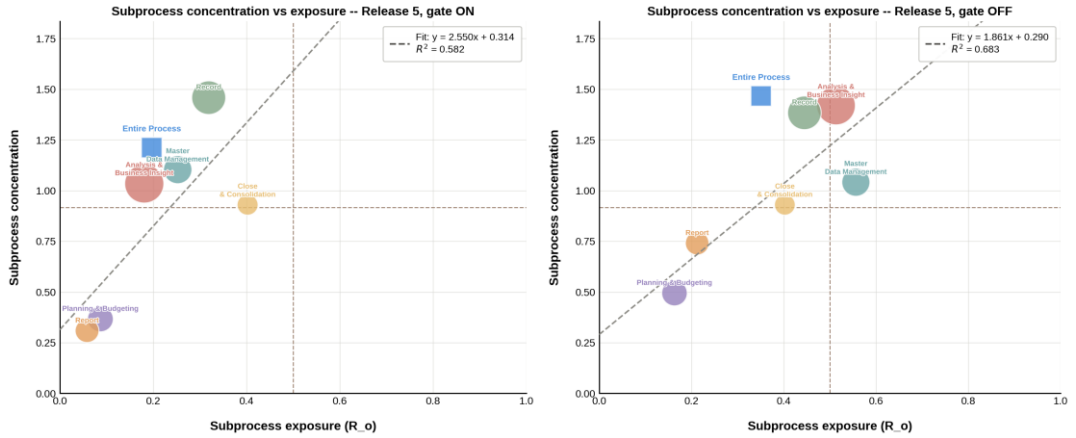


Figure 7: Subprocess concentration against subprocess exposure (R_o), Release 5, gate=ON (left) and gate=OFF (right). Marker size scales with subprocess conversation volume; dashed lines show the OLS fit, the sample concentration median, and the middle point between no and full exposure. Concentration is the time-fraction-weighted dispersion of task exposure around the subprocess mean, scaled by ln(1 + N) (Section 3.2.2); the square marks the entire process. With six subprocesses the fitted line is descriptive rather than inferential. Its positive slope indicates that subprocesses absorbing more exposure absorb it more unevenly across their tasks, the pattern Hampole et al. (2025) associate with reallocation rather than displacement.

The scatterplots in Figure 7 demonstrate that as a subprocess absorbs more AI usage, that usage does not spread across its tasks but narrows onto fewer of them. Across both gate conditions, subprocess exposure and concentration move together positively, with R^2 of 0.582 under gate=ON and

0.683 under gate=OFF. The slope, however, flattens in the opposite direction, from 2.550 to 1.861. Indeed, gating retains only production-scale tasks where concentration per unit of exposure is already at its sharpest, so admitting lower-volume tasks under gate=OFF distributes exposure more broadly, compressing the concentration increment per additional unit of mean exposure. Still, both gate settings agree on the directional conclusion that higher subprocess exposure is inseparable from greater internal task imbalance.

Subprocess-wise, the six subprocesses sort into two distinct regimes under gate=ON (Figure 7, left). Report and Planning & Budgeting fall squarely into the quadrant with low exposure and low concentration, and the gate moves neither. This is consistent with Section 4.1.2 finding that their exposure floor does not lift even when the volume threshold is removed, confirming the absence of structural automated AI imprint here. The remaining four subprocesses all exceed the concentration median, but their positioning within the high-concentration zone is not uniform. Master Data Management, Record, Close & Consolidation, and Analysis & Business Insights all land in the quadrant with moderate-to-low subprocess exposure paired with high concentration, meaning that a few tasks carry disproportionate exposure while the remainder sit largely untouched. Yet the mechanism differs. For Master Data Management and Analysis & Business Insights, as the gate filters out most of their activity, it leaves behind a concentrated residual of tasks that pass. Record's concentration reflects a different dynamic, where a few high-volume automated tasks dominate, and the gate changes little because those tasks already clear the volume threshold individually. Close & Consolidation sits at the boundary, with the highest subprocess exposure under gate ON and lower concentration than Record, suggesting its tasks are both production-grade and more evenly distributed internally.

The shift from gate=ON to gate=OFF is where the most interpretively consequential movement occurs (Figure 7, right). Master Data Management and Analysis & Business Insights undergo the largest exposure jumps of any subprocess, which, when paired with their sustained high concentration, moves both toward the targeted automation quadrant where only a few critical, highly important tasks are massively exposed, while others are untouched. However, this positioning must be read against their augmentation-heavy usage composition (Section 4.1.1). This gate=OFF positioning, therefore, does not signal substitution at scale, but rather augmentative concentration at scale as imbalanced AI engagement moves toward capability enhancement rather than task replacement. Record and Close & Consolidation, in contrast, shift little between gate conditions, reinforcing the fact that their tasks individually sustain production volume.

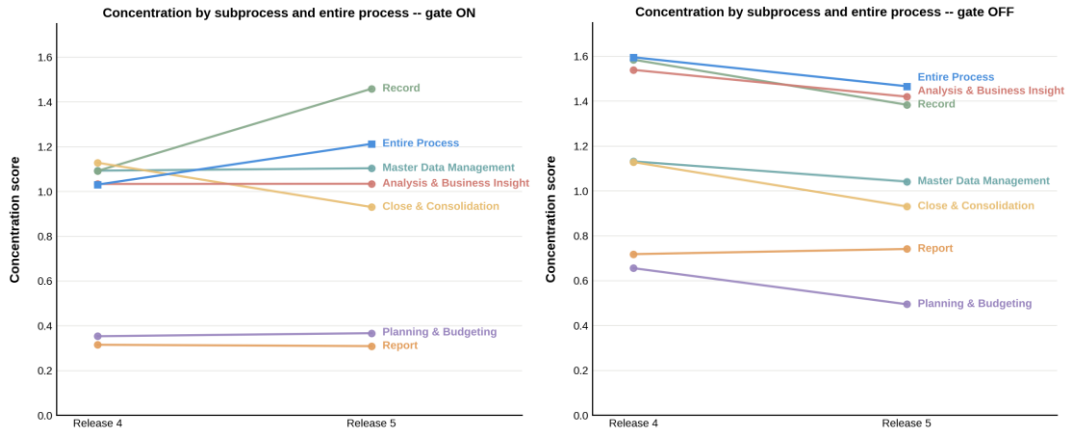


Figure 8: Longitudinal change in subprocess concentration from Release 4 to Release 5, gate=ON (left) and gate=OFF (right). Only Releases 4 and 5 are shown because the work-use classification required for WorkUsage_t, and hence for observed exposure and concentration, is published from Release 4 onwards. Levels differ between panels because the gate changes which tasks enter the dispersion sum; comparisons are therefore meaningful within, not across, panels.

On a longitudinal point of view**,** Figure 8 reveal that under gate=OFF, the dominant directional movement from Release 4 to Release 5 is convergence and decline, particularly with the Entire Process concentration decreasing from 1.59 to 1.47. This pattern tells the story of a maturing adoption dynamic, where engagement diffuses across a wider task range within each subprocess as users grow more familiar with Claude. The gate=ON panel, on the contrary, tells a different and sharper story. Close & Consolidation moves in the opposite direction from 1.13 to 0.93, suggesting its production-grade tasks became more evenly distributed in Release 5, with exposure spreading slightly rather than concentrating further. Reading against its slightly decreasing exposure from Section 4.1.2, this further consolidates the wonderings if the exposure pattern is actually taking hold. Master Data Management and Analysis & Business Insight are essentially flat, representing a stable but structurally uneven task footprint. The Entire Process rises modestly from 1.04 to 1.21, but this is largely an aggregation effect driven by Record's surge pulling the process-level dispersion upward.

Record's surge from approximately 1.10 to 1.46 (Figure 8, left) represents the largest single-subplot movement in either chart and a full divergence from the flat or declining trajectories of every other subprocess. In Release 5, production-scale AI usage within Record became markedly more concentrated on a bounded task core, and the gap between those tasks and the remainder of the subprocess has widened. However, a specific caveat applies to Record's concentration score. Many tasks within this subprocess share substantially similar underlying execution but carry distinct O*NET descriptions, and the Clio classification methodology tends to resolve Claude

conversations to a single specific task, with agreement declining further at lower taxonomic levels (Handa, Tamkin, et al., 2025). Consequently, observed task diversity within Record can be narrower than the true task set, inflating concentration as a result. If actual task diversity is broader than Clio's classification, Record's true concentration would be lower than what is recorded here. Paired with its production-grade, gate-robust exposure, AI could be broadly more relevant across this subprocess rather than focused on a bounded task core. Record's score is, therefore, best read as an upper bound on true within-subprocess concentration. Nevertheless, concentration and exposure are affected asymmetrically by the same classification behaviour, as classification consistency means that similarly executed tasks carry consistent usage signals and pass the volume gate together, which actually stabilises $R_{o,sp}$ rather than biasing it.

Another observation is a level effect, where gate=OFF produces systematically higher concentration scores across all subprocesses than gate=ON, with the gap most pronounced for Report and Planning & Budgeting (Figure 8). This is mechanically consistent with the concentration score construction under ungated volume scalar. The small amount of exploratory or incidental Claude usage these subprocesses attract is enough to register as dispersion when ungated, but not enough to pass the volume threshold and carry structural weight as it is compressed to near-zero structural concentration by the gate settings. These dynamics are not contradictory but reflect how two distinct adoption behaviours are manifesting in parallel. Exploratory users broaden their task range across the process, while automated pipelines deepen their footprint in the tasks they already target.

Dropping to the task level sharpens the differences among subprocess concentration patterns. According to Figure 9, most tasks cluster in the low-dispersion, low-exposure quadrant regardless of gate setting, demonstrating that the baseline condition across R2R is not uneven exposure but no exposure at all. A handful of outliers qualify this without overturning it, precisely representing the high-dispersion anchors that drive high concentration scores across most of the R2R cycle. Task 21321 *Develop or analyse information to assess the current or future financial status of firms.* stands apart from Analysis & Business Insights exposure under gate=ON, suggesting AI automation specific to that task rather than the broader subprocess. Under gate=OFF, tasks 1306 *Develop data models describing data elements and how they are used, following procedures and using pen, template, or computer software.* and 89 *Set up cost monitoring and reporting systems and procedures.* surface in the high-dispersion, moderate-exposure region, but their volumes are insufficient to pass through the volume scalar and register at the subprocess level. Regardless, gate=ON still witnesses a handful of outliers departing from their parents' exposure

profile across subprocesses, consolidating the earlier point that concentration is driven by isolated spikes against an otherwise low-exposure base. However, that pattern does not hold with Record, where its tasks cluster tightly in the low-to-moderate dispersion range, indicating that their exposure diverges modestly from their subprocess mean. Record's high concentration is thus not generated by a few anomalous tasks pulling far from the subprocess norm; rather, its targeted tasks are uniformly exposed, pass gating as a set, and are subsequently scaled up by volume. The concentration score is high because a subset of tasks jointly sustains that profile, not because any one task departs sharply from it. This is also consistent with the Clio classification caveat raised above, where tasks with similar underlying execution but distinct O*NET descriptions would produce exactly this kind of tight cluster rather than one outlier task carrying the subprocess.

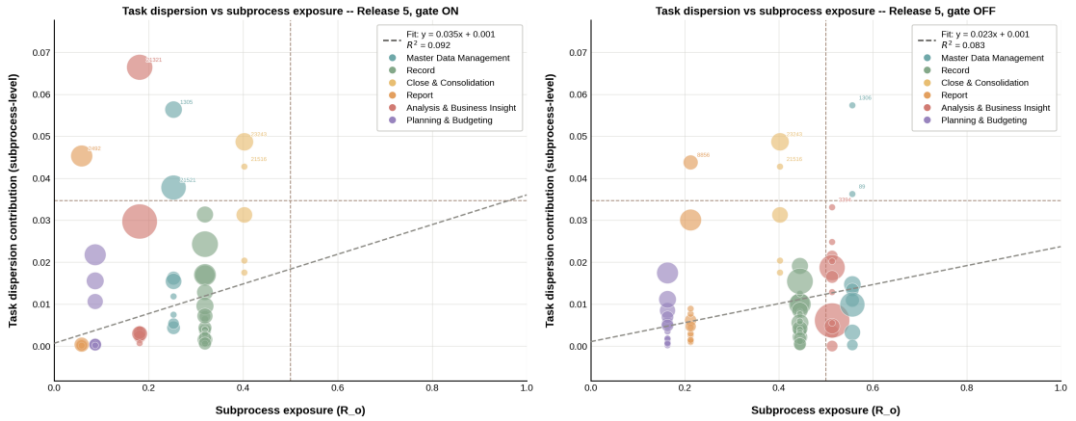


Figure 9: Task-level dispersion contribution against subprocess exposure, Release 5, gate=ON (left) and gate=OFF (right). Labelled points identify the highest-dispersion tasks by O*NET task ID. Each bubble is a task, coloured by subprocess and sized by usage; the y-axis is the task's weighted squared deviation from its subprocess mean, i.e. its contribution to subprocess concentration. Labels refer to the task register in Appendix A. The weak fit shows that a subprocess's exposure level says little about which of its tasks become concentration anchors, so concentration has to be read task by task.

The slope being positive means that AI usage, as it grows within a subprocess, doesn't gradually equalise across tasks but pulls further toward already-favoured tasks, incrementally reinforcing the imbalance rather than spreading outward to new ones. But the low R^2 of 0.092 at gate=ON and 0.083 at gate=OFF (Figure 9) means this tendency is weak and inconsistent across the task population, making subprocess exposure level a poor predictor of any individual task's dispersion contribution. The direction is informational, but the mechanism is idiosyncratic. Which specific tasks become the concentration anchor varies enormously across subprocesses. More exposure does mean slightly more internal polarisation on average, but *which tasks* get polarised is not predictable from exposure alone.

The concentration findings compound rather than merely echo the exposure findings. While observed exposure within R2R is structurally limited and uneven, even within the subprocesses that do register meaningful exposure, that exposure does not distribute evenly across their constituent tasks but pulls toward a bounded core. What the two patterns share is a logic of self-reinforcement rather than diffusion in exposure. AI in R2R is, at this stage, less a rising tide than a series of drill points.

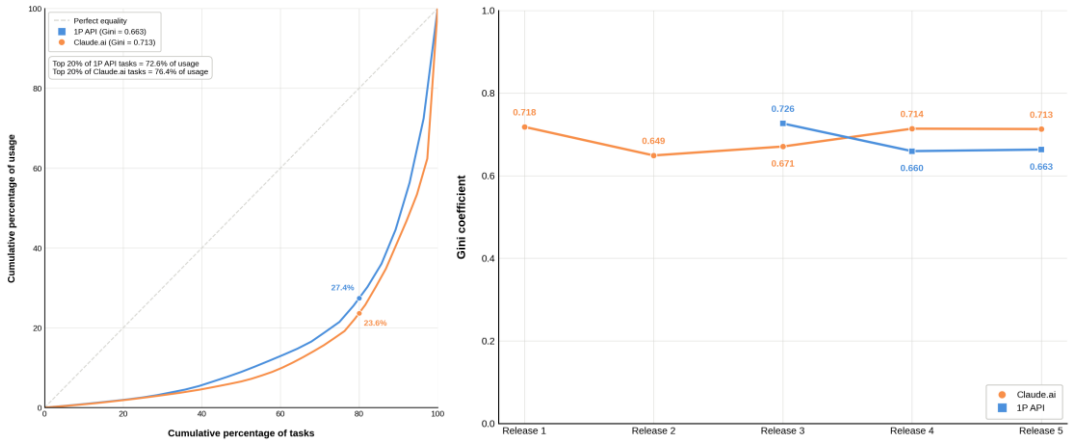


Figure 10: Lorenz curves of task-level usage concentration by platform, Release 5 (left), and the corresponding Gini coefficient by platform across releases (right). Lorenz curves rank tasks by usage share; the further a curve lies below the line of equality (45 degrees), the more usage is concentrated on only a few tasks, and the Gini coefficient summarises that distance per release. Unlike the concentration score, this measure uses raw usage and is independent of time fractions and gates, so its agreement with Figures 7 to 9 shows that the concentration finding is not an artefact of the weighting scheme.

The Lorenz curves and Gini coefficients confirm the same picture in aggregate. At R5, the top 20% of tasks account for 72.6% of 1P API usage and 76.4% of Claude.ai usage, with Gini coefficients of 0.663 and 0.713 respectively (Figure 10; the ten most-used tasks alone carry 85.3% and 82.6% of usage, Table D4, Appendix D). The near-flat Claude.ai trajectory of Gini coefficient consolidates the pattern identified above, one where usage deepens their footprint in tasks they already target, with enterprise adoption on this platform neither broadening nor diversifying across time. The marginally declining 1P API signals some experimental breadth, but such diffusion remains modest. Neither channel is approaching an evenly distributed adoption pattern, confirming that exposure remains concentrated rather than dispersing as usage matures.

These four measures (Figures 7 to 10) converge on a single answer to where GenAI exposure concentrates within R2R: a narrow, structurally consistent core of tasks instead of the process as a whole. According to Hampole et al. (2025), elevated mean exposure depresses employment

growth within an occupation-firm pair, while concentration of that exposure in a subset of tasks offsets this by enabling workers to reallocate effort toward unaffected work. R2R resists a clean reading on this basis, as its subprocesses split across a range of exposure-concentration combinations, from high exposure paired with rising concentration, to comparable exposure paired with flat or declining concentration, to exposure too low or too augmentative for either variable to carry much economic weight. GenAI in R2R is, thus, a small number of drill points in which exposure implications diverge sharply across subprocesses. Read together, Figures 1 to 10 answer RQ1: GenAI exposure in R2R is located in a narrow core of tasks, realised mainly in Record and Close & Consolidation, and concentrated rather than diffused within the subprocesses that register it.

4.2 Expertise Level of GenAI Exposure and Concentration

Autor and Thompson (2025) argue that automation reshapes the average expertise level of the tasks that remain within an occupation, which means what matters is not simply how concentrated exposure is, but the expertise composition of that concentration. This section accordingly investigates the expertise distribution of tasks exposed to GenAI in the R2R process. Figures 11 and 12 address the expertise composition of task coverage (Section 4.2.1); Figures 13 to 18 that of observed exposure and concentration (Section 4.2.2).

4.2.1 Expertise Composition of Task Coverage

This section first investigates expertise distribution on task coverage by comparing the education-years required by the subset of R2R tasks showing up in Claude usage against the education-years required by the full R2R task population, per platform and release, using the task-level education requirement predicted on the Ridge regression (Section 3.1.2). A negative gap indicates the platform is disproportionately engaging with higher-complexity tasks, thus leaving workers with the comparatively simpler residual, in other word, *deskilling*. A positive gap indicates the opposite, with coverage concentrated on lower-complexity tasks and leaving workers to handle the harder residual, in other word, *upskilling*.

Claude.ai displays a consistently negative net gap across all five releases, moving only marginally from -0.29 years in R1 to -0.27 years in R5 (Figures 11 and 12), which indicates that user-driven interactions on the consumer platform persistently target the higher-complexity, higher-education segment of the R2R task space. The 1P API tells a different story, beginning with a negative gap of -0.19 years in R3 and widening to -0.29 years in R4 before turning positive by R5 at +0.10 years (Figures 11 and 12). Programmatic API deployments have shifted, unevenly, toward lower-complexity tasks, moving the overall worker impact from deskilling to upskilling over time.

Nevertheless, such upskilling/deskilling patterns might simply reflect the way each platform is used as established across RQ1. Claude.ai usage skews toward tasks requiring more expertise and more time (Table 6), while 1P API usage skews toward less of both. If Claude.ai is disproportionately absorbing complex, high-education R2R work, a negative gap follows almost mechanically from usage patterns alone, and the inverse holds for 1P API's lighter task mix. Accordingly, the deskilling signal on Claude.ai, and vice versa, upskilling, in 1P API, is largely a restatement of their usage patterns.

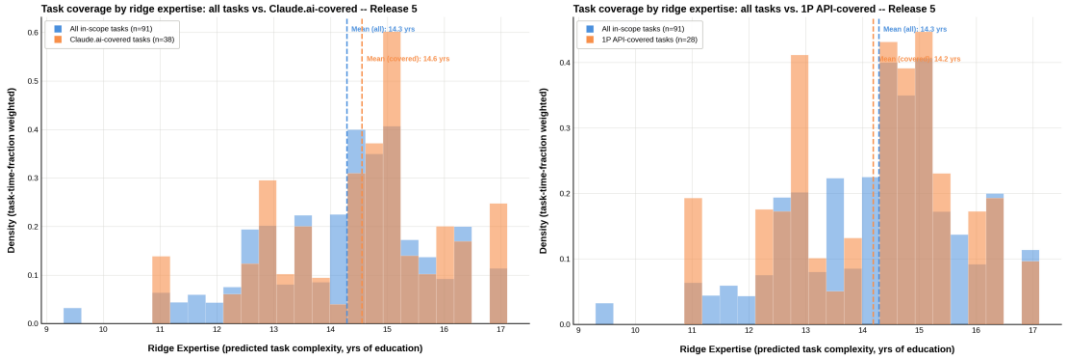


Figure 11: Time-fraction-weighted distribution of predicted task complexity (ridge-estimated years of education) for all in-scope R2R tasks versus tasks covered, Claude.ai (left) and 1P API (right), Release 5. Histograms of ridge-estimated education years, weighted by task time fraction, for all in-scope tasks and for tasks with any usage on the platform; dashed lines mark the weighted means. The difference between the two means is the net skill-coverage gap tracked over time in Figure 12.

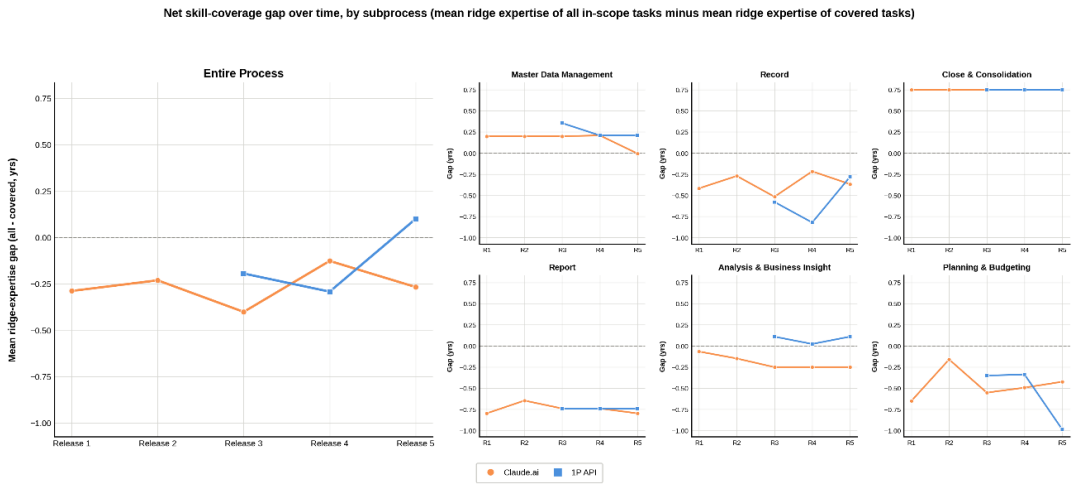


Figure 12: Net skill-coverage gap (mean complexity of all in-scope tasks minus mean complexity of covered tasks) over time, by subprocess and platform. Negative values mean that covered tasks are more complex than the subprocess as a whole, so the residual left to workers is simpler (deskilling); positive values indicate the reverse (upskilling). A flat line, as for Close & Consolidation, reflects an unchanged set of covered tasks across releases rather than a stable underlying process.

4.2.2 Expertise Composition of Observed Exposure and Concentration

Task coverage alone does not guarantee displacement, and more comprehensive automation exposure and concentration measures have been set in place to understand the characteristics of how GenAI is currently used in the R2R cycle. This section continues that investigation, asking at what expertise level does GenAI exposure concentrate within R2R. The true structure is bimodal, and the subprocess- and task-level analysis together

reveal why the process-wide concentration observed in the middle of the complexity range is the composite of two distinct forces operating at opposite ends of the expertise spectrum instead of a genuine central tendency.

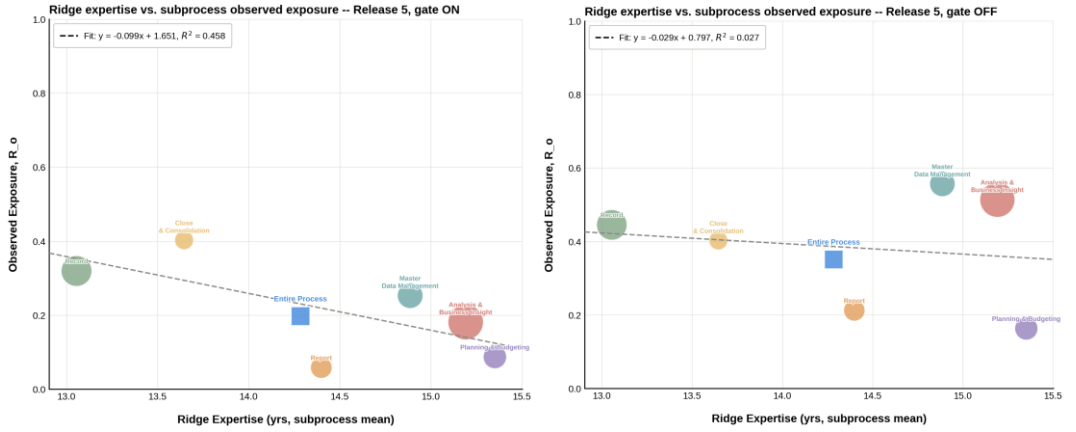


Figure 13: Ridge-estimated subprocess expertise against subprocess observed exposure (R_o), Release 5, gate=ON (left) and gate=OFF (right). Subprocess expertise is the time-fraction-weighted mean of task-level ridge estimates; bubbles are sized by conversation volume and the square marks the entire process. Fits across six points are descriptive. The contrast between panels shows that the negative relationship under gate = ON rests largely on Record and disappears once low-volume tasks are admitted.

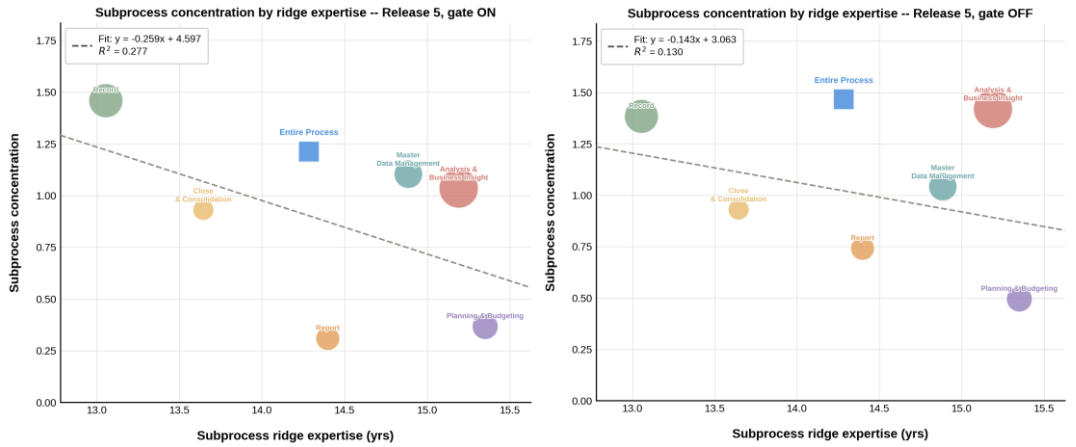


Figure 14: Subprocess concentration against ridge-estimated subprocess expertise, Release 5, gate=ON (left) and gate=OFF (right). As Figure 13, with concentration on the y-axis; the square marks the entire process. The split of the high-expertise subprocesses between high (Master Data Management, Analysis & Business Insight) and low (Planning & Budgeting) concentration is what prevents expertise from predicting concentration at this level.

In Figures 13 and 14, the fitted slope of observed exposure and concentration on expertise is negative under both gate conditions, with gate=ON presenting a steeper slope, meaning higher-expertise subprocesses trend toward lower exposure, though the relationship is weak across only six

subprocesses. Still, the positioning of individual subprocesses can be meaningful, with the lowest-expertise subprocess carrying the highest exposure score while the highest-expertise subprocesses are internally split rather than uniformly elevated. Record requires least expertise to execute yet represents the highest exposure and concentration score in the dataset, an outlier that drives much of the apparent negative slope, reflecting an internal structure consisting of a bounded set of high-volume automation-prone tasks that are gate-robust. At the other end of the expertise spectrum, Master Data Management, Analysis & Business Insights, and Planning & Budgeting occupy nearly the same expertise range but land at opposite extremes of the concentration axis, with the first two among the highest-concentration subprocesses while Planning & Budgeting among the lowest.

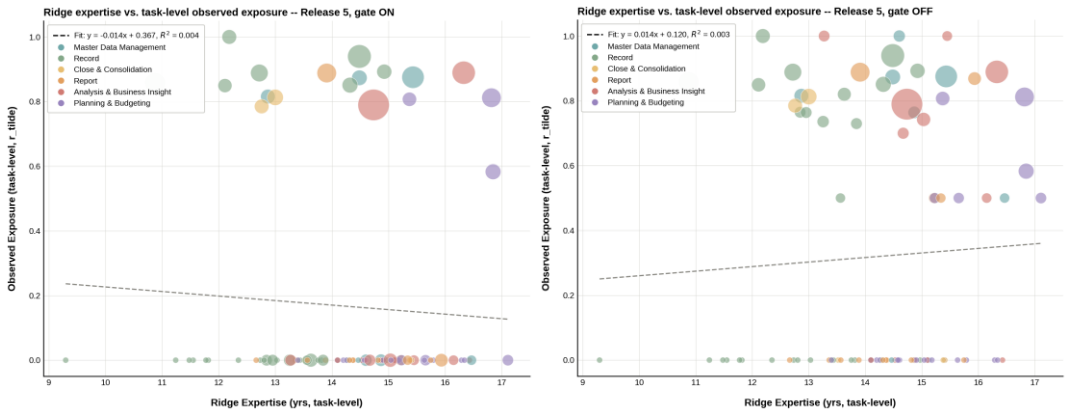


Figure 15: Ridge-estimated task expertise against task-level observed exposure, Release 5, gate=ON (left) and gate=OFF (right). Task-level exposure is zero for tasks failing either gate and lies between 0.5 and 1 otherwise, which produces the floor at zero and the band above 0.5. The near-zero R^2 shows that required expertise does not sort tasks into exposed and unexposed, consistent with near-universal theoretical capability.

The task-level dispersion charts provide a more granular picture than the subprocess-level regressions can offer. In Figure 15, expertise by observed exposure, under gate=ON, meaningful automation exposure concentrates almost entirely in low-expertise tasks, consistent with Autor and Thompson (2025). Removing the gate settings surfaces two further patterns confined to the high-expertise band: a jump from 0 to 100 exposure under gate=OFF, signalling experimental automation, and a handful of tasks sitting at a baseline, fully augmented exposure of 0.5. In Figure 16, expertise by concentration, the outliers that depart from their subprocess's observed exposure replicate exactly the pattern established in Section 4.1.3, representing the anchors that drive concentration scores upward at the subprocess level, and their placement differs sharply by subprocess. Record follows a conforming pattern, where concentration tracks close to the subprocess's overall exposure profile, whereas in subprocesses that are not themselves uniformly automated, it is the higher-education tasks that are

more likely to diverge from that profile and become the concentration anchors. Yet, the low R^2 itself across all these figures, once again, exactly strengthens the interpretation of how subprocess- and task-specific this pattern is, and it is not possible to predict which task will diverge from its subprocess mean exposure based on how complex they are.

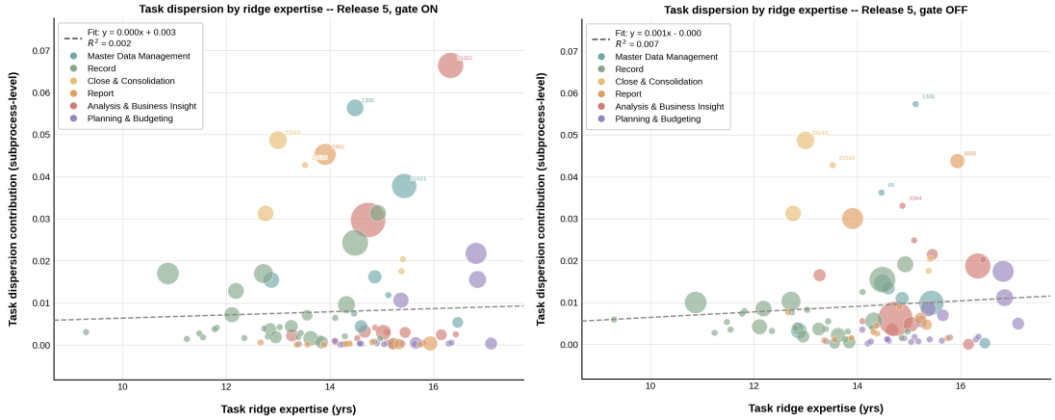


Figure 16: Task-level dispersion contribution against ridge-estimated task expertise, Release 5, gate=ON (left) and gate=OFF (right). As Figure 9, with task expertise on the x-axis. The labelled concentration anchors spread across the expertise range rather than clustering at either end, which is the task-level basis of the bimodal reading in the text.

These results answer the second research question by first rejecting the premise that exposure operates at a single expertise level. A purely positional reading of the data, that concentrated tasks sit at roughly 14 years of ridge-estimated required education (14.1 years when each task is weighted by its contribution to process-level concentration under gate = ON, against 14.3 years for the process as a whole; Figure 14), would describe a midpoint that does not correspond to the actual concentration mechanism in R2R found in Section 4.1.3. The true structure is *bimodal*, and the subprocess- and task-level analysis reveals the composite of two distinct forces operating at opposite ends of the expertise spectrum. Record, the lowest-complexity subprocess in R2R, anchors the low end, where its concentration conforms to a bounded set of transactional tasks. Master Data Management and Analysis & Business Insights sit at the opposite end of the complexity spectrum, and their concentration is anchored rather than conforming, with a small number of high-dispersion tasks carrying the exposure inside subprocesses that otherwise remain largely untouched. This is selective delegation rather than holistic substitution, and the distinction tracks complexity directly, where a high-complexity subprocess does not offer a large homogeneous block of tasks that could be uniformly automated. What it offers instead are isolated points of sufficiently bounded judgment for Claude to be trusted, and/or tested, with, consequently spiking the concentration measures. Figures 13 to 16 therefore answer RQ2 at the level

of exposure and concentration: the expertise level of GenAI exposure in R2R is bimodal rather than central.

Looking at this from a value perspective inspired by Hampole et al. (2025) reinforces the same picture, where the reallocation dynamics differ among subprocesses' value. Hampole et al. (2025) find that while substitution effects are strongest for higher-paid occupations, these same jobs also benefit from the strongest reallocative gains because their AI exposure is often highly concentrated. The wage-concentration relationship at the subprocess level (Figure 17) closely mirrors the expertise-concentration findings discussed earlier, where the downward directional signal is present but not structurally meaningful. The connection to Hampole et al. (2025) reallocation logic where concentration scales with wage is, hence, partial and process-specific here. Master Data Management and Analysis & Business Insights replicate the high-value, high-concentration pattern the economy-wide framework predicts, while Planning & Budgeting's combination of low exposure and low concentration per RQ1 means the reallocation mechanism never activates there in the first place, and Record stands as a direct counter-example to the framing altogether. Task-level dispersion reinforces lack of connection to theory at finer grain, where dispersion mildly decreases as task value rises (Figure 18). However, the fit is weak, and high-dispersion outliers appear across the entire wage range with no clustering at either end. Whether a task functions as a concentration spike within its subprocess is, thus, not only process-specific, but also task-specific, and independent of where the worker performing it sits in the wage distribution.

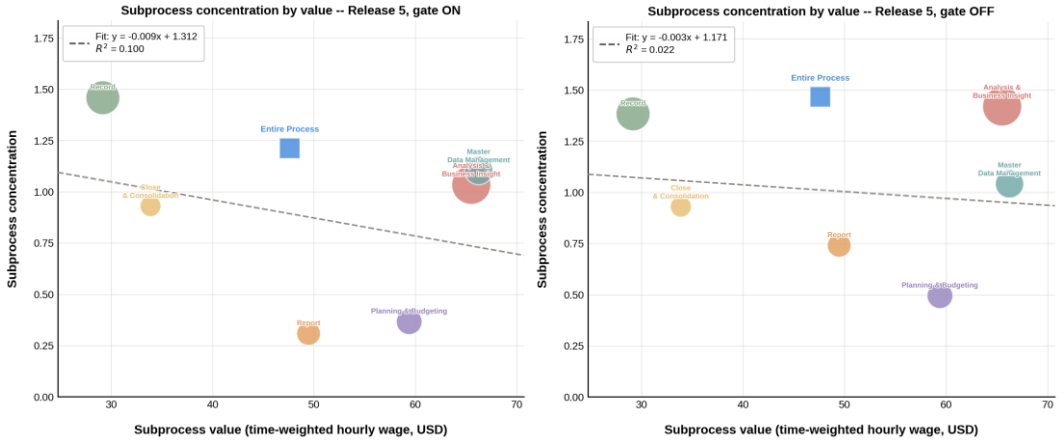


Figure 17: Subprocess concentration against subprocess value (time-fraction-weighted hourly wage), Release 5, gate=ON (left) and gate=OFF (right). Subprocess value is the time-fraction-weighted mean hourly wage (BLS OEWS, May 2025) of the occupations performing its tasks. The figure tests the wage dimension of Hampole et al.'s (2025) reallocation argument at process level; the weak negative fit is driven by Record's combination of low value and high concentration.

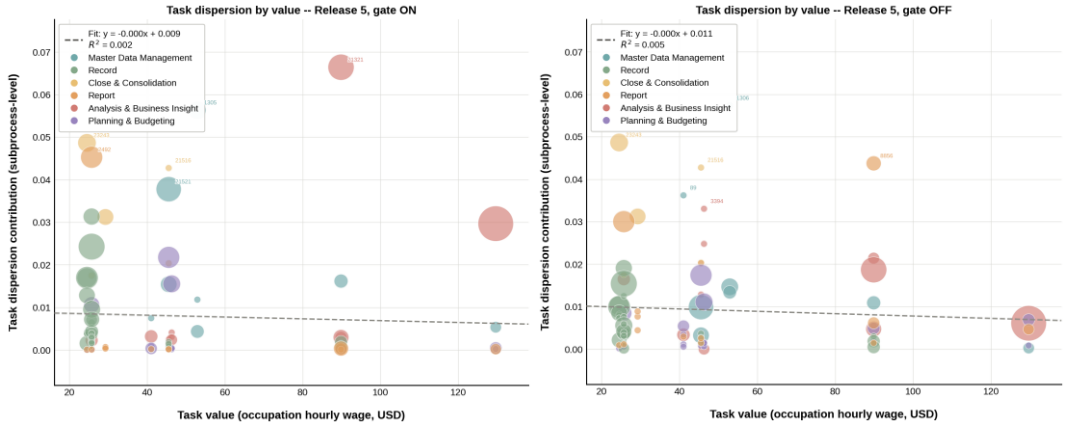


Figure 18: Task dispersion contribution against task value (occupation hourly wage), Release 5, gate=ON (left) and gate=OFF (right). Subprocess value is the time-fraction-weighted task value within its subprocess. Because task value is the wage of the task's occupation, tasks of the same occupation share an x-value and form vertical columns. High-dispersion tasks appear across the wage range, so task value does not identify which tasks become concentration anchors.

In conclusion, the expertise level of exposure in R2R is split between a low-complexity conforming regime and a high-complexity anchoring regime, and each regime's concentration mechanism is a function of the expertise profile of the subprocess it occurs in. The process-wide appearance of moderate-expertise exposure is an artifact of averaging across these two regimes rather than evidence that either exists on its own.

4.3 Effectiveness of GenAI Exposure

This section answers RQ3. Sections 4.1 and 4.2 locate GenAI exposure within R2R and characterise its expertise composition, but both treat an exposed task as a task taken over. This section tests that assumption by asking whether the exposure observed in R2R actually converts into successful performance. Figures 19 and 20 and Tables 4 and 5 compare effective exposure with task coverage and with observed exposure (Section 4.3.1), and Figures 21 to 26 and Tables 6 to 8 examine what determines task success on each platform (Section 4.3.2).

4.3.1 Effective Exposure Relative to Exposure Patterns

RQ1 and RQ2 establish where exposure sits and at what expertise level it concentrates, but both are silent on a question that determines whether either finding matters in practice: whether that exposure actually converts into performance. Nothing in current constructs speaks to whether Claude performs the task well, poorly, or anywhere in between, and a subprocess in either exposure profile can entirely compose of tasks Claude is failing. This gap is not neutral with respect to complexity. A conforming, low-complexity block could concentrate exposure precisely because it is easy to touch in terms of codifiability, not because it is being reliably executed; an anchoring,

high-complexity task could concentrate exposure because it is worth attempting even at low success rates, not because Claude has closed the judgment gap RQ2 identifies. Therefore, it is necessary to consider not only where exposure lands, but also whether it penetrates, and this is operationalised via effective exposure. Effective exposure is the share of total task-time that Claude performs successfully, calculated as a weighted sum of task-level success rates, with each task's weight given by its share of subprocess time. Constructed on the same time-fraction basis as exposure and concentration, the two are, thus, directly comparable to facilitate the investigation of whether exposure effectively displaces time-intensive work, or merely occupies its margins.

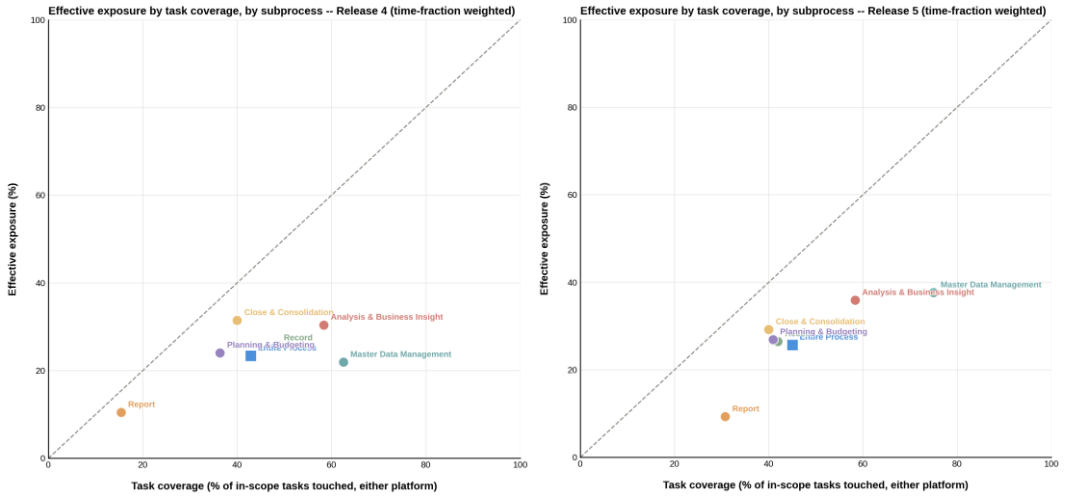


Figure 19: Effective exposure against task coverage, by subprocess, Release 4 (left) and Release 5 (right), time-fraction weighted. The dashed line marks the line of equality. Points below the line of equality are touched more widely than they are successfully performed in time-weighted terms; Table 4 reports the vertical distances. Comparing the two releases shows whether success keeps pace with expanding coverage.

Looking at effective exposure against task coverage, across all of R2R, every subprocess, and the process as a whole, sits below the 45-degree line. Claude touches roughly 45% of in-scope tasks but successfully performs only about 24 to 26% share of time needed for the process (Figure 19), representing a process-wide deficit of approximately -19.3 to -19.5 percentage points (Table 4). Decomposing this gap at the task level (Table D7 and Figure D5, Appendix D) shows that it stems almost entirely from low success rather than low weight: the touched tasks carry slightly more process time than their count share (a weight component of +1.2 and +1.6 percentage points in Releases 4 and 5), whereas the shortfall in success on those same tasks, which Claude completes in only about 52 to 53% of conversations on average, accounts for −20.7 and −21.0 percentage points. Master Data Management is the only subprocess in which low-weight tasks contribute

materially (−8.3 percentage points in Release 4). Master Data Management and Report present the most notable observation. In Master Data Management, both exposure and effective exposure actually increased from Release 4 to Release 5, thus the gap remained constant as a result of both axes rising together rather than one outpacing the other. In Report, task coverage roughly doubles, from about 15 to 31% (Figure 19), while effective exposure stays flat or slightly declines, from about 10 to 9%, thus widening the deficit from -4.9 percentage points to -21.5 percentage points (Table 4). Practitioners are expanding their experimental scope faster than success rates could follow; curiosity scaled, yet reliability did not.

Table 4: Effective exposure minus task coverage, in percentage points, by subprocess and release, time-fraction weighted.

Scope	Release 4	Release 5
Entire process	**−19.5**	**−19.3**
Master Data Management	−40.6	−37.3
Record	−22.7	−15.4
Close & Consolidation	−8.5	−10.8
Report	−4.9	−21.5
Analysis & Business Insight	−27.9	−22.4
Planning & Budgeting	−12.3	−14.0

Note: Effective exposure is the time-fraction-weighted sum of task success rates over all in-scope tasks, with untouched tasks contributing zero (Section 3.2.3); task coverage is the share of in-scope tasks that appear in the usage data on either platform. Negative values indicate that a subprocess is touched more widely than it is successfully performed. Values equal the vertical distance of each point from the line of equality in Figure 19.

Task coverage asks only "was this task touched at all", whereas observed exposure asks "was this task touched enough, in a sufficiently automated way, on tasks Claude is theoretically equipped to do, and how much of the subprocess's time do those tasks carry." Comparing effective exposure against observed exposure rather than raw task coverage is, therefore, a categorically stricter and consistent test, producing a more informative picture. With the exposure gate=ON, most subprocesses and the process as a whole sit above the 45-degree line (Figure 20, left; Table 5), demonstrating that where exposure clears the volume threshold, Claude tends to perform those tasks reasonably well, or those tasks tend to be the more central ones within the subprocess. With the gate=OFF, almost every subprocess drops back below the line (Figure 20, right; Table 5). Restoring the low-volume, low-trust tasks that the gate was built to exclude reintroduces exactly the tasks that failed to pass it in the first place, dragging the exposure-coverage relationship back down. The gate is not an arbitrary filter but is picking out

tasks where usage and reliability actually track each other and signal user behaviour. People are not deploying Claude uniformly across a subprocess's task list; they are converging, through repeated use, on the tasks it is actually capable of, and the gate is detecting that exact convergence.

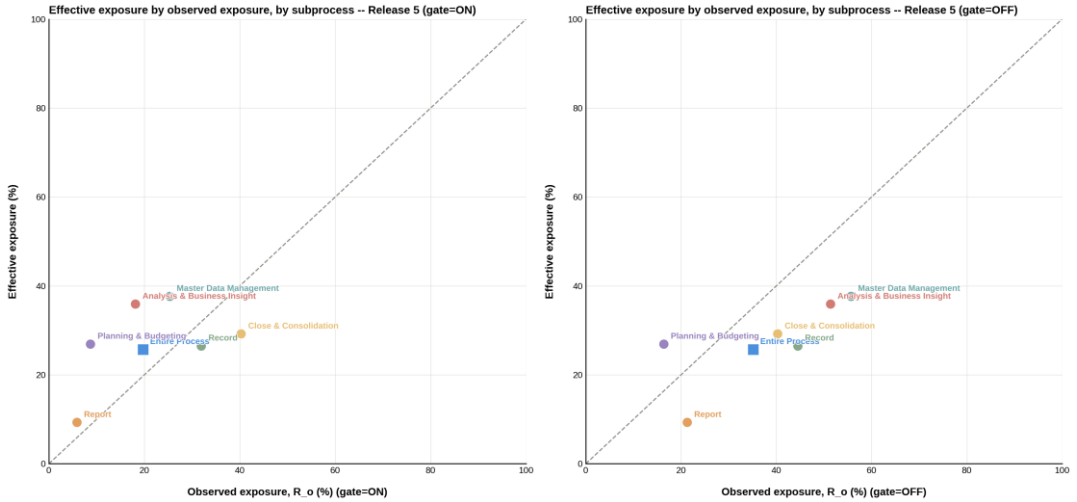


Figure 20: Effective exposure against observed exposure, by subprocess, Release 5, gate=ON (left) and gate=OFF (right), time-fraction weighted. Observed exposure is rescaled to percent (R_o × 100). Points above the line indicate that Claude's success-weighted performance exceeds its automation-weighted exposure; the contrast between gate settings shows how far this depends on excluding low-volume tasks. Table 5 reports the distances for both releases.

Table 5: Effective exposure minus observed exposure, in percentage points, by subprocess, release, and gate setting, time-fraction weighted.

	Gate = ON		Gate = OFF	
Scope	**Release 4**	**Release 5**	**Release 4**	**Release 5**
Entire process	**+7.2**	**+6.0**	**−9.2**	**−9.4**
Master Data Management	−3.6	+12.4	−28.6	−18.0
Record	+4.9	−5.4	−14.0	−18.0
Close & Consolidation	−12.8	−11.1	−12.8	−11.1
Report	+4.4	+3.5	−5.0	−11.9
Analysis & Business Insight	+12.1	+17.8	−16.1	−15.4
Planning & Budgeting	+15.4	+18.2	+4.0	+10.6

Note: Observed exposure (R_o) is multiplied by 100 so that both measures are expressed in percentage points of subprocess time. Under gate = ON, task-level exposure counts only for tasks with at least 100 work-related conversations; under gate = OFF the volume gate is removed, while the capability gate ($\beta \geq 0.5$) applies

in both settings. Positive values indicate that Claude's success-weighted performance on a subprocess exceeds its automation-weighted exposure. Release 5 values equal the distance of each point from the line of equality in Figure 20.

There are two exceptions to this finding. Record and Close & Consolidation present the most consistent exposure per Section 4.1.2, yet hold up poorly against effective exposure, remaining negative across gate settings (Table 5). Record exposure is consistent and conforming, yet is not reliably converting into delivered effective exposure. Close & Consolidation even manifests this outcome directly in its usage patterns as a steady increase in augmentation usage in Claude.ai across releases (Figure 3), perhaps as a response to the 17.71 percentage points decrease in success rates in 1P API (Table D2, Appendix D), which is the very platform driving its high exposure score. In both cases, then, the subprocess most visibly automated is simultaneously the one where that automation is least reliable.

This forces a reversal of RQ1's ranking. The subprocesses categorised as ad-hoc and bottlenecked, the ones that looked less saturated on exposure and concentration alone, come out ahead in whether such exposure is meaningful or not. On the contrary, the subprocess categorised as high-production and looked more saturated came out behind in effective exposure. Stopping at exposure and concentration without adding a success-weighted, time-fraction measure on top would have reported Record and Close & Consolidation as the frontier of R2R automation and missed that neither is yet delivering meaningful usage. At the same time, the spikey concentration pattern in subprocesses seeing otherwise better effective exposure outcomes may just reflect a narrow and self-selected sample rather than a subprocess ready to scale. The subprocesses in which exposure is most visible are not those in which it is most effective.

Effectiveness also bears back on the expertise findings of RQ2. Considering these effective exposure patterns against that of task coverage per Section 4.2.1, showing up on complex tasks might not be the same as reliably completing them; thus, a deskilling signal built on unreliable coverage is not necessarily a deskilling reality. On Claude.ai, the aggregate deskilling signal is harder to take at face value once effective exposure is factored in. In accordance with the difference in effective exposure across gate settings established above, if Claude.ai is only performing reliably on a subset of all the tasks feeding the deskilling signal, then a large share of that signal is being driven by tasks Claude merely touches instead of successfully completes. Furthermore, usage on this platform skews augmentation, and augmentation does not translate cleanly into deskilling, as partial involvement in a task is not the same as removing the worker entirely from it. Analysis & Business Insight only sharpens this observation. This subprocess shows a deskilling signal on Claude.ai in every release while showing upskilling on 1P API (Figure 12), suggesting the deskilling here is

less Claude.ai independently eroding worker skill and more users being unable to automate judgment-heavy work on 1P API and persistently falling back to Claude.ai's augmented form for the harder end instead. Similarly, while Report and Planning & Budgeting also show deskilling, given their low exposure profile, this reads less as a meaningful erosion of worker skill and more as a thin, hard-to-generalise slice of the task space.

Subprocesses more prone to 1P API reinforces the ambiguous deskilling reality. Record shows a consistent, sizable deskilling gap across every release (Figure 12). This subprocess' covered tasks are meaningfully more complex than its own average, yet, Record's success rates are declining sharply by Release 5 in this very platform (Table D2, Appendix D); at the same time, this subprocess holds a consistent negative effective exposure gap across gate settings shown above. Even where substitution is seemingly present, it isn't necessarily meaningful. This is the mirror image of what's happening in Close & Consolidation, which shows the largest and most stable upskilling gap in the process (Figure 12). Read alongside Record and its effective exposure profile, these are not separate phenomena but the same one suggesting that the more complex, judgment-heavy bookkeeping work is not being automated effectively. The fact that 1P API as a whole only recently bounced up toward upskilling, and that Record itself moves closer to upskilling region by R5, reinforces this interpretation, where usage is correcting back toward the tasks the platform can actually handle.

4.3.2 Determinants of Success and Effective Exposure

Having established that exposure converts only partially into performance, the second part of RQ3 asks why. What explains success rates and effective exposure measures? An intuitive explanation leans on a self-reinforcement effect: because this reflects actual usage data, users repeatedly bring Claude tasks it already succeeds at, pulling usage toward what it can already do well. The bimodal structure established along the expertise gradient in Section 4.2.2 demonstrates that expertise alone does not explain exposure, a pattern restated here in how task expertise fails to predict success rates on either platform across releases (Figure 21), which further aligns with AI having been able to do almost all of the R2R tasks (Section 4.1.2). Since usage patterns diverge between the two platforms (Section 4.1.1), and the descriptive statistics of the primitives differ correspondingly, the following section examines the reasons behind this discounted effective exposure separately for each platform.

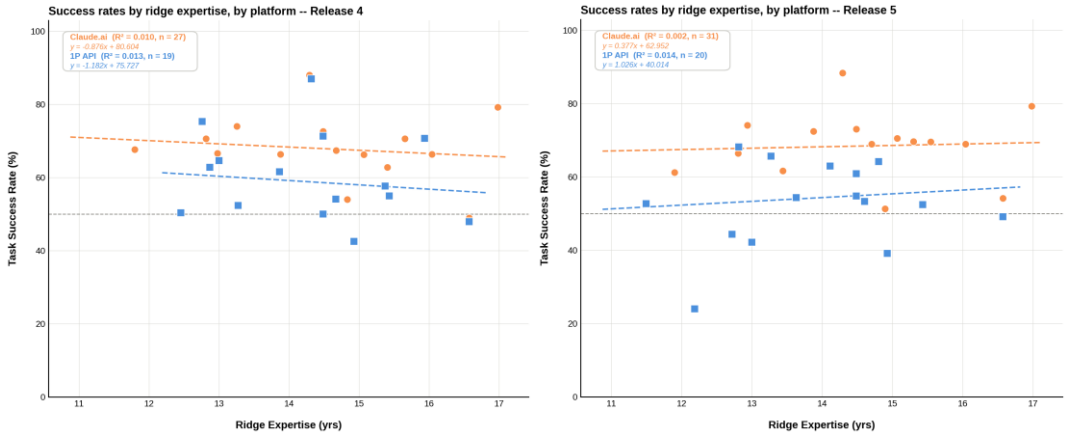


Figure 21: Task success rate against ridge-estimated task expertise, by platform, Release 4 (left) and Release 5 (right), with per-platform OLS fits. Each marker is a task with a success reading on the platform (n in the legend); dashed lines are per-platform OLS fits and the dotted line marks 50% success. Near-zero slopes and R^2 on both platforms and in both releases indicate that required expertise alone does not explain whether Claude completes a task, motivating the search for other determinants in the remainder of the section.

Over time, users have used *Claude.ai* for more difficult tasks, as evident in the marginal increase Human education years required to understand the prompt, as well as a 7.61 percentage point increase in the Human can't do alone rate (Table 6). Another clear evidence of rising difficulty pertains to the jump from 187.5 to 214.3 minutes (+26.7 minutes) in Human-only time between releases (Table 6), all pointing to the fact that users are handing Claude.ai more time-intensive problems than before. This primitive is also the mediator connecting expertise to success. Task expertise predicts human-only time on Claude.ai with an R^2=0.163 (slope=+0.458 hours per year of required education) (Figure 22, left), and human-only time in turn predicts success (R^2=0.293) (Figure 22, right) far better than task expertise is capable of directly (R^2=0.002) (Figure 21). At the same time, these more complex tasks are also the ones granted higher AI autonomy (R^2=0.149, slope of 0.078) (Figure 23, left). As required expertise rises, users are delegating more, not less, judgment to Claude at exactly where it is less reliable. According to Section 4.1.1, this platform's dominant usage is also the judgment-heavy, augmentation-leaning work, yet, autonomy is not substituting for human effort on harder tasks here, but accumulating next to it without the corresponding gain in success. Taken on its own, however, autonomy is a weak predictor of success on Claude.ai. When regressed directly against task success (Figure 23), AI autonomy explains almost none of the variance (R^2=0.050, slope -9.045), a far weaker relationship than the 1P API shows over the same range (R^2=0.299, slope -24.144).

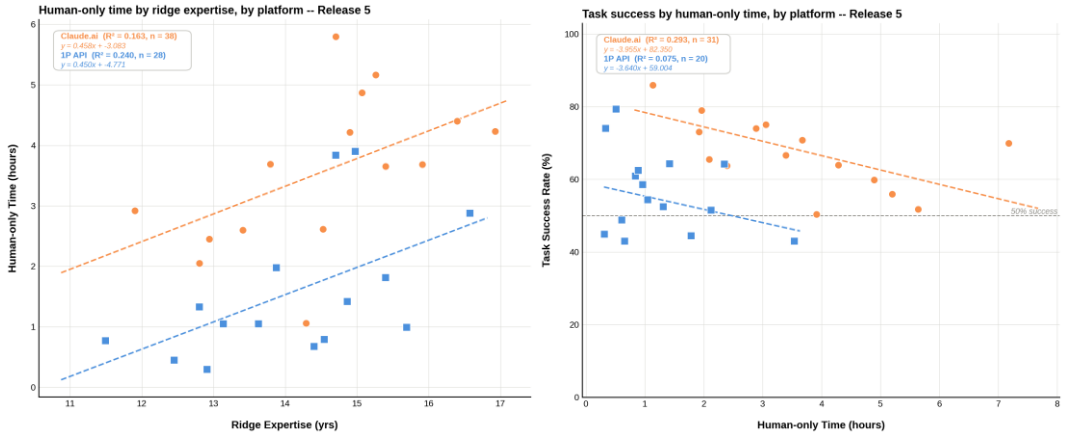


Figure 22: Human-only completion time against ridge-estimated task expertise (left) and task success rate against human-only time (right), by platform, Release 5, with per-platform OLS fits. Human-only time is the AEI estimate of the hours a competent professional would need without AI. Read left to right, the panels trace the path from expertise to task duration and from duration to success, which the regressions in Table 7 test with controls.

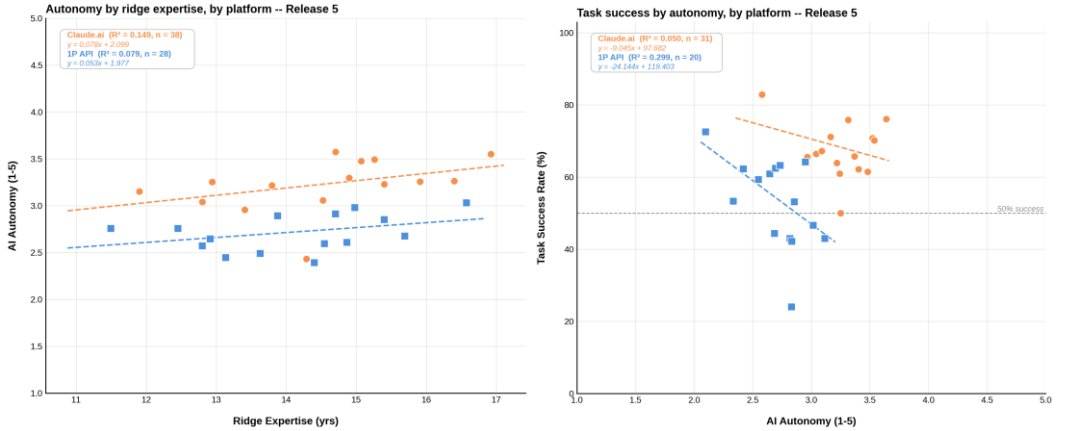


Figure 23: AI autonomy against ridge-estimated task expertise (left) and task success rate against AI autonomy (right), by platform, Release 5, with per-platform OLS fits. AI autonomy is the AEI classifier's 1 to 5 rating of the decision-making delegated to Claude. The left panel shows how delegation varies with task expertise, the right panel whether delegated tasks succeed; the much steeper autonomy–success slope on the 1P API anticipates the multivariate results in Table 8.

Table 6: Average task primitives for the entire R2R process, by platform, Release 4 versus Release 5.

	Claude.ai			1P API		
Primitive	**R4**	**R5**	**Δ**	**R4**	**R5**	**Δ**
Numeric facets (mean)						
Human education (years)	13.17	13.34	+0.17	13.56	13.77	+0.21
AI education (years)	12.94	13.25	+0.31	12.79	13.07	+0.28
AI autonomy (1 to 5)	3.13	3.23	+0.10	2.79	2.72	−0.07
Human-only time (minutes)	187.52	214.27	+26.74	79.35	94.22	+14.87
Human-with-AI time (minutes)	15.67	15.52	−0.15	3.22	3.35	+0.13
Task outcomes (%)						
Human cannot complete alone	9.7	17.3	+7.61	7.7	7.5	−0.19
Task success rate	67.8	68.4	+0.60	58.8	54.3	−4.52
Use case (%)						
Work	62.6	71.8	+9.20	87.3	88.8	+1.42
Personal	15.0	18.2	+3.22	6.9	5.9	−1.04
Coursework	17.1	6.8	−10.30	2.5	2.1	−0.39
Not classified	5.2	3.1	−2.12	3.2	3.2	+0.01

Note: Unweighted means across in-scope O*NET tasks with a reading on the respective platform and release. Δ is Release 5 minus Release 4, in the unit of the row (percentage points for shares). Primitives are defined in Section 3.1.4 and Appendix B.

Indeed, a stepwise regression on Claude.ai task success confirms that the bivariate picture between task success and task expertise is incomplete (Table 7; correlations among the regressors in Table D3, Appendix D). Expertise alone is a weak predictor (-0.217, R^2=0.001), and adding autonomy contributes almost nothing on its own (-6.943; R^2=0.022). The mediating addition is human-only time, increasing R^2 to 0.349 while turning both expertise's (2.082*) and flipping autonomy's coefficient (10.530*) to marginal significance. This reversal for autonomy, however, does not survive a further control for subprocess fixed effects. Once subprocess identity is added, both expertise and autonomy coefficients fall back to non-significance, and human-only time remains the only variable that is robust across every specification. This also holds with standard errors clustered by

task, with a release indicator, and within each release separately, whereas the marginal expertise and autonomy coefficients of specification (3) do not survive clustering (Tables D8 and D10, Appendix D). Nevertheless, the subprocess fixed effects impacts remain insignificant ($F(5,49)=1.81$, $p=0.128$), suggesting this pattern is directionally supportive and compatible with findings in Section 4.2.1. In brief, the only interpretation that survives across every specification is that human-only time is the mechanism actually driving success down. The real story on Claude.ai, then, isn't that harder tasks fail more, but one in which as tasks get longer and meatier, Claude is being handed a larger, less-supervised share of the judgment, yet success declines alongside that shift. From Section 4.1.1, users switch platforms for more complex tasks, but usage patterns show that even this strategy is not paying off. This mirrors Kwa et al. (2025) finding that human task-completion time, independent of any measure of cognitive difficulty, is strongly predictive of AI success.

Table 7: Stepwise OLS regressions of Claude.ai task success rate on task expertise, AI autonomy, human-only time, and subprocess fixed effects, Releases 4 and 5 pooled.

	(1) Expertise only	(2) + Autonomy	(3) + Human-only time	(4) + Subprocess FE
Ridge expertise (years)	−0.217	0.428	2.082*	−0.256
	(1.132)	(1.273)	(1.095)	(1.409)
AI autonomy (1 to 5)		−6.943	10.530*	6.289
		(6.316)	(6.188)	(6.453)
Human-only time (hours)			−5.741***	−5.815***
			(1.102)	(1.263)
Constant	71.293***	84.087***	23.876	70.105**
	(16.540)	(20.200)	(20.252)	(28.091)
Subprocess fixed effects	No	No	No	Yes
F-test of fixed effects				F(5, 49) = 1.81
				[p = 0.128]
Observations	58	58	58	58
R^2	0.001	0.022	0.349	0.451

Note: Dependent variable: task success rate (%) on Claude.ai. Unit of observation: task × release, Releases 4 and 5 pooled. Standard errors in parentheses. The F-test compares specification (4) with specification (3). Human-only time is the only regressor that is significant in every specification in which it enters; the expertise and autonomy coefficients in (3) lose significance once subprocess fixed effects are added. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are conventional

(homoskedastic) OLS standard errors; no release indicator is included. Task-clustered standard errors, with and without a release indicator, are reported in Table D8 and estimates by release in Table D10 (Appendix D).

Why does this same mechanism fail to appear on the 1P API? The slopes of success on time are nearly identical (Claude.ai -3.955 versus 1P API -3.640) (Figure 22, right), so a longer task fails at roughly the same rate on either platform. However, human only time's relationship with task success is much weaker on 1P API (R^2=0.075) than Claude.ai (R^2=0.293), likely because tasks brought to the API are shorter and vary less in absolute duration (mean 1.6 against 3.6 hours, standard deviation 1.3 against 1.6 hours; Table 6 and Figure D3, Appendix D). Autonomy, even when weak in Claude.ai, only reinforces the same story. On 1P API, it explains a lot more variance in success rates (R^2=0.299) with a much steeper slope (-24.144) (Figure 23, right), while also holding much weaker with task expertise (R^2=0.079). These suggest the API's flat education-success line is not evidence of indifference to difficulty, but the symptom of a shorter, more codifiable, automation-oriented task mix in which the same mediating mechanism in Claude.ai simply never gets engaged. This raises the question of what explains complexity registers so flatly on the API in the first place. Comparing success rates between two platforms only deepens the puzzle. Across releases, task success on Claude.ai rose by 0.60 percentage points (Table 6) even as expertise requirements climbed, while 1P API success fell 4.52 percentage points (Table 6).

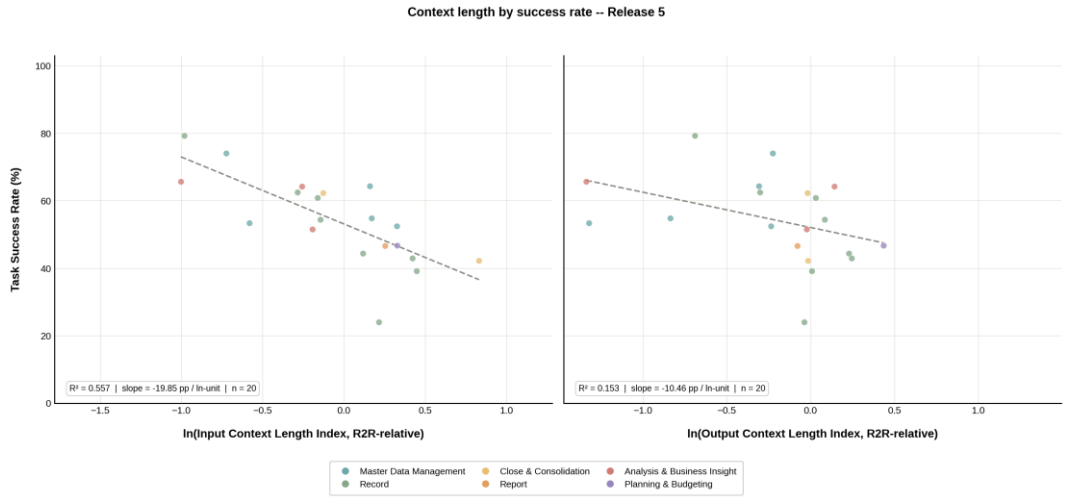


Figure 24: Task success rate against log-relative input and output context-length indices, by subprocess, 1P API, Release 5, with OLS fits. Context-length indices are the AEI prompt-token (input) and completion-token (output) indices re-indexed to the R2R task set and log-transformed, so zero marks the R2R average; they are available for the 1P API only. The steeper and tighter input relationship identifies context ingestion as the API's binding constraint, which Table 8 tests with controls.

1P API fails because it is context dependent, and this context is independent of complexity. By Release 5, input context length accounts for 55.7% of the variance in task success (Figure 24, left). For every 1-unit increase in log-relative input context, task success drops by 19.85 percentage points, and this effect has even intensified from Release 4 with an R^2=0.23 and a slope of -10.53 percentage point (Figure D4, Appendix D). On the other hand, the penalty for generating tokens is only half as severe as the penalty for ingesting input context at 10.46 percentage points for each unit increase in log-relative output while explaining only 15.3% of the variance (Figure 24, right). However, this relationship does not hold up once input context is controlled for (Table 8), suggesting the apparent output penalty largely reflects the fact that longer outputs tend to accompany longer inputs (see Figure D2, Appendix D, for the input index plotted against the output index). The primary operational failure point in API usage is thus context ingestion instead of output generation. Context length is also only weakly related to education (R^2=0.001) (Figure 25), and this weak relationship is also evident in how tasks of all complexity can vary in how much context they require. Tasks of all complexity register in the over-indexing region, with low-education ones in Record regularly registering high input indices, reflecting their requirement for reference material across vendor master files, account mappings, validation tables that has nothing to do with the cognitive difficulty of the task itself.

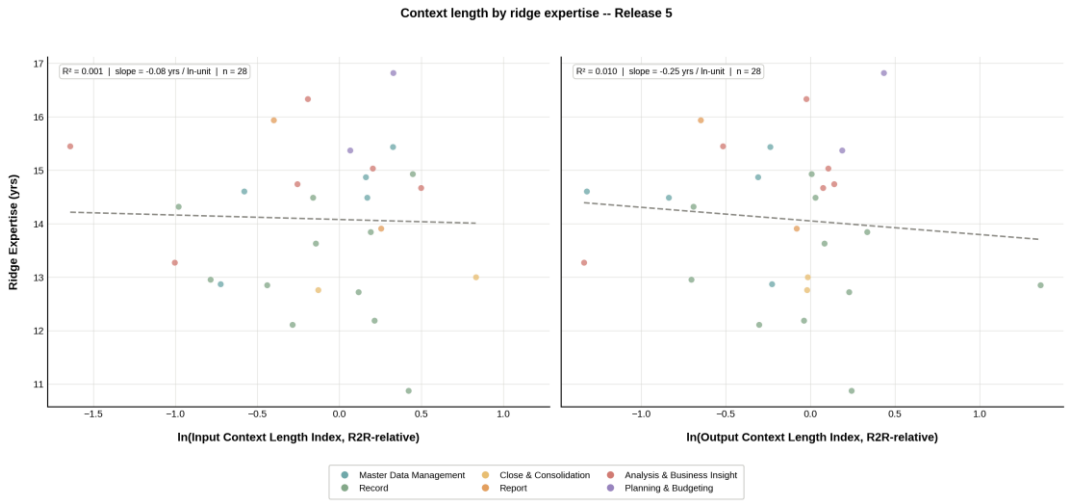


Figure 25: Ridge-estimated task expertise against log-relative input and output context-length indices, 1P API, Release 5. As Figure 24, with ridge-estimated expertise on the y-axis. The flat fits show that the context a task demands is unrelated to the expertise it requires, so the input-context penalty in Figure 24 is not a proxy for task difficulty.

This reframes what the flat education-success line on the API means, as success on 1P API depends on how much context a task demands to execute, a property that varies by subprocess and by workflow far more than by

expertise. The API's problem looks more mechanical than that on Claude.ai, where Claude is constrained by attention mechanisms decaying over long inputs rather than the cognitive difficulty of the accounting logic itself. This is consistent with Liu et al. (2024) finding that model performance on long-context tasks degrades due to a position-driven effect independent of the task's underlying reasoning difficulty.

Table 8: OLS regressions of 1P API task success rate on input and output context-length indices, AI autonomy, and subprocess fixed effects, Releases 4 and 5 pooled.

	(1) Input index	(2) + Output index	(3) + Autonomy	(4) + Subprocess FE
ln(input context-length index)	−14.198***	−15.471***	−12.261***	−16.406***
	(3.239)	(3.563)	(3.890)	(3.993)
ln(output context-length index)		3.964	5.601	9.367*
		(4.552)	(4.509)	(5.497)
AI autonomy (1 to 5)			−13.767*	−16.939**
			(7.648)	(8.043)
Constant	55.670***	56.215***	94.451***	107.734***
	(1.755)	(1.869)	(21.319)	(23.387)
Subprocess fixed effects	No	No	No	Yes
F-test of fixed effects				F(5, 30) = 1.98
				[p = 0.110]
Observations	39	39	39	39
R^2	0.342	0.355	0.410	0.557

Note: Dependent variable: task success rate (%) on the 1P API. Unit of observation: task × release, Releases 4 and 5 pooled. Context-length indices are the AEI prompt-token (input) and completion-token (output) indices re-indexed to the R2R task set and log-transformed. Standard errors in parentheses. The F-test compares specification (4) with specification (3). The input-index coefficient is stable in size and significance across all specifications. * p < 0.10, ** p < 0.05, *** p < 0.01. Standard errors are conventional (homoskedastic) OLS standard errors; no release indicator is included (task-clustered standard errors in Table D8). The output-index coefficient is marginally significant (p < 0.10) only in specification (4).

A multivariate check confirms this asymmetry directly. Regressing task success on input index, output index, and AI autonomy together (Table 8), ln(input index) remains large and highly significant (-14.2 to -16.4 percentage points, p<0.01) whether or not output, autonomy, or subprocess fixed effects are included, and the coefficient does not shrink under these

controls. That the subprocess fixed-effects block is only marginally significant ($F(5,30)=1.98$, $p=0.110$) indicates the input effect is a general property of context ingestion itself, not an artifact of certain high-context subprocesses. AI autonomy enters significantly negative in the pooled specification (-13.8* to -16.9** percentage points) even after controlling for both context measures, but this relationship does not reproduce within either release taken separately (Table D1, Appendix D), which is more consistent with a between-release baseline shift in autonomy and success levels. The autonomy coefficient also loses significance once standard errors are clustered by task, whereas the input-index coefficient remains significant at the 1% level (Table D8, Appendix D).

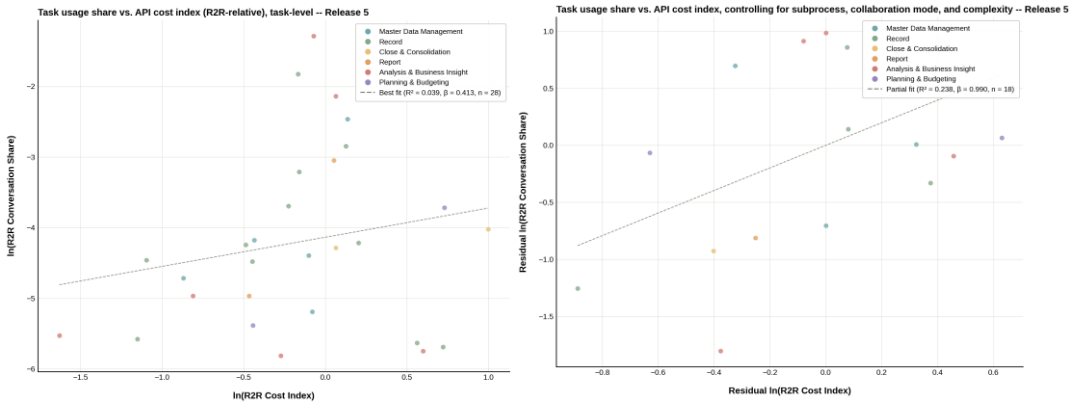


Figure 26: Task-level usage share against the R2R-relative API cost index, Release 5: raw relationship (left) and the partial relationship after controlling for subprocess, collaboration mode, and task complexity (ridge-estimated expertise) (right). Left: log share of R2R API conversations against the log R2R-relative cost index. Right: the same relationship after partialling out subprocess, collaboration mode, and task complexity from both variables (n = 18). The weak positive partial slope indicates that API usage is not steered towards cheaper tasks within the observed range.

Enterprises are not price-sensitive when it comes to API deployment either. Within the observed range, the raw cost-usage relationship is weak ($R^2=0.039$, $\beta=0.413$) (Figure 26, left), and even after controlling for subprocess, collaboration mode, and task complexity, the partial relationship strengthens directionally ($R^2=0.238$, $\beta=0.990$) but remains statistically weak (Figure 26). Still, the direction is suggestive. Since cost scales directly with the amount of input driving the success penalty, higher usage of costlier tasks is plausibly higher usage of exactly the tasks where the input-context bottleneck is most severe. Enterprises are, to an extent, paying a premium to deploy Claude precisely where it is least reliable, and doing so at increasing scale rather than pulling back. If this pattern persists as adoption widens, the tasks firms are most economically motivated to automate may also be the ones most structurally resistant to it, representing a mismatch between where economic value is concentrated and where the model can actually deliver. This is particularly pronounced as success rates of Record, the

platform dominant subprocess, are decreasing from Release 4 to Release 5 (Table D2, Appendix D).

Both platforms tell the same higher-order story, where expertise is not the variable driving success, but something more platform-specific is. On Claude.ai, it is human-only time; on 1P API, it is input context length. In parallel, success is governed by the volume of operational burden the task imposes on its platform's particular bottleneck instead of the cognitive difficulty of the underlying accounting logic. Claude.ai processes them through sustained human-AI interaction, so its bottleneck is human time; the API processes them through a context window, so its bottleneck is input token volume. This is not incidental to which subprocesses each platform attracts but the same property working in both directions. The judgment-heavy, iterative subprocesses that users route to Claude.ai are, by their nature, sustained-interaction tasks, and that same property is what generates the high human-only time eroding success as tasks get harder. Symmetrically, the codifiable, transactional subprocesses concentrated on 1P API, and even some usage of Analysis & Business Insight here, are exactly the tasks that pull in large reference and validation material, which is what drives the input-context load degrading success there. The platform's dominant use case and its dominant failure mode are, paradoxically, the same thing, where the very codifiability that concentrates this subprocess on 1P API is inseparable from the context load that degrades it. In both cases, the property that makes a subprocess a good fit for its platform and drives its usage pattern is also the very property driving its failure on that platform.

Figures 19 to 26 and Tables 4 to 8 answer the third research question. Exposure in R2R is only partly effective: Claude successfully performs about a quarter of the time the process requires, and the subprocesses whose exposure is most visible convert least reliably into performance, so the practical consequences of the two expertise regimes identified under RQ2 are substantially blunted by success gaps. What governs that effectiveness is not the expertise level of a task but the operational burden it imposes on its platform—human-only time on Claude.ai and input context on the 1P API.

5 Discussion

5.1 The Layer between Perception and Outcome

Existing research on GenAI in accounting sits at two ends of a chain. At one end is perception, what practitioners report about the technology and how they use it; at the other is outcome–productivity, time allocation, or labour demand. Choi and Xie (2026) reach both ends, pairing survey perception with productivity outcomes from a platform log, and Böke et al. (2024) and Awyong et al. (2025) reach the outcome end through investment and job-posting data. What none of them observe is the layer in between, where and how GenAI is actually being used, a layer that is more grounded than perception, since it records what people do rather than what they perceive, and sets the grounds for outcome. This study focuses on that middle layer on a different dataset, at a finer granularity, and in a process-based frame that tracks usage across the whole of R2R cycle rather than the record-keeping slice a platform can log, and it builds from that usage a multi-dimensional exposure profile for each subprocess in terms of observed exposure, concentration, expertise composition, and effective exposure. Doshi and Moore (2026) argue that understanding GenAI's heterogeneous effects requires framing it in the context of performing tasks, and the exploratory character of this study across five releases of the Anthropic Economic Index complements that framing empirically.

The three research questions traced a single line from location, to composition, to effectiveness. RQ1 found exposure concentrated on a narrow, structurally consistent core of tasks rather than spread across the process, a series of drill points rather than a broad diffusion across the process. RQ2 found that this core does not sit at one expertise level but splits into two regimes, a low-expertise conforming regime in Record where a bounded set of transactional tasks is exposed together, and a high-expertise anchoring regime in Master Data Management and Analysis & Business Insight where a few tasks carry the exposure inside subprocesses otherwise untouched. RQ3 found that, once success is factored in, the ranking reverses, with the conforming regime that looked most automated converting least reliably into effective automation, and success is governed by the operational burden a task places on its channel rather than by its expertise. None of these answers stands on its own. Concentration means little without knowing whether it sits on senior judgment or junior data entry; neither means much without knowing whether the exposed tasks are actually being completed.

Capability is not exposure, and exposure is not performance. The efficiency objective of the finance function (Bedford et al., 2025) is being automated but not reliably; the effectiveness objective (Bedford et al., 2025) is being augmented but not substituted. GenAI has entered R2R at both ends and stalled at both, for different reasons. Such ineffective exposure, together

with the collaboration profile that accompanies it, complicates the reallocation that labour-economics frameworks would predict. In brief, the R2R process is fully exposed to GenAI in capability, narrowly exposed in use, and effectively exposed for about a quarter of its time, with the exposure that does exist sitting either in an augmented mode that does not substitute or in an automated mode that does not yet succeed. Indeed, GenAI represents a different technology regime from the automation those frameworks were built to study, in which a task exposed to capital is a task taken from labour. The remainder of this chapter develops that reading in four steps. Section 5.2 sets out what the exposure profile predicts for the redistribution of work along the process. Section 5.3 shows why that prediction does not yet hold, by tracing the unfolding dynamics of real-world usage. Section 5.4 questions how, for a technology that admits augmentation, the task and the machine are not the whole allocation, and turns to the human half of the interaction. Section 5.5 returns to the task-level and process-based framing employed by the study to assess whether it served the analysis.

5.2 Distribution of Labour and Capital along the Record-to-Report Process

The exposure and concentration patterns of RQ1 and the expertise composition of RQ2 produce a clear theoretical reading of where R2R work would redistribute. Record, the lowest-complexity subprocess in the process, anchors the low end, and its concentration conforms to a bounded set of transactional tasks. Following Autor and Thompson (2025) logic, removing a coherent low-expertise chunk of an occupation's task list mechanically raises the average expertise of what remains, which signals substitution, and the logic applies here because Record's tasks were low-complexity and homogeneous enough in exposure to be displaced together. Awyong et al. (2025) finding that firm digitalisation reduces demand for financial clerks while leaving demand for financial specialists unchanged aligns with this pattern: the routine, transactional end of the profession is where substitution is predicted to land first.

Master Data Management and Analysis & Business Insight sit at the opposite end of the complexity spectrum, and their concentration is anchored rather than conforming–a small number of high-dispersion tasks carrying the exposure inside subprocesses that otherwise remain largely untouched. This is selective delegation rather than whole substitution. These high-complexity subprocesses do not offer a large homogeneous block of tasks that could be uniformly automated; what they offer instead are isolated points of sufficiently bounded judgment for Claude to be trusted, or tested, with, and these spike the concentration measures without shifting the average expertise of remaining work the way Record's pattern does. Since most of each subprocess's task mix is left intact, workers retain abundant

unaffected tasks to reallocate to, which is precisely the reallocation space Hampole et al. (2025) identify as the counterweight to displacement. Hampole et al. (2025) find that while substitution effects are strongest for higher-paid occupations, these same jobs also benefit from the strongest reallocative gains because their AI exposure is highly concentrated. At the subprocess level the correspondence is partial and process-specific. Master Data Management and Analysis & Business Insight replicate the high-value, high-concentration pattern the economy-wide framework predicts; Planning & Budgeting's combination of low exposure and low concentration means the reallocation mechanism never activates there in the first place; and Record, low in value yet high in concentration, stands as a counterexample to the framing altogether.

The two regimes describe a redistribution of labour and capital across the process that the process framing makes visible in a way an occupational frame could not. The R2R cycle must be completed regardless of which combination of human and machine executes it, so the subprocesses where GenAI leaves no structural imprint, Planning & Budgeting and Report, and the more complex tasks within Close & Consolidation that either do not appear or scale back, mark where the human share of the process would remain. The familiar promise that automating routine work frees time for strategic work would then manifest in two forms. Partially as migration within a subprocess toward its harder tasks, visible already in how automation of Record scaled back to its less complex tasks. Fully as migration across subprocesses toward higher-expertise ones, which is what Analysis & Business Insight's growing share of Claude.ai usage records. Choi and Xie (2026) observe both movements directly. Their platform log records the within-workflow shift, as accountants' attention moves from manual categorisation toward more complex entries such as accruals and closing-book analysis, while their panel survey records the across-task shift, from data entry toward business communication and quality assurance. Awyong et al. (2025) observe the across-occupation version in the falling demand for clerks. This study does not observe labour outcomes, yet reading the exposure profile through the theoretical predictions of Hampole et al. (2025) and Autor and Thompson (2025) arrives at the same direction of movement that Awyong et al. (2025) and Choi and Xie (2026) report. The exposure profile is informative about redistribution before any redistribution has been measured, thus the mechanism the theory proposes, interpreted through actual usage itself, can be considered a plausible account of what the outcome studies see. Nevertheless, considering how effective such exposures are (RQ3), how labour-economics mechanisms are actually activating is the subject of interrogation in the next section.

5.3 Self-Selection, Effectiveness, and the Latency of Those Channels

When almost every task is capable, why is there still heterogeneity in exposure? Actual usage data captures three things capability annotation cannot: which tasks professionals choose to bring to Claude, which platform they bring them to, and how those choices unfold across releases. It is this pattern of self-selection, and how it unfolds across releases, that produces the heterogeneity on which the prediction of Section 5.2 rests. Following these self-selection patterns surfaces where that prediction holds and where it does not, and therefore how far the mechanisms behind it are actually activating.

The Anthropic series' own explanation for why enterprise usage concentrates where it does is that institutional inertia and fixed costs of adoption steer early enterprise use toward specialised tasks where deployment is easy, capabilities are dependable, and the returns are high (Appel et al., 2025). That explanation presupposes that capability discriminates between tasks. However, when every task is capable, and there still exists a gap between capability and observed exposure, real-world usage seems to follow the user rather than the task. The platform divide is the first trace of that selection. 1P API carries automation in 77 to 87% of conversations; Claude.ai is augmentation-led at every release; and the gap is stable rather than narrowing. The selection operates on both platforms but in different ways, with API customers selecting for tasks amenable to automation and Claude.ai users selecting for tasks that benefit from iteration. Accordingly, Raisch and Krakowski (2021) automation-augmentation distinction becomes a property of the task embedded in the channel, with users consistently bringing tasks they deem suitable to the respective platform.

As users select, they also correct, and the unfolding runs in opposite directions on the two platforms. On 1P API, the progressive shift toward upskilling–usage correcting back toward tasks the platform can actually handle–is the aggregate signature of within-process reallocation. Enterprises are not persisting uniformly at the tasks where API performance is degrading; at the process level, the mix is adjusting. Record moving closer to upskilling territory by Release 5 is the specific subprocess-level expression of this. As success rates decline on its cognitively simple but context-heavy tasks, usage is being drawn back toward the simpler end of even Record's own task distribution, and, as Section 4.3.1 showed, the substitution that appeared present in Record is not thereby meaningful. This is reallocation operating within a subprocess in real time, driven by failure feedback. On Claude.ai, the response is the reverse. Harder and more time-intensive tasks are increasingly handed to Claude with higher autonomy, yet success declines alongside that shift, which means that even within the anchoring

subprocesses there is upward pressure along the cognitive spectrum that exposure and concentration, measuring the where on the surface rather than the where underneath, do not capture. The anchoring pattern in RQ2 was read as users finding isolated points of sufficiently bounded judgment in more complex subprocesses; Section 4.3.2 shows that the boundary of what counts as sufficiently bounded is being tested, and sometimes exceeded, with users pushing further into judgment-heavy territory than the current reliability warrants. Analysis & Business Insight demonstrates a combination of both, presenting deskilling on Claude.ai and upskilling on the API in the same releases. This pattern reads less as Claude.ai eroding worker skill than users unable to automate judgment-heavy work on the API, thus persistently falling back to Claude.ai's augmented form for the harder end. Compared to benchmarking exercises, such as Kwa et al. (2025) for coding tasks or Benchek et al. (2026) for accounting work, that assign a fixed set of tasks to AI models, real-world usage provides a view into not only the model's capability but also the user's perception of what works and what does not.

This is a meaningful nuance relative to Autor and Thompson (2025). Their framework treats expertise as a relatively stable property of a set of tasks that predicts whether automation raises or lowers the average skill level of remaining work; the longitudinal data show the property to be dynamic, with users constantly adjusting toward what works and what does not. Consequently, the mechanisms of Hampole et al. (2025) and Autor and Thompson (2025) are directionally correct yet only partially activated, because both require that automation actually removes tasks from the occupation's bundle. The low-complexity conforming regime looks like substitution on exposure measures but behaves like failed substitution on performance measures. Record's tasks were homogeneous and low-complexity enough to be exposed together, which is the precondition Autor and Thompson (2025) logic needs; however, their success rates are declining and usage is being drawn back toward the simpler end of even Record's own distribution. The reallocation that Autor and Thompson (2025) anticipate is still occurring, but it is arrived at through degradation rather than displacement. Rather than workers reallocating toward higher-expertise residual tasks because low-expertise work has been successfully automated away, the reallocation here resembles a dynamic correction driven by performance failure. Whether the average expertise of remaining work rises, as the framework predicts, depends on the level at which one looks. Within Record it does, since the harder Record tasks are what usage abandons and users are moving toward the more complex end of the subprocess; across subprocesses it may not, since nothing has been reliably absorbed and so nothing has been freed to move. The anchoring regime of Master Data Management and Analysis & Business Insight preserves the reallocation space Hampole et al. (2025) identify, because most of each subprocess's task

mix is untouched, but the boundary defining which tasks are bounded enough to delegate is itself moving, and moving faster than success. The expertise level at which concentration appears is drifting upward; the level at which it is operationally meaningful is not keeping pace. In both regimes, the condition on which the theory rests is unmet, and the reallocation channel each predicts remains latent rather than active.

Even automation usage might be overstated. This data cannot observe what users do with the output, whether they accept it as it is, or continue iterating on it outside of even an automated conversation. Anthropic's interview study found 65% of participants describing AI's primary role as augmentative against 35% automative (Handa, Stern, et al., 2025), a disconnect from usage data that plausibly reflects users refining outputs after the chat ends, using different providers for different tasks, or perceiving their use as more collaborative than their conversations look. Against the observed exposure construct, which was built in a way that aligns with the labour-economics theories by treating automation as the channel through which exposure produces consequences, that augmentation still surfaces despite the automation-leaning exposure construct throughout the multi-dimensional profile, in the platform divide, in the correction on Claude.ai, and in the tier of subprocesses that the construct understates by design, is evidence that the automation regime it was built to measure is not the only regime present in GenAI usage in the R2R cycle.

Collaboration mode, then, is not incidental to the exposure profile but a signal in its own right about the form any reallocation takes. Automated use removes human time from a task; augmented use redistributes human effort within it. Autor et al. (2003) and Acemoglu and Autor (2011) allocate tasks between labour and capital by comparative advantage, and the frameworks reviewed in Chapter 2 extend that allocation to AI without altering its binary form. Task complementarity, according to Acemoglu (2025), in the same tradition, is modelled between tasks, where the gain on a task the worker is freed to concentrate on is contingent on the additional time reallocated to it. However, the data show a further allocation within capital, between an automated and an augmented channel, and show the choice to be sticky across releases, which opens a view into complementarity within a task rather than only between tasks. Whether a task is used in an automated or an augmented manner is a dimension the task-based framework lacks, and it lacks it because the technologies those frameworks were built on did not have one, since a machine either performed a task or did not. This is a nuance that neither mixed methods present in Choi and Xie (2026) nor the posting-based measures of the labour economics literature carry, since both record the quantity of labour on a task and not the manner of its collaboration with capital. Consequently, where the worker is also assisted on that task, in other

word, augmented, the size of the gain depends in part on the person directing that assistance. The human is, accordingly, the other half to be explained.

5.4 The Human Half

Wood (1986) held that task completion should be assessed with the task separated from its performer, and this thesis, built entirely on the task side, makes the case by omission. Task expertise does not predict success on either platform once human-only time or input context is controlled for; the variables that do are task- or machine-specific. This thesis is silent on the users, and conforming to labour-economics theories in that respect means missing a context of particular importance in human-AI interaction, especially when every study able to observe the users finds that they matter. Massenkoff et al. (2026) find high-tenure users with a 10% higher success rate, iterating more, delegating less directively, taking on tasks requiring more education, and spreading usage more widely; Hallman et al. (2026) find senior employees' GenAI use more sophisticated; Hitzig et al. (2026) find 19% of sessions from novice users abandoned against 5 to 7% for everyone else; Choi and Xie (2026) find experienced accountants using the system more strategically and gaining more from it. Expertise as a property of the task fails to explain success here, while expertise as a property of the person explains it in those other streams of literature, and that asymmetry is the missing piece.

Why the performer matters, and why it matters particularly where use is augmented, follows from what that expertise consists of. Handa, Stern, et al. (2025) found workers with fifteen or more years of experience judging AI able to do roughly ten percentage points less of their work than those in their first year, and attributes the difference to accumulated tacit and context-specific expertise the model cannot mimic. Vendraminelli et al. (2025) locate the same boundary between conceptual tasks, which draw on explicit knowledge and general reasoning that GenAI supplies well, and execution tasks, which draw on tacit knowledge and context-specific judgment acquired through experience. It is this tacit knowledge and human-specific expertise that the augmented channel draws on when the machine fails. Augmentation, moreover, is what separates GenAI from earlier waves of automation, so such level of expertise itself helps the technology diffuse. The Index's third release records Korinek's observation that AI may accelerate its own diffusion by playing the role skilled workers historically played in learning to wield new technologies in novel settings (Goldin & Katz, 2008; Nelson & Phelps, 1966), and notes that whether the growth in directive use reflects improving capability or learning-by-doing carries different labour implications–the former expanding what can be automated outright and the latter rewarding the workers most able to adapt. The evidence here favours the latter, since capability is already near-universal while usage is what

moves. Diffusion of this kind runs through the pairing of the model's execution with the human's capacity to direct and verify it. Kremer (1993) O-ring now sits in that pairing, and the weak link becomes a capable model directed by a worker without the tacit knowledge to steer or check it.

The same expertise governs how far a worker can be reallocated. Vendraminelli et al. (2025) describe GenAI as a knowledge intermediary that supplies expertise across what would otherwise be rigid occupational lines, and Lave and Wenger (1991) found that this works better within a domain than across one, since shared context and schemas support near transfer. R2R subprocesses are adjacent in exactly that sense, and the migration of usage toward harder tasks within a subprocess and toward higher-value subprocesses across the process, in accordance with Section 5.2, restricts how much reallocation and reinstatement can actually unfold. The gains of the augmented regime, then, accrue disproportionately to those who already hold expertise. Appel et al. (2025) identifies a channel through which skill-biased technological change may already be unfolding, finding that early adopters with high-skill tasks have more successful interactions than later, less technical adopters, and observing that such users may be at once the most exposed to disruption and the most aided by it in these initial, waves of adoption. Doshi and Moore (2026) note that the benefits of GenAI accrue to either the lower end of the ability distribution in some contexts or to the most capable in others. In R2R, where augmentation is relied on in the high-expertise subprocesses, the benefit appears to favour the latter.

If the gains of the augmented regime accrue to those who already hold expertise, the question that follows for accounting is whether that advantage is fixed or can be acquired. Hallman et al. (2026) offer the closest available answer, as the only study to observe actual GenAI prompts within an accounting function. Across nearly 4,000 back-office employees of KPMG, sophisticated use, measured as the breadth of tasks to which the tool is applied, the deliberateness of prompting, and the clarity of instructions, rises with seniority, consistent with domain expertise complementing the tool. Sophistication does not improve over eight months of use, and the gain observed in the month of formal AI training does not persist. The concentration of gains among those who already hold expertise is, consequently, likely to persist rather than self-correct, and the advantage that does accrue is built through experience accumulated in the work rather than through instruction about the tool.

That accrual is compounded by the channels through which the two migrations run. Record's thinning runs through the API, which is enterprise-mediated and deployed from the top down, while the migration toward Master Data Management and Analysis & Business Insight runs through Claude.ai, which is individual and adopted from the bottom up. Bick et al. (2026) find worker adoption outpacing firm adoption across the economy,

and the process-level counterpart is that professionals are augmenting themselves into higher-value work faster than their firms are automating the routine work beneath them. Reallocation is, once again, relative to the strength of what each worker brings; its benefits stay with those who can already use the tool well, and no firm-level decision passes them on to those who cannot. Beyond the firm, this pattern has another societal dimension. If the gains of augmentation accrue to those who already hold expertise while routine entry-level work thins, the profession's pipeline for building that expertise, and the distribution of AI's benefits across workers, becomes a question for educators, professional bodies, and policy.

The human side of the interaction also points to where new tasks are likely to come from, which matters because reinstatement is the one mechanism this study cannot observe directly. Acemoglu and Restrepo (2019) distinguish a displacement effect, a productivity effect, and a reinstatement effect in which new tasks are created where labour holds comparative advantage and gets allocated to. However, following a static O*NET task list means that this study remains silent on this third effect. Doshi and Moore (2026) classify interactions that require an audit of both the process and the outcome as expert applications, where expert knowledge is needed to interpret how an outcome was reached as well as the outcome itself. Verification is, therefore, a candidate as a new task. Coordination is the second. Hitzig et al. (2026) find a clear division of labour in agentic coding, in which people decide what to build and the agent decides how, with domain expertise rather than coding proficiency amplifying effective use. If that transfers, the higher-value end of R2R will not simply absorb more human time but change in composition, with machines doing more execution and humans more direction.

The redistribution along R2R that emerges is, then, the one Section 5.2 predicted, corrected by Section 5.3, and conditioned by Section 5.4. The exposure profile predicts Record thinning, Master Data Management and Analysis & Business Insight densifying by selective delegation, and Planning & Budgeting and Report remaining human-held. Self-selection and effectiveness (RQ3) correct the prediction: the thinning of Record is arrived at through degradation rather than displacement, the delegation in the anchoring subprocesses runs ahead of reliability, and the channels through which either would produce labour-market consequences remain latent. The human half conditions what would activate them, not model capability, which is already near-universal, but the pairing of such capability with a performer whose tacit knowledge makes AI usage work, whose expertise decides who benefits, and whose verification and coordination are the tasks the fixed taxonomy yet register.

5.5 Evaluating the Task and Process Lens

These findings and interpretation demonstrate how the task-level unit of observation and the process-based unit of aggregation proposed in Sections 2.4 and 2.5 served the analysis. Task-level analysis enabled the investigation of exposure heterogeneity across platforms, subprocesses, and the expertise range within a subprocess. Every correlation and regression in Chapter 4 is estimated at this level. None of the heterogeneity described above could have been observed at the occupation level. Moreover, only at this granularity can redistribution along the process be seen at all. Tasks can dynamically move between the humans and machines that perform them, and between the subprocesses in which they sit, in a way that occupations cannot. An occupation-level frame lacks the ability to observe a clerk taking on analysis or a financial manager taking on the direction of agents. The task-level framework in Section 2.4 predicted exposure heterogeneity within an occupation; the findings show that the heterogeneity extends to which tasks move, in which direction, and in which mode.

Aggregation to a process-based frame, as Section 2.5 proposed, was what turned that granularity into structure. Without a frame to group tasks, the analysis would have shown one Record task degrading while another was handed more, with no way to tell noise from pattern; aggregated, those movements resolve into more or less homogeneous sets moving together– Record's success falling across its task set while Analysis & Business Insight's rises, the conforming and anchoring regimes, and the tiers in RQ1. Subprocess characteristics reflected those of their tasks in every analysis, in exposure, concentration, complexity, and effective coverage, which is the evidence that the aggregation surfaced structure rather than averaging it away. Aggregating to the process also registers a task moving between occupations as reallocation rather than losing it across a boundary, and it represents the continued presence of humans in a subprocess even where a particular occupation's presence declines. The falling demand for clerks that Awyong et al. (2025) document is, through an occupational frame, disappearance; through a process frame, it is the thinning of Record described above, with the residual tasks still present and still performed. This is the same adjacency that Lave and Wenger (1991) and Vendraminelli et al. (2025) describe from the human side in Section 5.4: the within- and across-subprocess reallocation that the process frame makes visible is the person being carried along the process by the technology. Task, process, capital, and labour fit together in a single frame, in which the nature of the technology allows augmentation, the nature of the human allows adjacent tasks to be taken up, and the nature of the process allows both to be seen as movement along it.

The inferences drawn in this Discussion chapter about redistribution and its beneficiaries rest on task-level usage and its success, and on external evidence about the performer, rather than on any direct observation of labour

outcomes. The lack of critical dimensions on the human side, rather than the task or the machine side, means that these inferences are early evidence rather than a confirmed trajectory, and that how humans and machines are redistributed along the process may be evolving and contingent on factors beyond the task itself. At the same time, that a study of observed usage, interpreted through the labour-economics mechanisms of Chapter 2 and read through a human-specific lens, arrives at the same reallocation outcome that Choi and Xie (2026) measure directly in time and platform activity means the signal is directionally meaningful. GenAI has the potential to reshape accounting not by simply displacing professionals, but by altering the mix of tasks they perform and by magnifying the value of expertise, and the findings of this study say where in the process, in which mode, and on what condition that reshaping is currently taking place.

6 Conclusions

This thesis set out to establish where GenAI is used within the Record-to-Report process, at what level of expertise, and how effectively, drawing on five releases of the Anthropic Economic Index rather than on capability ratings, job postings, or surveys. It finds that a process almost entirely exposed in capability is narrowly exposed in use. Exposure concentrates in a small core of tasks and is sorted by channel into API-based automation and chat-based augmentation (RQ1). Its expertise level is bimodal, with a low-expertise conforming regime in Record and a high-expertise anchoring regime in Master Data Management and Analysis & Business Insight (RQ2). Once success is accounted for, Claude effectively performs only about a quarter of process time, the subprocesses that look most automated convert least reliably into performance, and success is governed by human-only time on Claude.ai and input context on the API, rather than by the expertise a task requires (RQ3).

Theoretical contribution

Theoretically, the thesis carries the task-based account of technological change (Acemoglu & Autor, 2011; Autor & Thompson, 2025; Hampole et al., 2025) from the occupation into a single, causally ordered accounting process, contributing to current understanding of labour economics theories in four main ways. First, its concentration and expertise mechanisms hold only partially and process-specifically within R2R, and a process frame registers as reallocation what an occupational frame records as disappearance. Second, usage data reveal an allocation within capital, between an automated and an augmented channel, that is sticky across releases and that the framework's binary assignment of tasks to labour or capital cannot represent, pointing to complementarity within a task rather than only between tasks. Third, exposure is not performance: once exposure is discounted for success, the ranking of subprocesses reverses, so exposure measures, whether theoretical or observed, may overstate substitution until they account for reliability. Fourth, the expertise at which exposure concentrates is not a stable property of a task bundle but drifts as users self-select and correct, while success depends on the operational burden of a task rather than on its expertise, pointing to the performer as the variable the framework leaves out. For accounting research, the thesis supplies evidence on the layer between practitioners' perceptions and measured outcomes (Böke et al., 2024; Choi & Xie, 2026), together with a reproducible pipeline of multi-model task classification, task-level expertise estimation, and time-weighted exposure measures that other process-level studies can reuse.

Practical contribution

Practically, the findings caution finance leaders against reading adoption as delivery. Usage counts and task coverage overstate what GenAI contributes; monitoring exposure alongside success at the subprocess level would show that automation in Record and Close & Consolidation is visible but yet reliable, which argues for keeping review and reconciliation controls over AI-generated entries and for treating verification as a task around which roles can be designed. The channel to which tasks are directed should also be matched to the task, while simultaneously keep an eye out for the channel's own constraint. Codifiable, high-volume work suits API automation only where the context it requires can be bounded, so for context-heavy posting and reconciliation the investment that matters lies in data preparation and retrieval rather than in more autonomous deployment. Judgment-heavy analysis and planning benefit from augmentation directed by experienced staff. Since those gains accrue to people who already hold tacit expertise, and formal training alone may not change how well staff use the tool (Hallman et al., 2026), firms should protect the learning paths through which junior accountants acquire that expertise. Professional bodies and educators should emphasise process knowledge across adjacent subprocesses together with the direction and verification of AI output.

Limitations and Future Research

The principal limitation is the study's reliance on AI annotation, where almost every construct is derived from language models. Clio assigns each conversation to a single O*NET task, Claude-based classifiers determine collaboration mode and task success, and three models estimate the time fractions used as weights throughout. The constructs developed for this study were subject to annotations using model from multiple providers, incorporation of a human rater, and an external BLS benchmark. Still, inter-model agreement establishes reliability rather than validity, because models trained on overlapping data may share systematic biases. The success classifier is the least validated measure, as it is a Claude model assessing Claude conversations without verification by accountants. This reliance was, nevertheless, unavoidable—human annotation is infeasible at a scale that studies real-world records, and Clio's privacy design prevents access to conversation content. The results of this study thus place greater emphasis on directional accuracy than on exact precision. They should be best interpreted as early evidence of how GenAI is used in accounting work, which further data sources and methodologies can triangulate to paint a more comprehensive picture.

Second, the study's position as archival research both enables and constrains its contribution. According to Bloomfield et al. (2016), the Anthropic Economic Index is an archive with no dependent variable elicited,

no independent variable manipulated, and no setting controlled. Its strength lies in generalisation, as it captures naturally occurring usage across the whole R2R process. However, the archive records only usage. Labour outcomes are not observed, and the reallocation of work discussed in Chapter 5 is inferred from the task-based frameworks of labour economics rather than statistically tested. In addition, the static O*NET taxonomy registers only existing tasks, which leaves the reinstatement effect, the creation of new tasks in which labour holds a comparative advantage (Acemoglu & Restrepo, 2019), unobservable. The findings establish where and how GenAI is used, but not what that use causes.

Other research designs can address what archival data cannot. Field experiments that randomise access to an AI assistant, or its mode of use, across teams would provide attribution by turning the choice between automation and augmentation into a manipulated variable. Firm usage logs recording queries, acceptances, and overrides would provide contextualisation, linking usage to user characteristics such as tenure and expertise and to outcomes such as close time or error rates, as Choi and Xie (2026) demonstrate. Both designs could observe reinstatement and new tasks as they unfold, for instance whether verification and coordination emerge as distinct tasks. Decomposing O*NET tasks into Detailed Work Activities could further show not only which tasks shift but which underlying activities, and hence skills, they require. Because no single method achieves all the goals of an empirical literature (Bloomfield et al., 2016), triangulating these designs with archival usage data would enrich inferred reallocation into observed evidence of AI exposure in accounting, and the consequence that follows.

However, even the generalisation that such archive offers might be restricted. The thesis focuses only on Claude usage data, and Claude users may not be representative of all accounting professionals; API traffic classified as accounting work may originate from developers building tools rather than from accountants. This limitation bears most heavily on the subprocesses with near-zero exposure. Observed exposure of 0.06 for Report and 0.09 for Planning & Budgeting disagrees with the 31 percent adoption rate in financial planning and analysis reported by Brisbourne (2024) and with the GenAI-generated text in corporate filings detected by Blankespoor et al. (2026). Possible explanations include the use of other or private enterprise platforms. In addition, much of the methodological evidence supporting the Index is published by Anthropic and has not undergone independent peer review. Accordingly, besides triangulating across methodologies, triangulating across usage data from other providers, such as Google (Iscenko et al., 2026) and OpenAI (Chatterji et al., 2026), can paint a more comprehensive pictures of real-world AI usage in accounting.

Finally, the set of primitives is incomplete across releases, which makes the longitudinal findings early evidence. Observed exposure, concentration, and task success are available only for Releases 4 and 5. Consequently, while the persistence of self-selection and correction is supported according to simpler usage records from earlier releases, the interpretation that this correction responds to declining success rests on two releases of the least validated measure. Therefore, following the periodic nature of this initiative from Anthropic, tracking exposure in subsequent releases using the same pipeline could show whether the patterns observed here hold as more data becomes available. As further releases accumulate, the relationship between exposure, correction, and task success can be examined over a longer horizon, allowing the present findings to be confirmed, refined, or revised.

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Appendix

This part of the thesis contains the following appendices:

Appendix A. Task Categorization into Record-to-Report Subprocesses

Appendix B. Data Documentation

Appendix C. Prompts

Appendix D. Supplementary Results

All code used to produce the results of this thesis, covering the task categorisation, the data preparation in Section 3.1, and every table and figure in Chapter 4 and Appendix D, is available at:

https://github.com/vvinhhh/masters-thesis

Appendix A. Task Categorization into Record-to-Report Subprocesses

A.1 Procedure

Rationale. O*NET organises occupational tasks by job role, not by process. A Financial Manager's task list spans activities that, within the R2R framework, belong to entirely different subprocesses, from journal-entry preparation through variance analysis to forward-looking budget planning. Aligning O*NET task data with the R2R process structure is therefore a prerequisite for the exposure analysis. This mapping cannot be accomplished by keyword matching: a task description such as "Analyze financial data to ensure regulatory compliance" contains no subprocess-specific vocabulary, yet a domain-informed reader recognises it as an Analysis & Business Insight activity. The classification requires contextual reasoning, the capacity to read an abbreviated task description, activate knowledge of what that activity produces and for whom, and resolve it against a structured decision framework. The study responds to that requirement by using several large language models as annotators, as proxies for human judgement across the full task corpus.

A growing body of evidence confirms that LLMs are effective annotation instruments. Gilardi et al. (2023), Törnberg (2023), and Ziems et al. (2024) each find that LLMs match or exceed human annotators in accuracy, reliability, and inter-rater agreement across classification tasks ranging from relevance and stance detection to political ideology and emotion classification. The common argument is that LLMs encode sufficient contextual and domain background to reason about the meaning and intent behind a text rather than to match surface patterns, which makes them suited to structured categorisation work that would otherwise require extensive human annotation or supervised training data. The O*NET-to-R2R mapping falls squarely within this scope. An LLM-based pipeline is also version-controlled and replicable by any researcher with API access, addressing a recognised limitation of human annotation, and it made annotation at the required scale and domain specificity feasible within the resources of a master's thesis. Because Ziems et al. (2024) caution that LLMs can agree strongly even when inaccurate, a human rater was integrated as an independent annotator alongside the models.

Workflow. The classification proceeded in five stages. First, subprocess definitions and decision rules were written as a shared codebook, and the human rater classified all tasks before any model output was produced, to avoid anchoring. Second, a validity run on a calibration sample tested the robustness of the instrument across models, temperatures, tiers, and repeated runs. Third, informed by the validity run, a production run classified all 212 tasks, with each rater required to give a label and an explicit

chain of reasoning without seeing the other raters' outputs. Fourth, every task without unanimous agreement entered a consensus round in which each rater saw the anonymised reasoning of the others and produced a revised label; tasks reaching at least three-of-four agreement were resolved to that label. Fifth, remaining two-to-two splits were resolved by a final human decision. The output is the validated R2R task register used throughout the thesis.

Precedence. LLM classification of O*NET tasks has precedent in economics. Autor and Thompson (2025) used GPT-4.1 to classify tasks into an abstract/routine/manual taxonomy, reaching 89% agreement with a manual review of 100 tasks; Eloundou et al. (2024) used GPT-4 alongside human experts to rate more than 19,000 O*NET tasks by exposure; and Parshall and Lopez-Luzuriaga (2026) used Claude, GPT, and Gemini in a multi-model consensus pipeline over more than 23,000 tasks, with pairwise agreement rising from 75.0% before to 90.8% after consensus and classifications that track four external benchmarks. This study adapts their design to the domain-specific requirements of R2R subprocess mapping.

Input data. Task statements were drawn from Version 30.2 of the O*NET database. Guided by the extended R2R definition in Section 2.5, four major SOC groups were identified as relevant: Management (11-0000), Business and Financial Operations (13-0000), Computer and Mathematical (15-0000), and Office and Administrative Support (43-0000) occupations. Within these, ten occupations were selected for their substantive coverage of R2R activities: Chief Executives (11-1011.00), Financial Managers (11-3031.00), Treasurers and Controllers (11-3031.01), Cost Estimators (13-1051.00), Accountants and Auditors (13-2011.00), Budget Analysts (13-2031.00), Tax Preparers (13-2082.00), Database Administrators (15-1242.00), Bookkeeping, Accounting, and Auditing Clerks (43-3031.00), and Billing and Posting Clerks (43-3021.00). Their task statements yield a corpus of 212 tasks with occupation titles and task IDs.

Definitions, decision rules, and prompt design. Six subprocess labels were defined following the R2R framework of Section 2.5, combining the transactional and decision-support objectives of the finance function, and OUT_OF_SCOPE and AMBIGUOUS labels were added, giving eight allowed outputs. The decision rules served two functions. They acted as a scope filter, excluding tasks whose primary output lies outside the process boundary, such as audit and assurance, tax strategy, or general managerial coordination; and they provided a classification hierarchy that prioritises the primary purpose and output of a task over its vocabulary, with secondary rules for boundaries between adjacent subprocesses. The system prompt followed the guidelines of Ziems et al. (2024): instructions followed the context material, labels were listed separately, chain-of-thought reasoning preceded the label, and responses were returned as JSON. Each rater

assigned exactly one label, matching the single-assignment structure of the AEI data, and no subprocess examples were given (the response-format block contains a single format example, an OUT_OF_SCOPE task). The full definitions, decision rules, and prompt are reproduced in Appendix C.2.

Validity run. A 30% stratified sample of 64 tasks, reflecting the human rater's subprocess distribution, was classified by six models across two tiers of three providers (Claude Sonnet and Opus, GPT-mini and GPT flagship, Gemini Flash and Pro) at temperatures 0.2 and 1.0 with two runs each, giving 24 labels per task. Reliability was measured with Krippendorff's nominal α, chosen because subprocess categories are not evenly spaced on any underlying dimension; confusion matrices located the disagreements. The 24-rater instrument returned $\alpha = 0.739$ (95% CI [0.644, 0.818]), mean pairwise agreement of 0.842, and full agreement on 60.9% of tasks, with mean task-level consensus strength of 0.88. Intra-model stability at temperature 0.2 was near-perfect ($\alpha \geq 0.90$ for every model) against 0.785 to 0.908 at temperature 1.0, establishing 0.2 as the production setting. Mid-tier models were more internally consistent ($\alpha = 0.797$) than flagship models ($\alpha = 0.757$) and agreed more with the human rater for all three providers, because flagship models assigned OUT_OF_SCOPE 7 to 14 percentage points more often than the human (65 to 72% against 57.8%). Correlations between each model's subprocess volume profile and the human profile ranged from 0.988 to 0.995. Disagreement concentrated at the boundary between R2R and OUT_OF_SCOPE rather than within R2R; where raters agreed a task was in scope, they largely agreed on the subprocess, with residual confusion between Analysis & Business Insight and Planning & Budgeting and between Record and Close & Consolidation. Mid-tier models assigned OUT_OF_SCOPE at 53.1 to 62.5%, close to the human rater, whereas Claude Opus recorded the lowest human alignment of any model (70.3%, $\alpha = 0.538$). The scope-boundary disagreement is structurally explainable: the Master Data Management tasks of Database Administrators extend into general infrastructure management, some Record tasks combine recording with preceding clerical or physical steps, and some Report tasks involve consulting legal documentation before reporting, which Decision Rule 0 excludes as preparatory. The minimum task-level consensus strength of 0.32 marks a genuine boundary task rather than an instrument failure. The production run therefore used a single run at temperature 0.2 with the mid-tier model of each provider.

Production run. The three models classified all 212 tasks, and the human classification from the initial labelling exercise was carried forward as the fourth rater. Across the three models, the instrument returned $\alpha = 0.739$ (95% CI [0.674, 0.795]), mean pairwise agreement of 0.835, and full agreement on 76.4% of tasks; the stable α confirms that reliability did not degrade when scaling from the calibration sample to the full corpus, while

the higher full-agreement rate reflects the smaller number of raters. Agreement with the human declined by about 0.07 to 0.10 α units per model, as expected for a broader and harder corpus. Of the 212 tasks, 131 reached unanimous agreement and were resolved directly. The mean task-level consensus strength remained 0.87. The dominant axis of disagreement remained in-scope versus OUT_OF_SCOPE (70 dissenting rater votes), while the two within-R2R axes, Record versus Close & Consolidation (5 dissenting votes) and Analysis & Business Insight versus Planning & Budgeting (8), reflected task descriptions that state what is done rather than why: the former a divergence on where the transaction-to-general-ledger posting boundary sits, the latter descriptions carrying keywords from both subprocesses.

Consensus round. For the 81 disputed tasks, each rater received the anonymised, randomised reasoning, but not the labels, of the other raters (prompt in Appendix C.3); AMBIGUOUS was removed from the label set, and the human rater completed the round before seeing model outputs. Of the 81 tasks, 73 were resolved by majority, 34 with full agreement and 39 by a three-to-one majority, and 8 remained split (seven two-to-two and one two-one-one) and were resolved by human judgement. Krippendorff's α rose from 0.739 to 0.875 (95% CI [0.824, 0.919]), full agreement from 76.4% to 88.7%, and mean pairwise agreement from 0.835 to 0.925, with roughly symmetric improvement across models, indicating revision through shared reasoning rather than convergence on the most confident model. Residual disagreement fell into three patterns: three models applying the scope rule more conservatively than the human; adjacent-subprocess splits such as task 21531 ("Develop, maintain, or analyze budgets, preparing periodic reports that compare budgeted costs to actual costs"); and the eight split tasks whose descriptions span several subprocesses. In four further tasks (80, 2494, 7366 and 21529) the human rater overrode a three-model OUT_OF_SCOPE majority, and these overrides clustered into documentable model weaknesses: domain-prior contamination (a cost-estimation task read as construction work), misapplied audit carve-outs (preparer-side verification read as assurance), vague-purpose tasks read literally, and coordination-level tasks whose R2R relevance is implicit. These patterns support Ziems et al.'s (2024) case for including human annotators alongside LLMs.

A.2 Outcome

The final register contains 91 in-scope tasks across the six subprocesses and 121 OUT_OF_SCOPE tasks (Table A1). Record is the most populated subprocess (31 tasks), followed by Planning & Budgeting (22), Report (13), Analysis & Business Insight (12), Master Data Management (8), and Close & Consolidation (5). Of the 91 in-scope tasks, 39 were resolved unanimously in the production run and 52 in the consensus round: 18 with full agreement, 28 by a three-to-one majority (including the four human overrides of a three-

model OUT_OF_SCOPE majority), and 6 by human decision. Table A2 lists every in-scope task.

Table A1 reports the resolution path of the 212 tasks by final label.

Table A1: Resolution path of the classified O*NET tasks by final label.

Final label	Production run	Consensus round			Total
	Unanimous (4/4)	Unanimous	Majority (3/4)	Human decision (2/2)	
Master Data Management	5	0	2	1	8
Record	14	8	8	1	31
Close & Consolidation	2	1	2	0	5
Report	6	4	2	1	13
Analysis & Business Insight	6	1	3	2	12
Planning & Budgeting	6	4	11	1	22
In-scope total	**39**	**18**	**28**	**6**	**91**
OUT_OF_SCOPE	92	16	11	2	121
All tasks	**131**	**34**	**39**	**8**	**212**

Note: Raters: Claude, GPT, and Gemini mid-tier models at temperature 0.2 and one human rater. Source: final task register. The Majority (3/4) column includes four tasks (80, 2494, 7366, 21529) in which the human rater overrode a three-model OUT_OF_SCOPE majority; Human decision covers seven two-to-two splits and one two-one-one split (task 21323).

Table A2 lists the in-scope tasks by subprocess.

Table A2: Final register of in-scope O*NET tasks by R2R subprocess.

Task ID	Task statement (O*NET 30.2)	Resolution
Master Data Management (n = 8)		
89	Set up cost monitoring and reporting systems and procedures.	Unanimous, production run
1305	Specify users and user access levels for each segment of database.	Unanimous, production run
1306	Develop data models describing data elements and how they are used, following procedures and using pen, template, or computer software.	Majority (3/4), consensus round
1315	Revise company definition of data as defined in data dictionary.	Unanimous, production run

Task ID	Task statement (O*NET 30.2)	Resolution
8837	Coordinate the development or implementation of budgetary control systems, recordkeeping systems, or other administrative control processes.	Majority (3/4), consensus round
21323	Evaluate financial reporting systems, accounting or collection procedures, or investment activities and make recommendations for changes to procedures, operating systems, budgets, or other financial control functions.	Human decision (split vote)
21517	Establish tables of accounts and assign entries to proper accounts.	Unanimous, production run
21521	Develop, implement, modify, and document recordkeeping and accounting systems, making use of current computer technology.	Unanimous, production run
Record (n = 31)		
2486	Check figures, postings, and documents for correct entry, mathematical accuracy, and proper codes.	Unanimous, consensus round
2487	Operate computers programmed with accounting software to record, store, and analyze information.	Unanimous, consensus round
2489	Debit, credit, and total accounts on computer spreadsheets and databases, using specialized accounting software.	Unanimous, production run
2490	Classify, record, and summarize numerical and financial data to compile and keep financial records, using journals and ledgers or computers.	Unanimous, production run
2491	Calculate, prepare, and issue bills, invoices, account statements, and other financial statements according to established procedures.	Unanimous, consensus round
2493	Code documents according to company procedures.	Human decision (2/2 split)
2496	Reconcile or note and report discrepancies found in records.	Unanimous, consensus round
2497	Perform financial calculations, such as amounts due, interest charges, balances, discounts, equity, and principal.	Unanimous, production run
2499	Prepare bank deposits by compiling data from cashiers, verifying and balancing receipts, and sending cash, checks, or other forms of payment to banks.	Unanimous, production run
2500	Receive, record, and bank cash, checks, and vouchers.	Unanimous, production run
2501	Calculate and prepare checks for utilities, taxes, and other payments.	Unanimous, production run
2502	Compare computer printouts to manually maintained journals to determine if they match.	Majority (3/4), consensus round
2503	Reconcile records of bank transactions.	Unanimous, production run
2506	Transfer details from separate journals to general ledgers or data processing sheets.	Majority (3/4), consensus round
2509	Match order forms with invoices, and record the necessary information.	Unanimous, production run
2511	Maintain inventory records.	Unanimous, consensus round
8861	Receive, record, and authorize requests for disbursements in accordance with company policies and procedures.	Majority (3/4), consensus round

Task ID	Task statement (O*NET 30.2)	Resolution
8872	Receive cash and checks and make deposits.	Unanimous, production run
8873	Compute, withhold, and account for all payroll deductions.	Unanimous, consensus round
18561	Prepare and process payroll information.	Unanimous, consensus round
21326	Review collection reports to determine the status of collections and the amounts of outstanding balances.	Majority (3/4), consensus round
21515	Prepare adjusting journal entries.	Majority (3/4), consensus round
21529	Direct activities of personnel engaged in filing, recording, compiling, and transmitting financial records.	Human override, consensus round
21533	Process invoices for payment.	Unanimous, production run
23238	Verify accuracy of billing data and revise any errors.	Majority (3/4), consensus round
23240	Prepare itemized statements, bills, or invoices and record amounts due for items purchased or services rendered.	Unanimous, production run
23241	Perform bookkeeping work, including posting data or keeping other records concerning costs of goods or services or the shipment of goods.	Unanimous, production run
23245	Review documents, such as purchase orders, sales tickets, charge slips, or hospital records, to compute fees or charges due.	Unanimous, production run
23246	Keep records of invoices and support documents.	Majority (3/4), consensus round
23251	Compare previously prepared bank statements with canceled checks and reconcile discrepancies.	Unanimous, production run
23264	Create billing documents, shipping labels, credit memorandums, or credit forms.	Unanimous, consensus round
Close & Consolidation (n = 5)		
2504	Prepare trial balances of books.	Unanimous, production run
7366	Check data input or verify totals on forms prepared by others to detect errors in arithmetic, data entry, or procedures.	Human override, consensus round
21516	Review accounts for discrepancies and reconcile differences.	Unanimous, production run
21522	Evaluate taxpayer finances to determine tax liability, using knowledge of interest and discount rates, annuities, valuation of stocks and bonds, and amortization valuation of depletable assets.	Unanimous, consensus round
23243	Resolve discrepancies in accounting records.	Majority (3/4), consensus round
Report (n = 13)		
87	Prepare cost and expenditure statements and other necessary documentation at regular intervals for the duration of the project.	Unanimous, production run
2492	Compile statistical, financial, accounting, or auditing reports and tables pertaining to such matters as cash receipts, expenditures, accounts payable and receivable, and profits and losses.	Unanimous, production run

Task ID	Task statement (O*NET 30.2)	Resolution
2510	Complete and submit tax forms and returns, workers' compensation forms, pension contribution forms, and other government documents.	Unanimous, production run
7358	Compute taxes owed or overpaid, using adding machines or personal computers, and complete entries on forms, following tax form instructions and tax tables.	Majority (3/4), consensus round
7359	Prepare or assist in preparing simple to complex tax returns for individuals or small businesses.	Unanimous, consensus round
7362	Review financial records, such as income statements and documentation of expenditures to determine forms needed to prepare tax returns.	Unanimous, consensus round
8834	Prepare or present reports concerning activities, expenses, budgets, government statutes or rulings, or other items affecting businesses or program services.	Unanimous, consensus round
8855	Prepare and file annual tax returns or prepare financial information so that outside accountants can complete tax returns.	Unanimous, production run
8856	Prepare or direct preparation of financial statements, business activity reports, financial position forecasts, annual budgets, or reports required by regulatory agencies.	Unanimous, consensus round
21320	Prepare financial or regulatory reports required by laws, regulations, or boards of directors.	Unanimous, production run
21520	Report to management regarding the finances of establishment.	Human decision (2/2 split)
21526	Compute taxes owed and prepare tax returns, ensuring compliance with payment, reporting, or other tax requirements.	Unanimous, production run
23263	Compile reports of cost factors, such as labor, production, storage, and equipment.	Majority (3/4), consensus round
Analysis & Business Insight (n = 12)		
81	Assess cost effectiveness of products, projects or services, tracking actual costs relative to bids as the project develops.	Unanimous, production run
2494	Access computerized financial information to answer general questions as well as those related to specific accounts.	Human override, consensus round
3394	Analyze monthly department budgeting and accounting reports to maintain expenditure controls.	Unanimous, production run
3401	Review operating budgets to analyze trends affecting budget needs.	Unanimous, production run
3403	Perform cost-benefit analyses to compare operating programs, review financial requests, or explore alternative financing methods.	Human decision (2/2 split)
8825	Analyze operations to evaluate performance of a company or its staff in meeting objectives or to determine areas of potential cost reduction, program improvement, or policy change.	Unanimous, production run
8867	Analyze the financial details of past, present, and expected operations to identify development opportunities and areas where improvement is needed.	Unanimous, production run
8875	Determine depreciation rates to apply to capitalized items and advise management on actions regarding the purchase, lease, or disposal of such items.	Majority (3/4), consensus round
21314	Prepare operational or risk reports for management analysis.	Unanimous, production run

Task ID	Task statement (O*NET 30.2)	Resolution
21321	Develop or analyze information to assess the current or future financial status of firms.	Unanimous, consensus round
21506	Report to management about asset utilization and audit results, and recommend changes in operations and financial activities.	Majority (3/4), consensus round
21532	Prepare, analyze, or verify annual reports, financial statements, and other records, using accepted accounting and statistical procedures to assess financial condition and facilitate financial planning.	Human decision (2/2 split)
Planning & Budgeting (n = 22)		
80	Analyze blueprints and other documentation to prepare time, cost, materials, and labor estimates.	Human override, consensus round
84	Prepare estimates used by management for purposes such as planning, organizing, and scheduling work.	Unanimous, production run
91	Conduct special studies to develop and establish standard hour and related cost data or to reduce cost.	Unanimous, consensus round
2507	Compile budget data and documents, based on estimated revenues and expenses and previous budgets.	Majority (3/4), consensus round
2508	Calculate costs of materials, overhead, and other expenses, based on estimates, quotations and price lists.	Majority (3/4), consensus round
3395	Direct the preparation of regular and special budget reports.	Unanimous, consensus round
3396	Consult with managers to ensure that budget adjustments are made in accordance with program changes.	Majority (3/4), consensus round
3397	Match appropriations for specific programs with appropriations for broader programs, including items for emergency funds.	Majority (3/4), consensus round
3398	Provide advice and technical assistance with cost analysis, fiscal allocation, and budget preparation.	Majority (3/4), consensus round
3399	Summarize budgets and submit recommendations for the approval or disapproval of funds requests.	Unanimous, production run
3402	Examine budget estimates for completeness, accuracy, and conformance with procedures and regulations.	Majority (3/4), consensus round
3404	Interpret budget directives and establish policies for carrying out directives.	Unanimous, consensus round
3405	Compile and analyze accounting records and other data to determine the financial resources required to implement a program.	Unanimous, production run
8823	Direct or coordinate an organization's financial or budget activities to fund operations, maximize investments, or increase efficiency.	Human decision (2/2 split)
8827	Prepare budgets for approval, including those for funding or implementation of programs.	Unanimous, production run
8865	Coordinate and direct the financial planning, budgeting, procurement, or investment activities of all or part of an organization.	Unanimous, production run
8868	Advise management on short-term and long-term financial objectives, policies, and actions.	Majority (3/4), consensus round
20846	Collect historical cost data to estimate costs for current or future products.	Unanimous, consensus round
21315	Evaluate data pertaining to costs to plan budgets.	Unanimous, production run

Task ID	Task statement (O*NET 30.2)	Resolution
21519	Analyze business operations, trends, costs, revenues, financial commitments, and obligations to project future revenues and expenses or to provide advice.	Majority (3/4), consensus round
21531	Develop, maintain, or analyze budgets, preparing periodic reports that compare budgeted costs to actual costs.	Majority (3/4), consensus round
23260	Review compiled data on operating costs and revenues to set rates.	Majority (3/4), consensus round

Note: Task IDs and statements from O*NET 30.2. Resolution indicates the stage at which the final label was reached (Table A1); “Human override” marks a human decision against a three-model OUT_OF_SCOPE majority.

Appendix B. Data Documentation

B.1 Data Sources and Structure

The analysis combines two tables that join on task_id. The task master holds one row per O*NET task statement with its subprocess label, occupation, wage, ridge-estimated expertise, and time fraction (Table B1). The AEI panel is a long-format consolidation of the Anthropic Economic Index releases, built so that every release, whatever its native schema, is expressed in the same columns (Table B2). The panel is already restricted to in-scope tasks: rows whose task was labelled OUT_OF_SCOPE, or whose text did not resolve to any task in the task master, were dropped during the build. For Releases 1 to 5, panel rows were matched to task_id by normalised-text matching on the task statement (lower-cased, whitespace collapsed, trailing period stripped) rather than by any release-specific identifier, and ambiguous or unmatched text was dropped upstream.

The columns of both tables are documented below.

Table B1: Columns of the task master table (one row per O*NET task).

Column	Type	Description
task_id	string	Canonical O*NET task identifier and join key to the panel
task_statement	string	Original O*NET task text
subprocess	string	R2R subprocess from the final register (Appendix A), or OUT_OF_SCOPE
occupation_id	string	O*NET-SOC code of the occupation the task belongs to
occupation_title	string	Title of that occupation
wage_hourly_mean	float (USD)	BLS OEWS mean hourly wage of the occupation (task value)
wage_annual_mean	float (USD)	BLS OEWS mean annual wage of the occupation
ridge_task_years	float (years)	Ridge-estimated task expertise (Section 3.1.2); distinct from the AEI education primitives
task_time_fraction	float	Elicited weekly hours spent on the task (mean of three models), renormalised to a share within each scope at the point of use (Section 3.1.1)
frequency_occ_per_year	float	Occurrences of the task per year in the occupation; not used in the final constructs (Section 3.2.3)

Note: Each task maps to exactly one occupation.

Table B2: Columns of the long-format AEI panel (one row per metric value).

Column	Type	Description
release	integer	AEI release from which the row originates
platform	string	claude_ai or 1p_api
task_id	string	Join key to the task master
task_statement	string	Task text as it appeared in the release; may differ trivially from the task master text
dimension	string	Analysis facet, e.g. usage, collaboration, use_case, task_success, ai_autonomy (Table B3)
category	string	Value within a categorical dimension; blank for numeric dimensions
level	integer	Hierarchy depth within the facet; 0 = most granular
variable	string	Metric type: count, pct, index, mean, median, stdev, or confidence-interval bound
value	float	Metric value; pct is normalised to a 0 to 1 fraction across all releases; index uses 1.0 as baseline

Note: Confidence-interval bounds (mean_ci_lower/upper, median_ci_lower/upper) are 95% intervals. The panel is the output of the build and refine scripts described in the data documentation.

B.2 Dimensions and Release Coverage

Table B3 lists the dimensions in the panel and the releases in which each is published. Two features of release coverage shape the analysis. First, constructs that require the work-use classification, observed exposure and concentration, can only be computed from Release 4. Second, the panel extends to Release 6, but Release 6 does not publish the task-success facet required for effective exposure and publishes the human-only-ability and multitasking facets only as single shares, so the analysis uses Releases 1 to 5.

Coverage by dimension:

Table B3: Dimensions of the AEI panel and the releases in which they are available.

Dimension	Categories / unit	Releases	Used in
usage	count, pct of usage	1 to 6	Usage share; WorkUsage; Gini
collaboration	directive, feedback_loop, task_iteration, validation, learning, none (R2: filtered)	2 to 6	Collaboration mode; α_t
collaboration_bucket	automation, augmentation	6	—
use_case	work, personal, coursework, not_classified	4 to 6	WorkUsage; Table 6
human_only_ability	yes, no, not_classified (R6: yes share only)	4 to 6	Table 6
multitasking	as human_only_ability	4 to 6	—
task_success	yes, no, not_classified	4 to 5	$success_t$; effective exposure; Tables 7 to 8
human_only_time	hours	4 to 6	Figure 22; Table 7
human_with_ai_time	minutes	4 to 6	Table 6
ai_autonomy	1 to 5 scale	4 to 6	Figure 23; Tables 7 to 8
human_education_years	years (0 to 20)	4 to 6	Table 6
ai_education_years	years (0 to 20)	4 to 6	Table 6
artifact	32 artifact types	6	—
prompt_tokens, completion_tokens, cost	count, index (1P API only)	3 to 6	Figures 24 to 26; Table 8

Note: Collaboration categories are named differently in Release 2 (filtered instead of none/not_classified) and must be harmonised before pooling across releases.

B.3 Economic Primitives

From Release 4, each conversation in the AEI sample is annotated by a classifier prompted separately for each economic primitive, and the published values are task-level aggregates of those annotations (Appel et al., 2026). The primitives describe not whether a task is performed with Claude but how: how long it would take a professional, how much of the decision-making is delegated, how demanding the exchange is, and whether it succeeds. Table B4 defines each primitive and its role in this thesis. Two cautions apply when reading them. First, they are model-generated judgements about a conversation, not measurements; human-only time, for instance, is the classifier's estimate of how long a competent professional would need, not observed working time. Second, the education primitives describe the prompt and the response, not the task, which is why Section 3.1.2 constructs an independent task-level expertise measure.

Definitions of the primitives:

Table B4: AEI economic primitives and their use in this thesis.

Primitive	Definition	Scale	Use in this thesis
Human-only time	Hours a competent professional would need to complete the task without AI, assuming full domain knowledge, context, and tool access	Hours	Mediator of success on Claude.ai (Figure 22; Table 7)
Human-with-AI time	Minutes the user spent completing the task within the conversation	Minutes	Descriptive (Table 6)
Human-only ability	Whether the user could have completed the task without AI	Yes / no	Difficulty indicator (Table 6)
Multitasking	Whether the conversation covers a single task or several	Yes / no	Not used
Human education years	Years of formal education needed to understand the user's prompts	0 to 20	Descriptive (Table 6)
AI education years	Years of formal education needed to understand the assistant's responses	0 to 20	Descriptive (Table 6)
Use case	Whether the conversation serves work, coursework, or personal purposes	Category	Work filter for WorkUsage_t (Section 3.2.1)
AI autonomy	How much decision-making the assistant exercised	1 to 5	Figure 23; Tables 7 to 8
Task success	Whether the assistant completed the requested task successfully; averaged to a task-level rate	Share	success_t and effective exposure (Section 3.2.3); Tables 7 to 8
Token and cost indices	Prompt-token, completion-token, and cost indices relative to the economy-wide average, re-indexed to the R2R task set	Index	Figures 24 to 26; Table 8 (1P API only)

Note: Definitions follow Section 3.1.4 and the AEI release documentation. Automation comprises the directive and feedback-loop collaboration categories and augmentation the task-iteration, validation, and learning categories (Handa et al., 2025).

Appendix C. Prompts

C.1 Time-Fraction Elicitation Prompt

The prompt was sent as a single user message, without a system prompt, to claude-sonnet-4-6, gpt-5.4-mini and gemini-3.1-flash-lite on 15 July 2026: one call per model per occupation, each containing the occupation's complete O*NET task list (Core and Supplemental; 212 tasks across the ten occupations). No temperature was set (provider defaults); Claude was called with max_tokens = 4096. Replies were parsed as JSON, and the final estimate of each task is the unweighted mean of the three models' weekly hours, renormalised within the relevant scope at the point of use (Section 3.1.1). The task list is inserted as one “task_id: task statement” line per task.

```
You are estimating how much time workers in the occupation "<O*NET occupation
title>" spend on each of their job tasks. Below is the complete list of tasks for
this occupation. For each task, estimate how many hours per week a typical worker
spends on it.

Important: Don't worry about making the hours sum to exactly 40 or any specific
total - we'll normalize the results afterward. Just give your best estimate for
each task independently based on what seems realistic.

Tasks: <task_id>: <task statement>
<task_id>: <task statement>
...

Return ONLY a JSON object mapping each task_id to your estimated hours per week,
with no additional text, explanations, or commentary. Format: {"task_id_1": hours,
"task_id_2": hours, ...}
```

C.2 Subprocess Definitions, Decision Rules, and Classification Prompt

Production run (June 2026): claude-sonnet-4-6 (max_tokens = 512), gpt-5.4-mini (structured output) and gemini-3.1-flash-lite (JSON response schema), all at temperature 0.2, one call per model per task (636 calls). The system prompt below contains the six subprocess definitions, the AMBIGUOUS and OUT_OF_SCOPE labels, Decision Rule 0 (scope, with the Master Data Management carve-in and the audit/assurance and tax-planning carve-outs), the primary purpose rule and the output-type fallback, the secondary boundary rules (Rule 3, Master Data Management; Rule 4, Record versus Close & Consolidation) and the single-label rule. Each task was then sent as the user message shown after it.

System prompt:

```
You are an expert in Record-to-Report (R2R) process classification. Your task is to
assign each O*NET task statement to exactly one of the six R2R subprocesses below,
or flag it as AMBIGUOUS or OUT_OF_SCOPE.

SUBPROCESS DEFINITIONS:

1. Master Data Management — Establishing and maintaining the structural and
governance foundation of the financial architecture: the chart of accounts,
reporting entity hierarchies, cost and profit centre structures, company-specific
attributes, currency rate tables, system interfaces, and user access rights.
Provides the infrastructure on which all downstream subprocesses depend, without
```

itself processing period transactions.

2. Record — The ongoing accumulation and initial processing of financial transactions during the fiscal period, per accepted accounting principles: processing journal entries, posting transactions to subledgers, recording accruals, and performing transaction-level reconciliations as activity occurs.

3. Close & Consolidation — The period-end procedures that finalise and aggregate recorded transactions into group-level financial results: closing subledgers, executing general ledger accounting, finalising accruals against the close deadline, reconciling GL accounts and resolving variances, managing intercompany accounting and eliminations, calculating tax liability (different from tax filings), and consolidating entity results into preliminary group financial statements.

4. Report — Translating the closed and consolidated ledger into structured outputs for stakeholders: producing internal financial management reports, external financial statements, and tax filings, in order to meet the organisation's internal and external disclosure obligations.

5. Analysis & Business Insight — Interrogating reported financial results to generate interpretive, decision-useful intelligence: performing variance and performance analysis, developing and maintaining KPIs, and providing insight intended to influence management decisions, including post-close review and identification of needed adjustments.

6. Planning & Budgeting — Translating analytical findings into forward-looking financial direction: developing long-range strategic plans, annual budgets and operating plans, rolling forecasts, product costing, and performance targets that govern the next iteration of the cycle.

7. AMBIGUOUS — Use only when both of the following conditions are met:
(a) The task is clearly within R2R scope — it constitutes financial recording, consolidation, disclosure, analysis, or planning work (Rule 0 is satisfied).
(b) Rules 1–4, applied in sequence, still do not resolve the task to a single subprocess — because the task description genuinely and equally spans two or more subprocesses with no resolvable priority between them.

AMBIGUOUS is not a hedge for difficult tasks. If the rules resolve the task, even with effort, do not use AMBIGUOUS. Reserve it for cases where the task description is constitutively split — where assigning a single label would misrepresent what the task actually does.

8. OUT_OF_SCOPE — Use when Rule 0 is not satisfied: the task does not itself perform R2R work, regardless of its organisational context or eventual purpose.

DECISION RULES:

Apply Rules 0 through 4 strictly in numerical order, one rule at a time. Fully resolve or fully rule out each rule before moving to the next — do not jump ahead to a later rule to justify a classification that an earlier rule would have already determined differently. Begin every task with Rule 0.

Rule 0 — Before applying this rule, read the subprocess definitions carefully. The scope of R2R in this study is determined exclusively by those definitions, not simply general intuitions about what constitutes "financial" work. Do not exclude a task on the grounds that it resembles operational, engineering, or non-accounting work if the subprocess definitions would encompass it. A task is in scope if it constitutes activity within any of the six subprocess definitions provided. If the task instead consists of data-gathering, or cross-functional communication that precedes and informs subprocess activity — but does not itself constitute such activity as defined — classify as OUT_OF_SCOPE.

Exception — Master Data Management carve-in: A task that maintains, establishes, or governs a structural element of the financial architecture (chart of accounts, entity hierarchy, cost/profit centres, currency tables, system access) IS

constitutive R2R activity, even though it does not itself process a transaction, figure, report, or plan. Treat it as in-scope and proceed to Rule 3.

Carve-out — Audit/assurance activity: A task describing audit, assurance, or internal control work — testing controls, verifying compliance, examining records or processes for accuracy as an independent check, assessing the adequacy of a control environment — is OUT_OF_SCOPE, even when it involves reviewing financial records or transactions. Its purpose is independent verification of work already performed, not the performance of the recording, consolidation, disclosure, analysis, or planning itself. (O*NET groups "Accountants and Auditors" under shared occupational codes; this carve-out separates the two distinct job functions.)

Carve-out — Tax planning/strategy: A task describing tax planning, tax strategy, or general tax advisory (e.g., developing tax-minimisation strategies, advising on entity tax structuring) is OUT_OF_SCOPE. It does not itself produce a recorded entry, consolidated tax provision, filed disclosure, or budget. Distinguish from tax provision calculation (→ Close & Consolidation) and tax filing/return preparation (→ Report), which remain in scope. Distinguish from other financial planning and strategic activities, which remain in scope.

Rule 1 — Purpose of output (primary rule). Classify by why the output is being produced, not by the activity that produces it. A task that prepares reports for management analysis is Analysis & Business Insight, not Report, because the purpose is decision support, not disclosure.

Note: This rule routes a task to Analysis & Business Insight only when the analysis interrogates the organisation's own recorded financial or accounting data (e.g., variance to budget, financial KPI trend, post-close review). Analysis of external information — market research, competitor benchmarking, macroeconomic or industry forecasting — does not qualify, even if the task is described as "financial analysis"; such tasks fail Rule 0 and are OUT_OF_SCOPE.

Rule 2 — Output type (fallback when Rule 1 leaves no clear purpose). Classify by the type of artefact produced:
- A recorded or reconciled entry → Record
- A consolidated group figure → Close & Consolidation
- A disclosure document or regulatory filing → Report
- A budget, forecast, or performance target → Planning & Budgeting

Rule 3 — Maintenance vs. execution (Master Data Management boundary). A task framed as maintaining, establishing, or governing a structural element (chart of accounts, entity hierarchy, cost/profit centres, currency tables, system access) → Master Data Management, regardless of which subprocess uses that structure. A task that uses such structure to process a period's transactions → the relevant operational subprocess per Rules 1–2.

Rule 4 — Transaction-level vs. period-end (Record vs. Close & Consolidation). Is the task processing an individual transaction, or closing out a period? Transaction-level reconciliations and adjustments → Record. Account-balance or period-end reconciliations, adjustments, and reviews → Close & Consolidation.

Rule 5 — Single-label enforcement. Apply Rules 0–4 in sequence to arrive at exactly one label, even where a task description spans multiple subprocesses. Multi-subprocess phrasing alone is not grounds for AMBIGUOUS. Reserve AMBIGUOUS only for genuine unresolved contestation after all rules are exhausted, where both purpose and output rules cannot resolve which subprocess to put the task in.

ALLOWED LABELS — use exactly as written:
- Master Data Management
- Record
- Close & Consolidation
- Report
- Analysis & Business Insight
- Planning & Budgeting
- AMBIGUOUS
- OUT_OF_SCOPE

```
RESPONSE FORMAT: Return raw JSON only — no other text before or after, and no
markdown code fences (do not wrap the JSON in ```json ... ```). Your response must
start with the character { and end with the character }.

{
"reasoning": "<in 1 to 2 sentences, strictly maximum 60 words: name the
determinative rule (e.g. 'Rule 0 exception' or 'Rule 4') and the fact that
triggered it — do not restate rule text or walk through every rule considered>",
"classification": "<exactly one of the allowed labels>"
}

Example (illustrative of format only — not a subprocess example):
Input task: "Liaise with business units to collect headcount data for inclusion in
the annual workforce plan."
Output:
{
"reasoning": "Data-gathering that feeds planning but produces no record itself;
Rule 0, out of scope.",
"classification": "OUT_OF_SCOPE"
}
```

User message:

```
Task ID: {task_id}
Task Statement: {task_statement}

Classify this task.
```

C.3 Consensus-Round Prompt

Consensus round: the same three models, 81 disputed tasks, 243 calls; no temperature was set (provider defaults). For each task and model, the production-round reasoning of the other three raters – the human rater and the two other models – was shown without their labels, anonymised as Rater A to C in an order shuffled separately for every task–model pair (the orders used are logged in consensus_prompts.csv). AMBIGUOUS was removed from the label set. The prompt is reproduced as sent; Rule 0 contains a typo (“defnition”) and Rule 5 still refers to AMBIGUOUS.

System prompt:

```
You are an expert in Record-to-Report (R2R) process classification, participating
in a deliberative consensus round.

In an earlier independent round, multiple raters classified the same O*NET task
statements into subprocesses of the R2R accounting process. You will be shown the
reasoning from those other raters — but not their labels — for tasks where the
initial round produced a split.

Your task is to read each argument carefully, weigh its merits against the
subprocess definitions and decision rules below, and provide your own independent
classification.

SUBPROCESS DEFINITIONS:

1. Master Data Management — Establishing and maintaining the structural and
governance foundation of the financial architecture: the chart of accounts,
reporting entity hierarchies, cost and profit centre structures, company-specific
attributes, currency rate tables, system interfaces, and user access rights.
Provides the infrastructure on which all downstream subprocesses depend, without
itself processing period transactions.
```

2. Record — The ongoing accumulation and initial processing of financial transactions during the fiscal period, per accepted accounting principles: processing journal entries, posting transactions to subledgers, recording accruals, and performing transaction-level reconciliations as activity occurs.

3. Close & Consolidation — The period-end procedures that finalise and aggregate recorded transactions into group-level financial results: closing subledgers, executing general ledger accounting, finalising accruals against the close deadline, reconciling GL accounts and resolving variances, managing intercompany accounting and eliminations, calculating tax liability (different from tax filings), and consolidating entity results into preliminary group financial statements.

4. Report — Translating the closed and consolidated ledger into structured outputs for stakeholders: producing internal financial management reports, external financial statements, and tax filings, in order to meet the organisation's internal and external disclosure obligations.

5. Analysis & Business Insight — Interrogating reported financial results to generate interpretive, decision-useful intelligence: performing variance and performance analysis, developing and maintaining KPIs, and providing insight intended to influence management decisions, including post-close review and identification of needed adjustments.

6. Planning & Budgeting — Translating analytical findings into forward-looking financial direction: developing long-range strategic plans, annual budgets and operating plans, rolling forecasts, product costing, and performance targets that govern the next iteration of the cycle.

DECISION RULES:

Apply Rules 0 through 4 strictly in numerical order, one rule at a time. Fully resolve or fully rule out each rule before moving to the next — do not jump ahead to a later rule to justify a classification that an earlier rule would have already determined differently. Begin every task with Rule 0.

Rule 0 — Before applying this rule, read the subprocess definitions carefully. The scope of R2R in this study is determined exclusively by those definitions, not simply general intuitions about what constitutes "financial" work. Do not exclude a task on the grounds that it resembles operational, engineering, or non-accounting work if the subprocess definitions would encompass it. A task is in scope if it constitutes activity within any of the six subprocess definitions provided. If the task instead consists of data-gathering, or cross-functional communication that precedes and informs subprocess activity — but does not itself constitute such activity according to the defnition — classify as OUT_OF_SCOPE.

Exception — Master Data Management carve-in: A task that maintains, establishes, or governs a structural element of the financial architecture (chart of accounts, entity hierarchy, cost/profit centres, currency tables, system access) IS constitutive R2R activity, even though it does not itself process a transaction, figure, report, or plan. Treat it as in-scope and proceed to Rule 3.

Carve-out — Audit/assurance activity: A task describing audit, assurance, or internal control work — testing controls, verifying compliance, examining records or processes for accuracy as an independent check, assessing the adequacy of a control environment — is OUT_OF_SCOPE, even when it involves reviewing financial records or transactions. Its purpose is independent verification of work already performed, not the performance of the recording, consolidation, disclosure, analysis, or planning itself. (O*NET groups "Accountants and Auditors" under shared occupational codes; this carve-out separates the two distinct job functions.)

Carve-out — Tax planning/strategy: A task describing tax planning, tax strategy, or general tax advisory (e.g., developing tax-minimisation strategies, advising on entity tax structuring) is OUT_OF_SCOPE. It does not itself produce a recorded entry, consolidated tax provision, filed disclosure, or budget. Distinguish from tax provision calculation (→ Close & Consolidation) and tax filing/return

preparation (→ Report), which remain in scope. Distinguish from other financial planning and strategic activities, which remain in scope.

Rule 1 — Purpose of output (primary rule). Classify by why the output is being produced, not by the activity that produces it. A task that prepares reports for management analysis is Analysis & Business Insight, not Report, because the purpose is decision support, not disclosure.

Note: This rule routes a task to Analysis & Business Insight only when the analysis interrogates the organisation's own recorded financial or accounting data (e.g., variance to budget, financial KPI trend, post-close review). Analysis of external information — market research, competitor benchmarking, macroeconomic or industry forecasting — does not qualify, even if the task is described as "financial analysis"; such tasks fail Rule 0 and are OUT_OF_SCOPE.

Rule 2 — Output type (fallback when Rule 1 leaves no clear purpose). Classify by the type of artefact produced:
- A recorded or reconciled entry → Record
- A consolidated group figure → Close & Consolidation
- A disclosure document or regulatory filing → Report
- A budget, forecast, or performance target → Planning & Budgeting

Rule 3 — Maintenance vs. execution (Master Data Management boundary). A task framed as maintaining, establishing, or governing a structural element (chart of accounts, entity hierarchy, cost/profit centres, currency tables, system access) → Master Data Management, regardless of which subprocess uses that structure. A task that uses such structure to process a period's transactions → the relevant operational subprocess per Rules 1–2.

Rule 4 — Transaction-level vs. period-end (Record vs. Close & Consolidation). Is the task processing an individual transaction, or closing out a period? Transaction-level reconciliations and adjustments → Record. Account-balance or period-end reconciliations, adjustments, and reviews → Close & Consolidation.

Rule 5 — Single-label enforcement. Apply Rules 0–4 in sequence to arrive at exactly one label, even where a task description spans multiple subprocesses. Multi-subprocess phrasing alone is not grounds for AMBIGUOUS. Reserve AMBIGUOUS only for genuine unresolved contestation after all rules are exhausted, where both purpose and output rules cannot resolve which subprocess to put the task in.

ALLOWED LABELS — use exactly as written:
- Master Data Management
- Record
- Close & Consolidation
- Report
- Analysis & Business Insight
- Planning & Budgeting
- OUT_OF_SCOPE

IMPORTANT INSTRUCTIONS:
- AMBIGUOUS is NOT valid at this stage. You must commit to one of the six subprocesses or OUT_OF_SCOPE. The purpose of this round is to resolve uncertainty, not carry it forward.
- Focus on the quality of the arguments, not the number of raters who held a position.
- Do not anchor to your own prior classification. Re-examine the task fresh against the rules.
- Apply the rules in order (Rule 0 first); stop at the first rule that resolves the case.

RESPONSE FORMAT: Respond with a valid JSON object only — no other text before or after.

```
{
"reasoning": "<1-2 sentences: engage with the other raters' arguments, name the rule that resolves the case, state why>",
```

```
"classification": "<exactly one of the allowed labels>"
}
```

User message:

```
Task ID: {task_id}
Task Statement: {task_statement}

--- Reasoning from other raters (anonymised, labels withheld) ---

Rater A:
{reasoning of 1st shuffled rater}

Rater B:
{reasoning of 2nd shuffled rater}

Rater C:
{reasoning of 3rd shuffled rater}

---

Having read the arguments above, provide your own independent classification of
this task. Remember: AMBIGUOUS is not valid at this stage.
```

Appendix D. Supplementary Results

This appendix collects Release 4 counterparts of Release 5 results cited in Chapter 4, per-release robustness checks, and diagnostics of the constructed variables.

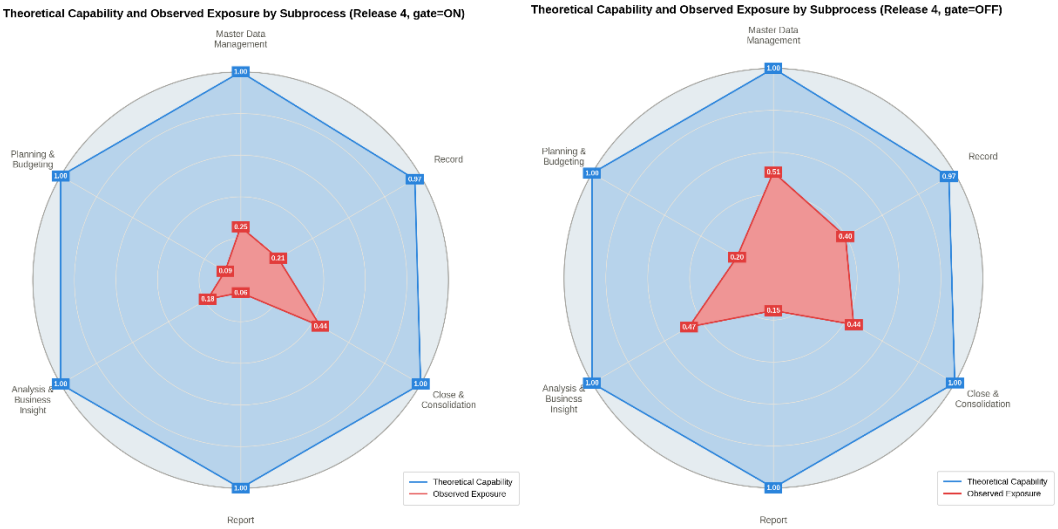


Figure D1: Theoretical capability and observed exposure by subprocess, Release 4, gate = ON (left) and gate = OFF (right).

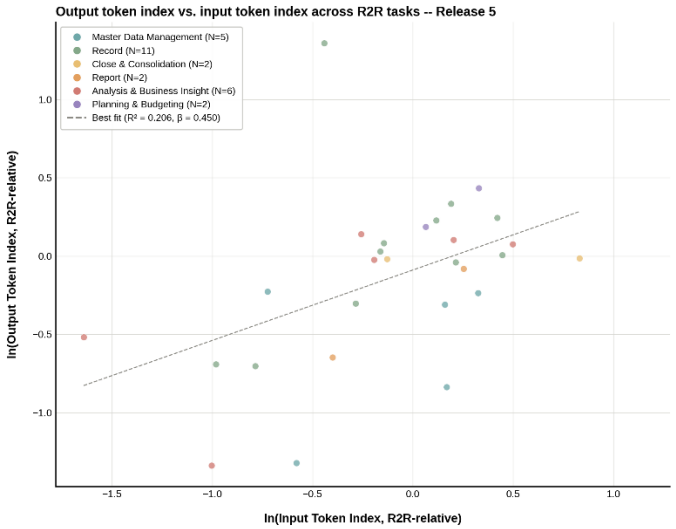


Figure D2: Input against output context-length index, 1P API, Release 5.

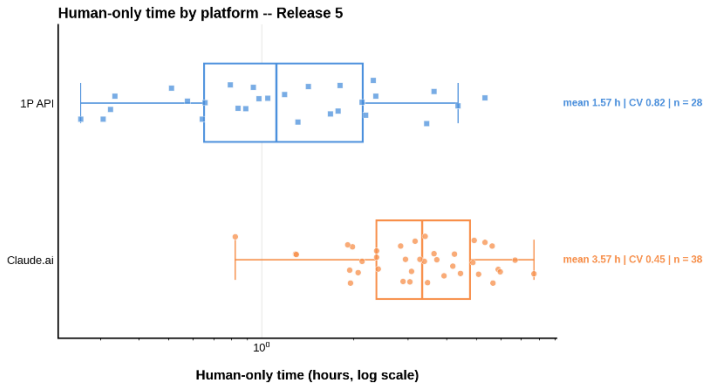


Figure D3: Distribution of human-only time by platform, Release 5.

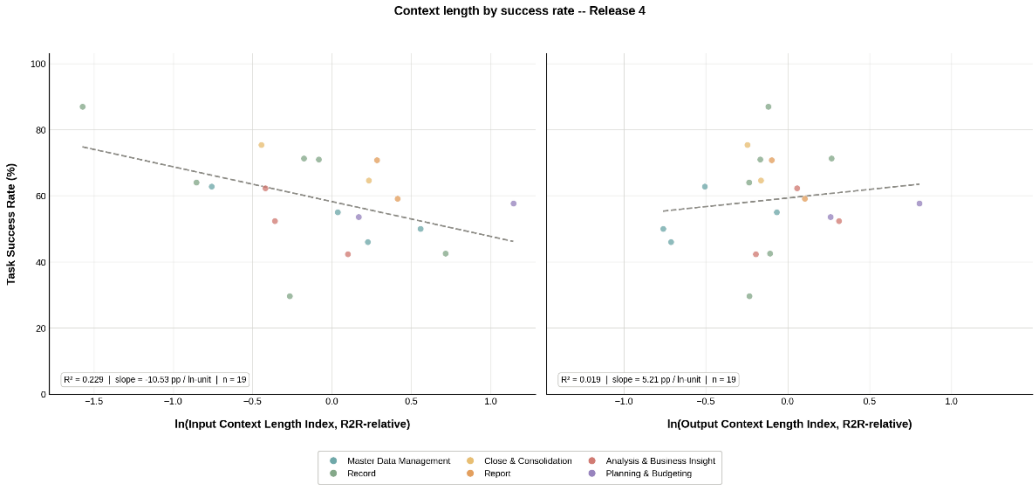


Figure D4: Task success rate against log-relative input and output context-length indices, 1P API, Release 4.

Table D1: OLS regressions of 1P API task success rate, estimated separately by release (specifications as in Table 8).

Panel A. Release 4

	(1) Input index	(2) + Output index	(3) + Autonomy	(4) + Subprocess FE
ln(input context-length index)	−10.532**	−11.869**	−8.171	−11.559*
	(4.681)	(4.742)	(5.157)	(5.720)
ln(output context-length index)		9.857	6.159	5.728
		(8.054)	(8.105)	(15.610)
AI autonomy (1 to 5)			−19.274	−22.868
			(12.599)	(17.464)
Constant	58.222***	59.091***	112.840***	122.434**
	(2.823)	(2.872)	(35.243)	(46.416)
Subprocess fixed effects	No	No	No	Yes
F-test of fixed effects				F(5, 10) = 1.24
				[p = 0.359]
Observations	19	19	19	19
R^2	0.229	0.295	0.390	0.624

Panel B. Release 5

	(1) Input index	(2) + Output index	(3) + Autonomy	(4) + Subprocess FE
ln(input context-length index)	−19.855***	−21.439***	−20.250***	−24.744**
	(4.177)	(5.372)	(6.460)	(8.240)
ln(output context-length index)		2.621	3.608	11.950
		(5.397)	(6.204)	(8.144)
AI autonomy (1 to 5)			−4.131	−7.404
			(11.681)	(16.372)
Constant	53.102***	53.567***	64.989*	83.220
	(1.991)	(2.249)	(32.381)	(47.106)
Subprocess fixed effects	No	No	No	Yes
F-test of fixed effects				F(5, 11) = 1.02
				[p = 0.453]
Observations	20	20	20	20
R^2	0.557	0.563	0.566	0.703

Note: Dependent variable: task success rate (%) on the 1P API; unit of observation: task. Conventional OLS standard errors in parentheses; the F-test compares specification (4) with (3). * p < 0.10, ** p < 0.05, *** p < 0.01.

Table D2: Task success rate by subprocess, platform, and release.

Subprocess	Platform	Release 4 (n)	Release 5 (n)	Change (pp)
Entire Process	Claude.ai	67.8 (27)	68.4 (31)	+0.60
Entire Process	1P API	58.8 (19)	54.3 (20)	−4.52
Master Data Management	Claude.ai	74.4 (3)	69.5 (4)	−4.83
Master Data Management	1P API	53.5 (4)	59.8 (5)	+6.34
Record	Claude.ai	69.7 (8)	67.1 (10)	−2.53
Record	1P API	60.9 (6)	51.0 (8)	−9.98
Close & Consolidation	Claude.ai	57.1 (2)	64.8 (2)	+7.65
Close & Consolidation	1P API	70.0 (2)	52.3 (2)	−17.71
Report	Claude.ai	59.2 (2)	58.4 (2)	−0.77
Report	1P API	64.9 (2)	46.6 (1)	−18.29
Analysis & Business Insight	Claude.ai	58.6 (5)	68.7 (5)	+10.15
Analysis & Business Insight	1P API	52.3 (3)	60.5 (3)	+8.15
Planning & Budgeting	Claude.ai	75.0 (7)	72.7 (8)	−2.35
Planning & Budgeting	1P API	55.7 (2)	46.7 (1)	−8.94

Note: Unweighted mean of task-level success rates (%) across tasks with a reading on the platform, number of tasks in parentheses; change is Release 5 minus Release 4 in percentage points. Same convention as Table 6.

Table D3: Pearson correlations among the task-level regressors of Tables 7 and 8 (ridge expertise, AI autonomy, human-only time, ln input and ln output context-length indices) and task success, by platform; task × release, Releases 4 and 5 pooled (same samples as Tables 7 and 8).

	Ridge expertise	AI autonomy	Human-only time	ln input index	ln output index	Task success
Claude.ai (N = 58)						
Ridge expertise	1.00	0.46	0.49	—	—	−0.03
AI autonomy	0.46	1.00	0.65	—	—	−0.14
Human-only time	0.49	0.65	1.00	—	—	−0.50
Task success	−0.03	−0.14	−0.50	—	—	1.00
1P API (N = 39)						
Ridge expertise	1.00	0.30	—	0.20	0.12	0.03
AI autonomy	0.30	1.00	—	0.54	0.38	−0.49
ln input index	0.20	0.54	—	1.00	0.41	−0.58
ln output index	0.12	0.38	—	0.41	1.00	−0.13
Task success	0.03	−0.49	—	−0.58	−0.13	1.00

Table D4: Share of usage carried by the ten most-used tasks, by platform and release (%).

Release	Claude.ai	1P API
Release 1	**83.8**	**—**
Release 2	77.2	—
Release 3	84.4	93.2
Release 4	84.3	88.1
Release 5	82.6	85.3

Note: Usage shares are renormalised to the in-scope R2R tasks of each platform and release; the 1P API is available from Release 3.

Diagnostics of the task-expertise model (Section 3.1.2):

Table D5: Diagnostics of the ridge regression used to estimate task-level expertise.

Item	Value
Merged O*NET–BLS corpus	18,217 tasks across 893 occupations
Training pool	18,005 tasks across 883 occupations (corpus excluding all tasks of the 10 study occupations)
Prediction set	212 tasks of the 10 study occupations (91 in scope)
Text representation	all-mpnet-base-v2 sentence embeddings
Target	BLS typical entry education, mapped to years
Regularisation strength (α)	1.0 (grid of 25 log-spaced values from 0.01 to 10,000, 5-fold GroupKFold by occupation)
Grouped cross-validation (by occupation): R^2	0.570 (SD 0.034)
Grouped cross-validation: RMSE (years)	1.852
Grouped cross-validation: Spearman ρ	0.751
Random K-fold cross-validation: R^2	0.608
Leakage gap (random − grouped R^2)	0.038
Occupation-level aggregation: R^2 against BLS (training pool, in-sample)	0.753

Note: Values as reported in Section 3.1.2. Cross-validation uses 5-fold GroupKFold by occupation; the random K-fold comparison uses 5 shuffled folds (seed 42). The occupation-level aggregation R^2 is computed in-sample on the training pool. Source: code/data_preparation/02_ridge_task_expertise.ipynb (outputs/02_ridge_expertise/tableD5_ridge_diagnostics.csv).

Table D6: Cross-model agreement in the time-fraction elicitation.

Item	Value
Models	Claude Sonnet 4.6, GPT-5.4-mini, Gemini-3.1-flash-lite
Rows rated	212 task–occupation rows
Krippendorff's α (ordinal)	0.681
Mean weekly hours per task: Claude / GPT / Gemini	2.73 / 2.13 / 3.76
Share of rows where Gemini gives the highest estimate	71.7% (91.5% including ties)

Note: Final time fractions are the mean of the three models' estimates, renormalised within scope (Section 3.1.1). Mean weekly hours are over all 212 task–occupation rows; over the 91 in-scope tasks they are 3.38 (Claude), 2.66 (GPT) and 4.46 (Gemini), and Gemini gives the highest estimate for 69.2% of them. Summed over each occupation's task list, the three models average 57.9, 45.2 and 79.7 hours per week.

Table D7: Decomposition of effective exposure minus task coverage (Table 4) into a weight component (touched tasks carrying less or more process time than their count share) and a success component (touched tasks not reliably completed), in percentage points, by subprocess and release, with counts of touched tasks that are low-weight, low-success, or both.

Scope	Task coverage (%)	Effective exposure (%)	Gap (pp)	Weight component (pp)	Success component (pp)	Mean success of touched tasks (%)
Release 4						
Entire Process	42.9	23.4	−19.5	+1.2	−20.7	51.7
Master Data Management	62.5	21.9	−40.6	−8.3	−32.3	48.3
Record	48.4	25.7	−22.7	+4.5	−27.2	43.7
Close & Consolidation	40.0	31.5	−8.5	+6.0	−14.6	63.6
Report	15.4	10.5	−4.9	+0.9	−5.8	62.0
Analysis & Business Insight	58.3	30.4	−27.9	+1.6	−29.5	49.4
Planning & Budgeting	36.4	24.1	−12.3	+0.3	−12.6	65.2
Release 5						
Entire Process	45.0	25.7	−19.3	+1.6	−21.0	53.5
Master Data Management	75.0	37.7	−37.3	−1.7	−35.6	52.5
Record	41.9	26.5	−15.4	+5.0	−20.4	51.3
Close & Consolidation	40.0	29.2	−10.8	+5.8	−16.6	58.5
Report	30.8	9.3	−21.5	+0.8	−22.3	27.6
Analysis & Business Insight	58.3	35.9	−22.4	+2.2	−24.6	57.9
Planning & Budgeting	40.9	26.9	−14.0	+0.6	−14.6	64.2

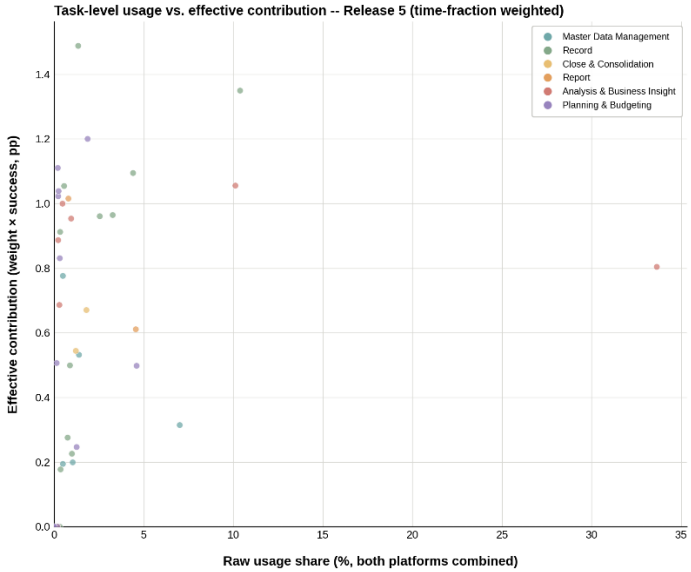


Figure D5: Task-level raw usage share against effective contribution (time-fraction weight × success rate), Release 5. Dashed lines mark the medians.

Table D8: Specifications of Tables 7 and 8 with standard errors clustered by task (CR1), without and with a Release 5 indicator.

Panel A. Claude.ai (specifications of Table 7)

	(1) Expertise only	(2) + Autonomy	(3) + Human-only time	(4) + Subprocess FE
Clustered SE (task)				
Ridge expertise (years)	−0.217	0.428	2.082	−0.256
	(1.257)	(1.624)	(1.234)	(1.799)
AI autonomy (1 to 5)		−6.943	10.530	6.289
		(7.907)	(7.661)	(7.870)
Human-only time (hours)			−5.741***	−5.815***
			(1.217)	(1.191)
Constant	71.293***	84.087***	23.876	70.105**
	(17.723)	(21.413)	(16.957)	(28.442)
Subprocess fixed effects	No	No	No	Yes
Observations	58	58	58	58
R^2	0.001	0.022	0.349	0.451
Clustered SE (task) + release indicator				
Ridge expertise (years)	−0.211	0.484	2.129*	−0.212
	(1.268)	(1.645)	(1.249)	(1.795)
AI autonomy (1 to 5)		−7.412	10.106	5.681
		(7.902)	(7.685)	(7.928)
Human-only time (hours)			−5.736***	−5.835***
			(1.242)	(1.207)
Release 5 indicator	0.578	1.312	1.140	1.533
	(2.112)	(2.203)	(2.151)	(2.138)
Constant	70.902***	84.064***	23.911	70.623**
	(17.857)	(21.355)	(16.695)	(28.239)
Subprocess fixed effects	No	No	No	Yes
Observations	58	58	58	58
R^2	0.001	0.025	0.351	0.454

Panel B. 1P API (specifications of Table 8)

	(1) Input index	(2) + Output index	(3) + Autonomy	(4) + Subprocess FE
Clustered SE (task)				
ln(input context-length index)	−14.198***	−15.471***	−12.261***	−16.406***
	(3.216)	(2.949)	(3.251)	(2.678)
ln(output context-length index)		3.964	5.601	9.367
		(3.336)	(4.259)	(6.081)
AI autonomy (1 to 5)			−13.767	−16.939
			(8.859)	(10.056)
Constant	55.670***	56.215***	94.451***	107.734***
	(2.147)	(2.242)	(23.816)	(29.646)
Subprocess fixed effects	No	No	No	Yes
Observations	39	39	39	39
R^2	0.342	0.355	0.410	0.557
Clustered SE (task) + release indicator				
ln(input context-length index)	−14.213***	−15.218***	−11.365***	−15.431***
	(2.977)	(2.893)	(3.357)	(2.672)
ln(output context-length index)		3.135	4.781	8.670
		(3.356)	(4.307)	(6.535)
AI autonomy (1 to 5)			−16.169	−18.924
			(9.549)	(11.117)
Release 5 indicator	−4.575*	−4.209	−5.609*	−4.406
	(2.463)	(2.558)	(2.711)	(2.776)
Constant	58.015***	58.259***	103.845***	115.327***
	(2.870)	(2.892)	(26.404)	(33.695)
Subprocess fixed effects	No	No	No	Yes
Observations	39	39	39	39
R^2	0.372	0.381	0.453	0.583

Note: Each task can appear twice in the pooled sample, so conventional standard errors may overstate precision. Standard errors in parentheses; p-values on G − 1 degrees of freedom (G = number of tasks). * p < 0.10, ** p < 0.05, *** p < 0.01.

Table D9: Descriptive statistics of task-level variables, in-scope R2R tasks, Release 4 (Release 4 counterpart of Table 3).

Variable	Platform	N	Mean	SD	Min	Median	Max
Task time fraction (hours/week, raw)	—	91	3.50	2.13	0.58	3.23	11.00
Ridge task expertise (years)	—	91	14.23	1.46	9.30	14.47	17.11
Task value (USD/hour)	—	91	49.44	29.48	24.55	45.56	129.63
Theoretical exposure (beta)	—	91	0.66	0.25	0.00	0.50	1.00
Work usage (conversations)	Pooled	32	262.2	508.9	16.0	95.5	2601.0
Automation factor (alpha)	Pooled	32	0.83	0.15	0.50	0.85	1.00
Task observed exposure (r~), gate = ON	Pooled	91	0.15	0.33	0.00	0.00	1.00
Task success (%)	Claude.ai	27	67.82	13.29	37.42	67.77	94.12
Task success (%)	1P API	19	58.81	13.56	29.69	59.13	86.96
Human-only time (hours)	Claude.ai	35	3.12	1.68	0.80	2.82	7.28
Human-only time (hours)	1P API	26	1.32	0.87	0.28	1.06	3.50
AI autonomy (1 to 5)	Claude.ai	35	3.13	0.30	2.41	3.16	3.72
AI autonomy (1 to 5)	1P API	26	2.79	0.28	2.27	2.79	3.41
ln(input context-length index)	1P API	26	−0.20	0.66	−1.57	−0.13	1.14
ln(output context-length index)	1P API	26	−0.07	0.38	−0.76	−0.14	0.80
ln(cost index)	1P API	26	−0.14	0.54	−1.11	−0.18	0.97

Table D10: Stepwise OLS regressions of Claude.ai task success rate (specifications of Table 7), estimated separately for Release 4 and Release 5.

	(1) Expertise only	(2) + Autonomy	(3) + Human-only time	(4) + Subprocess FE
Release 4				
Ridge expertise (years)	−0.876	−0.787	2.147	−0.844
	(1.762)	(2.110)	(1.892)	(2.370)
AI autonomy (1 to 5)		−0.864	19.820*	12.769
		(10.739)	(10.394)	(12.217)
Human-only time (hours)			−6.789***	−5.828**
			(1.856)	(2.360)
Constant	80.604***	82.025**	−3.421	57.098
	(25.834)	(31.730)	(34.781)	(52.883)
Subprocess fixed effects	No	No	No	Yes
F-test of fixed effects				F(5, 18) = 1.08 [p = 0.404]
Observations	27	27	27	27
R^2	0.010	0.010	0.374	0.519
Release 5				
Ridge expertise (years)	0.377	1.409	2.239	0.496
	(1.506)	(1.631)	(1.390)	(1.870)
AI autonomy (1 to 5)		−12.024	3.134	−0.075
		(8.094)	(8.023)	(8.140)
Human-only time (hours)			−5.062***	−6.109***
			(1.422)	(1.616)
Constant	62.952***	86.883***	43.639	81.337**
	(21.955)	(26.876)	(25.640)	(34.707)
Subprocess fixed effects	No	No	No	Yes
F-test of fixed effects				F(5, 22) = 1.39 [p = 0.268]
Observations	31	31	31	31
R^2	0.002	0.075	0.370	0.521

Note: Conventional OLS standard errors in parentheses; the F-test compares specification (4) with (3). * p < 0.10, ** p < 0.05, *** p < 0.01.